SOFTWARE REQUIREMENT SPECIFICATION

(SRS)

UNIWISE: A Web-Based Personalised Educational Recommendation System for SPM Graduates Using RIASEC Personality and Academic Eligibility Matching

Version 1.0

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Revision History

Date	Version	Author	Revision Details
	Version 1.0		Initial Document Creation

Document Approval

The following Software Requirements and Specification has been accepted and approved by the following:

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	Dr Harnani Binti Mat Zin	

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CHAPTER 1: INTRODUCTION

1.0 Introduction

This Software Requirements Specification (SRS) describes the requirements for the UniWise system. The purpose of this document is to explain the functional and non-functional requirements of the system.

UniWise is a web-based educational recommendation system developed to help SPM graduates choose suitable higher education programmes based on their academic results and RIASEC personality assessment.

This document will be used as a reference throughout the system development process to ensure that all requirements are clearly defined and implemented.

1.1 Purpose

The purpose of this UniWise system is to:

1. Provides Malaysian SPM graduates who are looking for suitable higher education programmes.

2. Allows users receive programme recommendations based on academic eligibility and personality matching.

3. Generated using a rule-based approach. The system focuses only on Diploma, Foundation, and Certificate programmes offered by Malaysian public higher education institutions (IPTA).

1.2 Scope

The system serves two main user groups:

1. **Users:** To help SPM graduates discover suitable higher education programmes based on their academic qualifications and personality. The system provides personalised recommendations to support users in making informed study choices.
2. **Administrator:** To manage programme information, user data, RIASEC assessment questions, and recommendation rules for the UniWise system.

1.3 Definitions, Acronyms and Abbreviations

Term	Description
SRS	Software Requirements Specification
SPM	Sijil Pelajaran Malaysia
IPTA	Public Higher Education Institution in Malaysia
RIASEC	Holland's personality theory consisting of Realistic, Investigative, Artistic, Social, Enterprising, and Conventional personality types
Rule-Based Recommendation	A recommendation method that generates results based on predefined rules
React	Frontend framework used to develop the user interface
Laravel	Backend framework used to develop the system
MySQL	Database management system used to store system data
API	Application Programming Interface used for communication between the frontend and backend
User	A registered user or administrator who uses the UniWise system

1.4 Reference

Rosli, M. A. H. (2026). *Requirement Analysis & Design Specification for UniWise: Final Year Project Archive*. Universiti Pendidikan Sultan Idris (UPSI).

1.5 Overview

This Software Requirements Specification (SRS) is organised into several chapters to describe the UniWise system.

Chapter 1 introduces the purpose, scope, definitions, references, and overview of the document.

Chapter 2 describes the general description of the system, including the product perspective, product functions, user characteristics, operating environment, and design constraints.

Chapter 3 presents the specific requirements of the system, including functional requirements, non-functional requirements, external interface requirements, and system features.

The remaining chapters describe the system design, database design, diagrams, and other supporting information related to the development of UniWise.

Chapter 2: General Description

2.1 Product Perspective

UniWise serves as a centralized web-based educational recommendation platform designed to assist Malaysian SPM graduates in selecting suitable public higher education (IPTA) programmes.

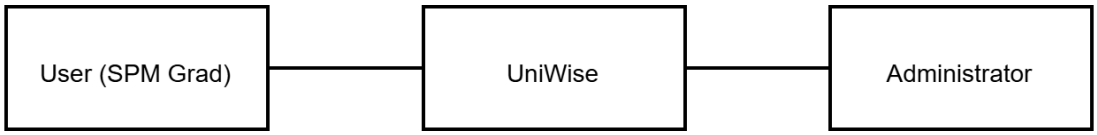


Figure2.1 UniWise Context Diagram

Figure 2.1 illustrates the context diagram for the UniPeace/UniWise application. The system interacts with two main external user entities: User (SPM Graduate) and Administrator. The User interacts with the UniWise system to enter their SPM examination results and complete the RIASEC personality assessment. Based on the inputs provided, the system automatically evaluates academic eligibility and personality trait alignment to generate personalized programme recommendations. The Administrator manages the core database records and rules within the UniWise platform, including IPTA programme details, RIASEC assessment questions, recommendation rules, and user accounts.

The application is developed using React for the frontend user interface, Laravel for backend logic and API services, and MySQL for data persistence. Authentication is handled via Firebase Authentication, and the system is accessible to users through modern web browsers with active internet connectivity.

2.2 Product Functions

The main functions provided by UniWise are described as follows:

- **User Authentication and Profile Management:**

  The system provides a secure authentication subsystem powered by Firebase Authentication. This subsystem enables new users to register an account, log in securely using encrypted credentials, and manage active user sessions. Authenticated users can modify profile parameters such as full name, profile image, and account password and securely terminate sessions via the logout function.

- **Academic Assessment and Eligibility Validation:**

The system facilitates the input of Sijil Pelajaran Malaysia (SPM) examination results by allowing users to select their academic stream alongside universal and elective subject grades. An automated validation module processes these entries to verify subject completeness, prevent duplication, and evaluate minimum academic entry criteria. This process computes an overall merit score percentage to ensure that only academically suitable higher education programmes are evaluated for recommendation.

- **RIASEC Personality Assessment and Trait Computation:**

The platform incorporates Holland's RIASEC vocational personality theory to evaluate user psychological traits. Users answer a series of standardized assessment items presented sequentially. The backend processes the user's responses, aggregates category scores across six personality dimensions (Realistic, Investigative, Artistic, Social, Enterprising, and Conventional), and identifies the user's dominant personality trait to support rule-based matching.

- **Personalized Programme Recommendation Engine:**

Using a rule-based matching algorithm, the system synthesizes the user's calculated academic merit score, specific subject grades, and dominant RIASEC personality code. By matching these parameters against predefined institution database rules, the system filters and generates a list of eligible Malaysian public higher education (IPTA) programmes. Users can inspect detailed programme profiles, including level of study (Diploma, Foundation, or Certificate), offering institution, and specific entry requirements (Syarat Kemasukan).

- **Recommendation History and Audit Logging:**

To ensure data persistence and eliminate redundant assessment sessions, the system logs a comprehensive session snapshot upon each recommendation execution. This snapshot captures the timestamp, evaluated academic scores, dominant RIASEC trait, and generated list of recommended programmes into the MySQL relational database. Users can access their historical assessment logs at any time to review previous recommendations.

2.3 User Characteristics

2.3.1 User Class Definition

User Class	Description
Guest	Can access the UniWise website and view basic information about the system with limited access to features.
User	Uses the system to obtain suitable higher education programme recommendations based on academic results and RIASEC personality assessment.
Administrator	Manages programme data, user information, RIASEC assessment questions, and recommendation rules within the system.

2.3.2 User Access Functions

Function	Guest	User	Administrator
Register	/		
Login		/	/
Logout		/	/
View Landing Page	/	/	/
View System Information	/	/	/
Manage Profile		/	/
Enter SPM Results		/	
Edit SPM Results		/	
Answer RIASEC Assessment		/	
View RIASEC Result		/	
Generate Programme Recommendation		/	
View Programme Recommendation		/	
View Programme Information		/	/

Function	Guest	User	Administrator
View Recommendation History		/	
Manage User Accounts			/
Manage Programme Information			/
Manage RIASEC Questions			/
Update Recommendation Rules			/
Maintain System Data			/

2.4 Constraints and Assumptions

2.4.1 Constraints

CR_ID	Constraint Requirement
CR001	Internet connectivity is required to access the UniWise system and use all system functions.
CR002	The system must be compatible with modern web browsers such as Google Chrome, Microsoft Edge, and Mozilla Firefox.
CR003	Users must register an account and log in before accessing the recommendation features of the system.
CR004	Passwords must follow basic security requirements to protect user accounts and personal information.
CR005	The system recommendation is generated using a rule-based approach based on academic eligibility and RIASEC personality matching.
CR006	The system only supports Malaysian public higher education institution (IPTA) programmes.
CR007	Only Diploma, Foundation, and Certificate programmes are included in the recommendation.

CR_ID	Constraint Requirement
CR008	Users must enter their SPM examination results and complete the RIASEC assessment before programme recommendations can be generated.
CR009	Administrator access is required to manage programme data, RIASEC questions, user accounts, and recommendation rules.

2.4.2 Assumptions

The assumptions and dependencies used in this system are as follows:

1. **Users will have basic computer literacy.**

- Users and administrators are expected to have basic knowledge of using a computer or mobile device, such as navigating web pages, entering information into forms, and interacting with buttons or menus in the system.

2. **Users will provide accurate SPM examination results.**

- The system assumes that users enter their actual SPM results correctly, as these results are used to determine academic eligibility and generate suitable programme recommendations.

3. **Users will answer the RIASEC assessment honestly.**

- The recommendation generated by UniWise depends on the responses provided in the RIASEC personality assessment. Honest answers are important to ensure that the recommended programmes match the user's interests and personality.

4. **Programme information and recommendation rules will be maintained by the administrator.**

- The administrator is expected to update programme details, entry requirements, RIASEC questions, and recommendation rules so that the system can provide accurate and up-to-date recommendations.

5. **Users will access the system through a supported web browser with internet connection.**

- Since UniWise is a web-based system, users are expected to use a supported browser such as Google Chrome, Microsoft Edge, or Mozilla Firefox, together with a stable internet connection.

CHAPTER 3: Specific Requirements

3.1 External Interface Requirements

3.1.1 System Interface

The UniWise system will be developed as a web-based application and can be accessed through modern web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. The system interface is designed to be simple and user-friendly so that users and administrators can navigate the system easily.

UniWise will provide interfaces for user registration, login, profile management, SPM result entry, RIASEC personality assessment, programme recommendation viewing, and recommendation history. The system will also support responsive design to ensure that it can be accessed from different devices such as desktops, laptops, tablets, and smartphones.

3.1.2 Hardware Interface

The hardware requirements for accessing the UniWise system include a device capable of running a modern web browser and connecting to the internet.

For desktop or laptop users, the recommended minimum requirements are:

- 2GB RAM
- 1.0 GHz processor or above
- Stable internet connection

For mobile users, the recommended minimum requirements are:

- Android 9 or later
- iOS 12 or later
- Stable internet connection

These requirements ensure that users can access the system and use its features smoothly.

3.1.3 Communication Interface

The UniWise system uses HTTPS to provide secure communication between users and the system. This helps protect user data such as login credentials, SPM results, and recommendation records.

Communication between the frontend and backend is handled through RESTful APIs, where the React frontend sends requests to the Laravel backend and receives responses in JSON format. This interface allows the system to process user data, generate recommendations, and retrieve programme information efficiently.

3.2 User Interface

The user interface of UniWise is designed to be clean, consistent, and easy to navigate so that both users and administrators can complete their tasks with minimal effort. The interface follows a simple visual hierarchy, clear labelling, and responsive layouts that adapt to desktop, tablet, and mobile screens.

3.2.1 Screenflow Diagram and description

3.2.1.1 Login

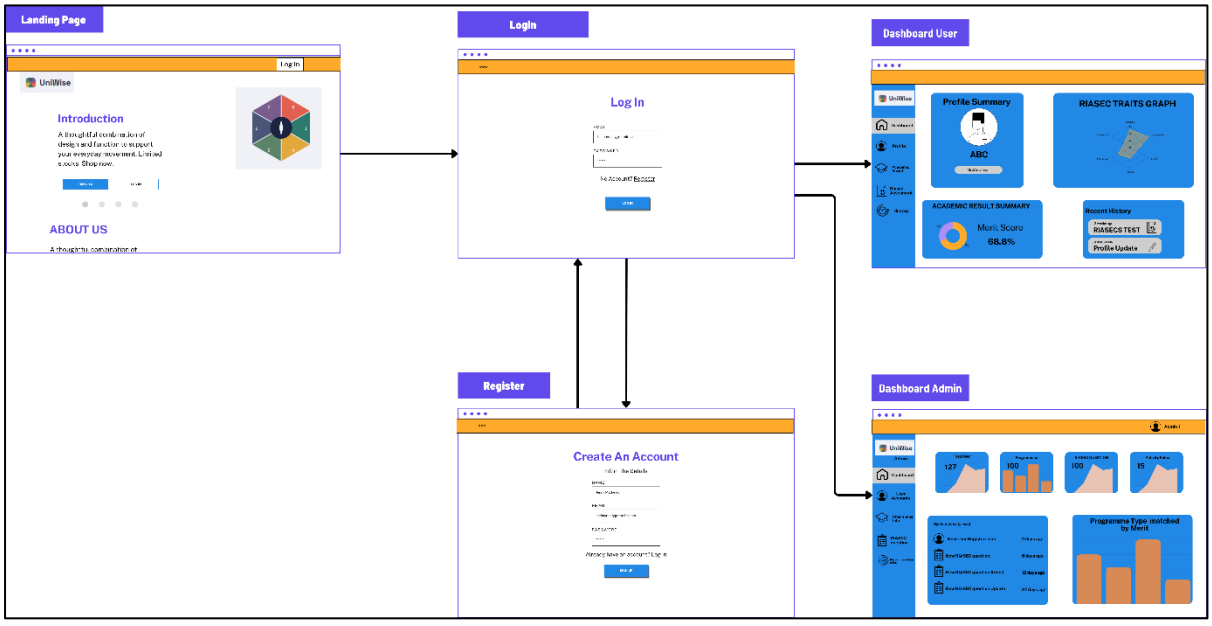


Figure 3.2.1.1.1: Login Screenflow

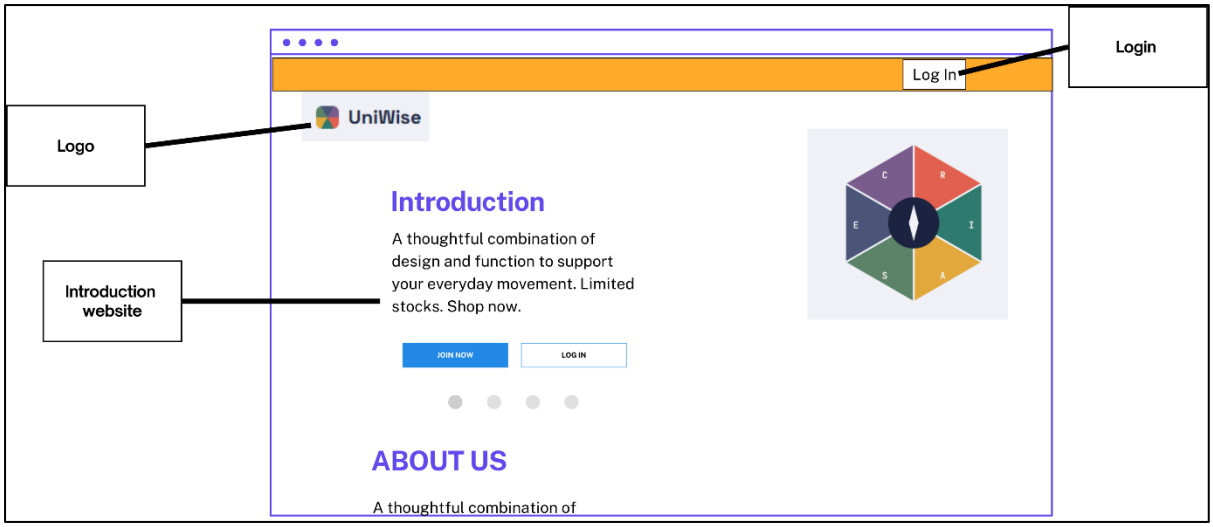


Figure 3.2.1.1.2: Home Page Description

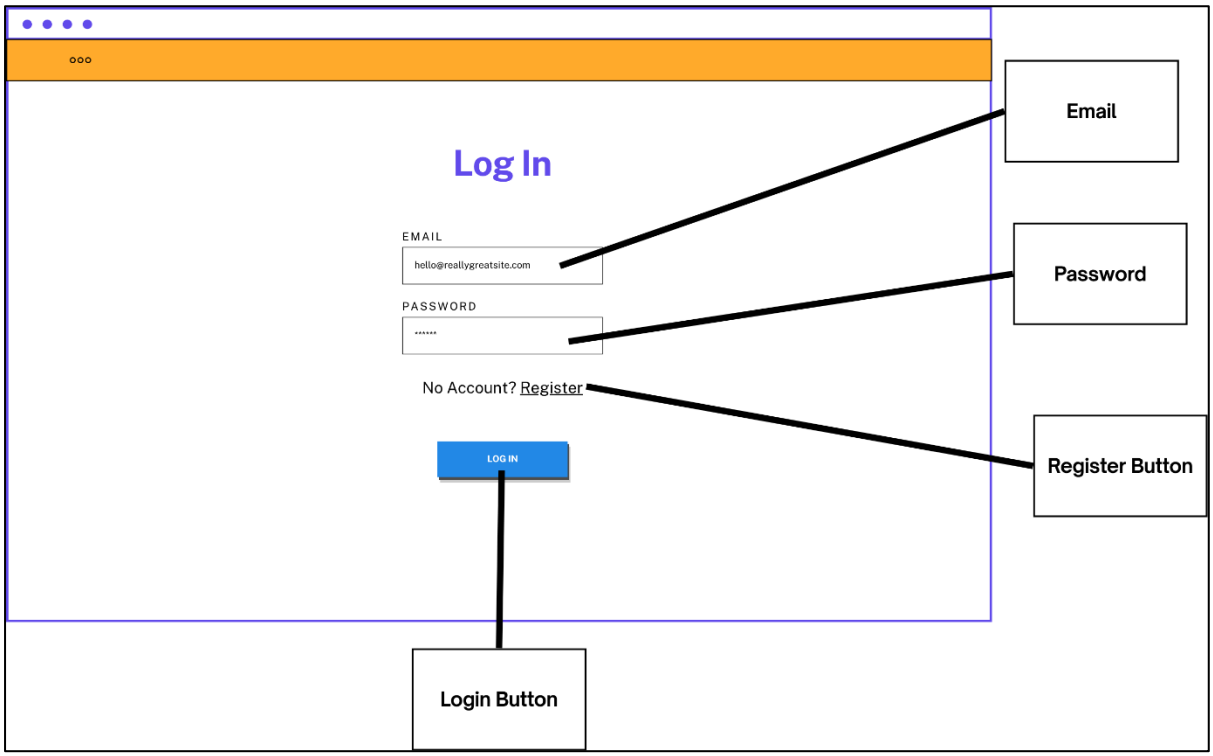


Figure 3.2.1.1.3: Login Page Description

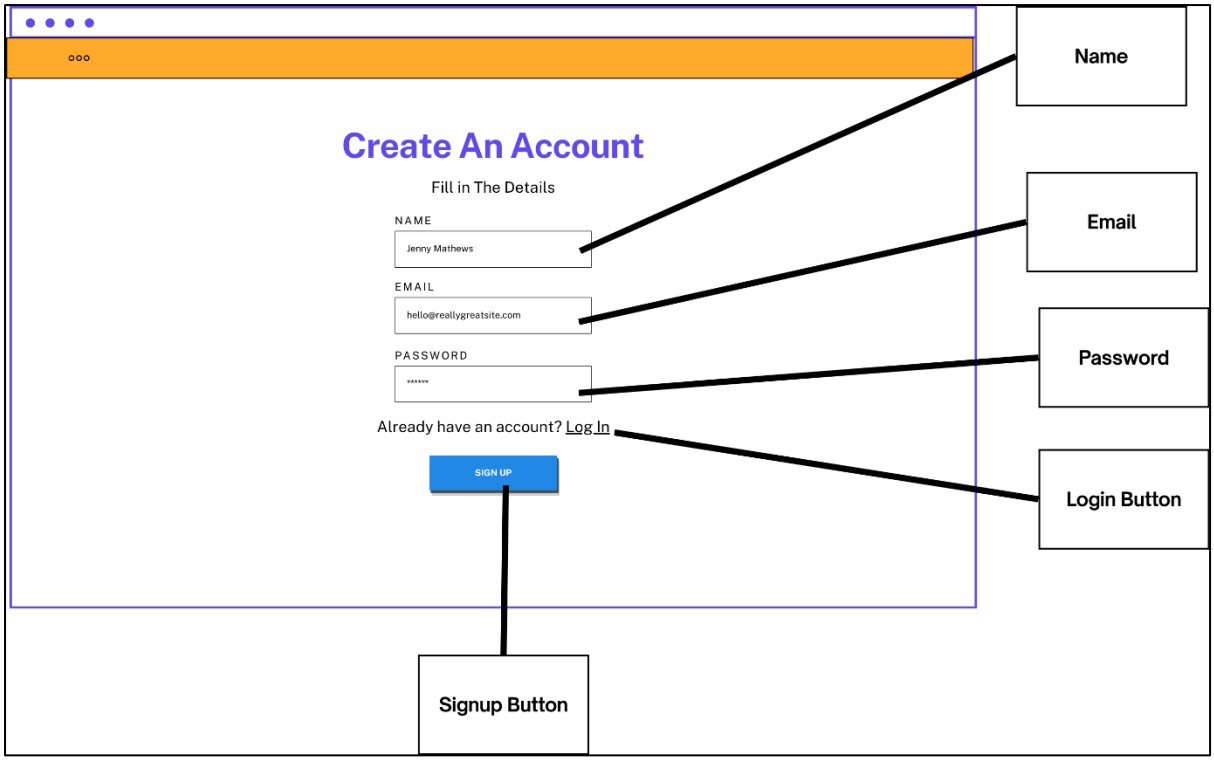


Figure 3.2.1.1.4: Register Page Description

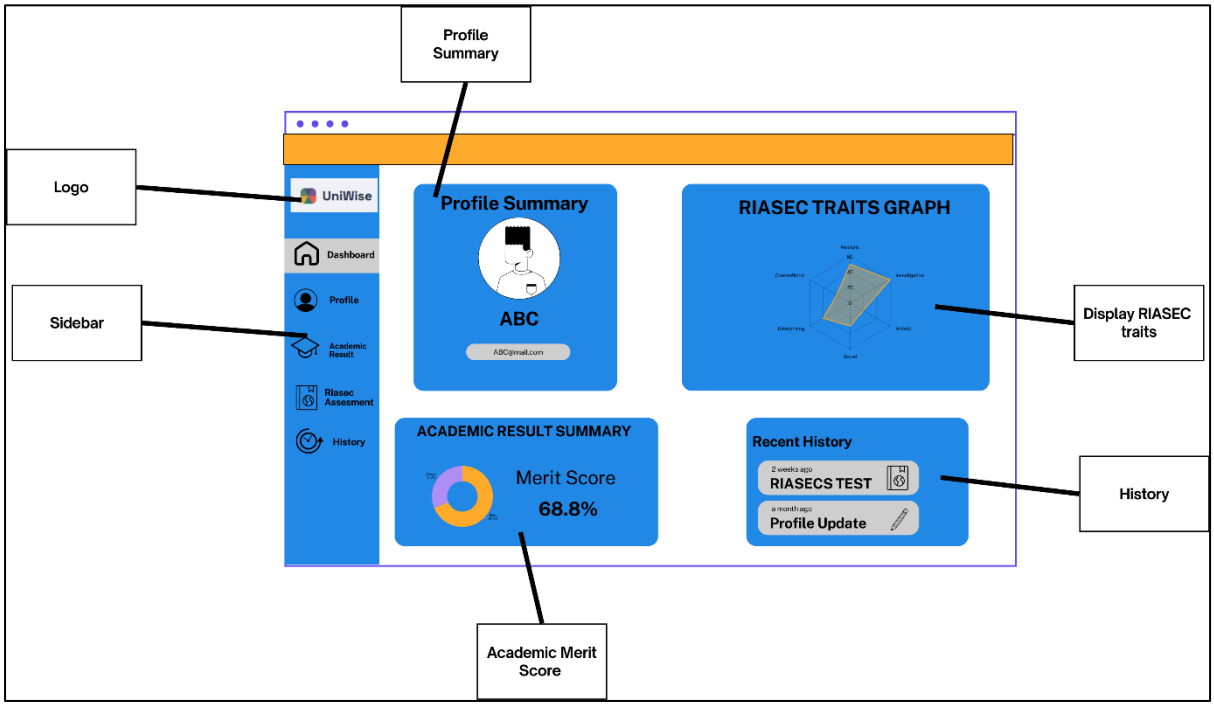


Figure 3.2.1.1.5: User Dashboard Description

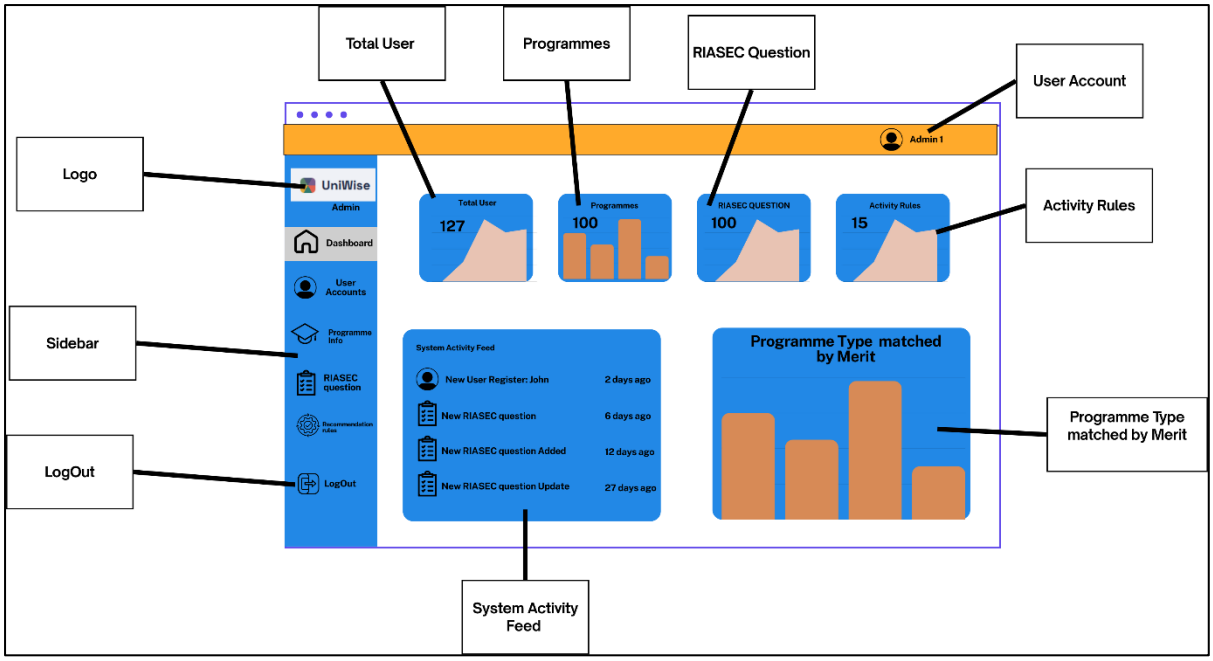


Figure 3.2.1.1.6: Admin Dashboard Description

3.2.1.2 User Profile

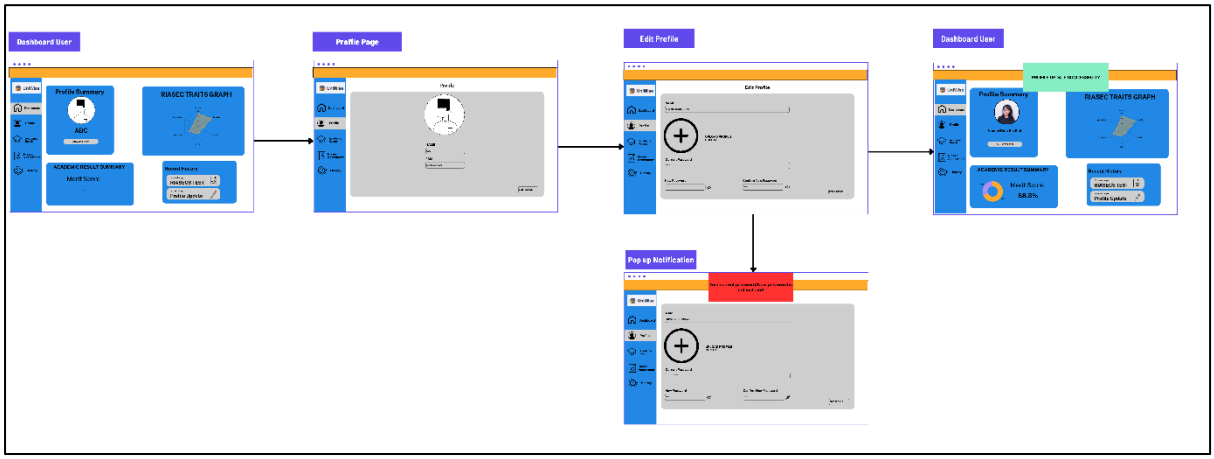


Figure 3.2.1.2.1: User Profile Screenflow

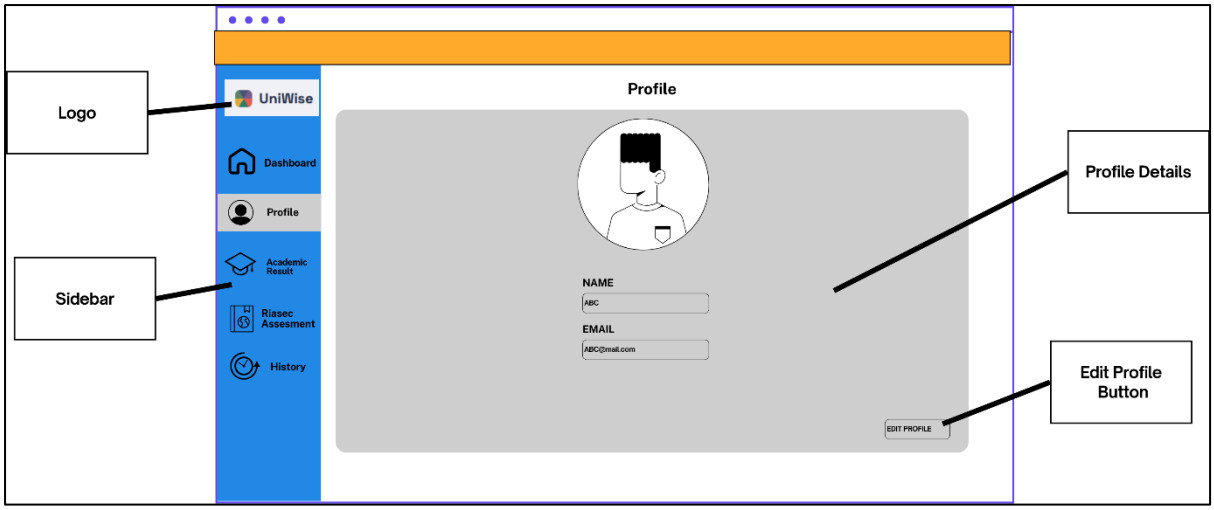


Figure 3.2.1.2.2: User Profile Page Description

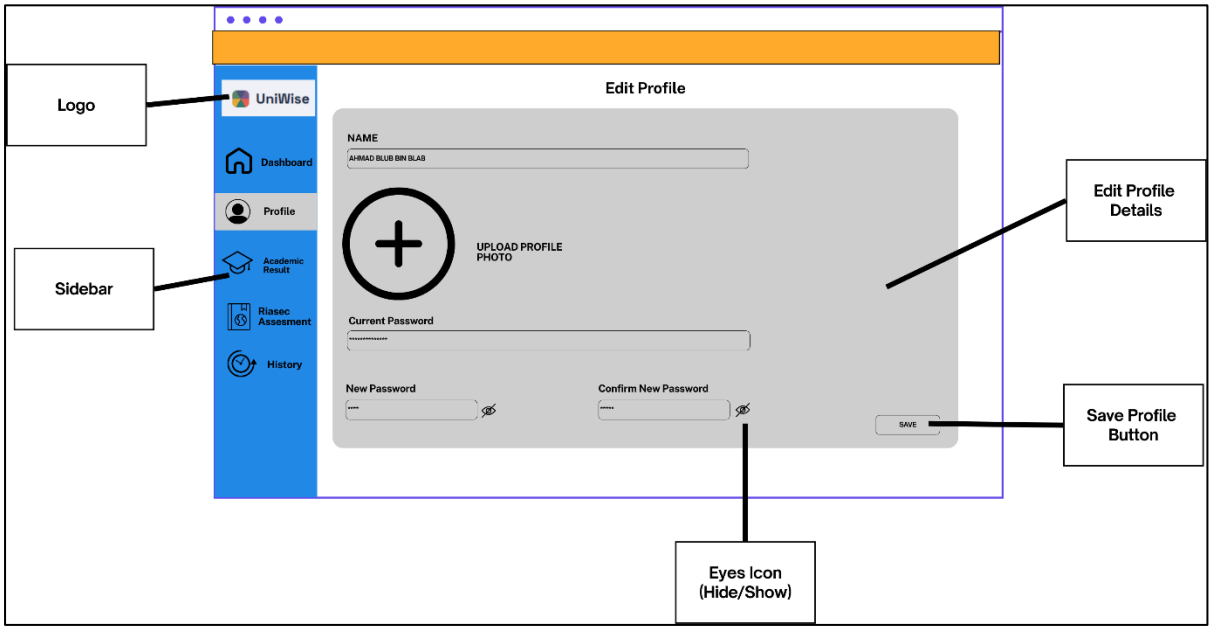


Figure 3.2.1.2.3: User Edit Profile Description

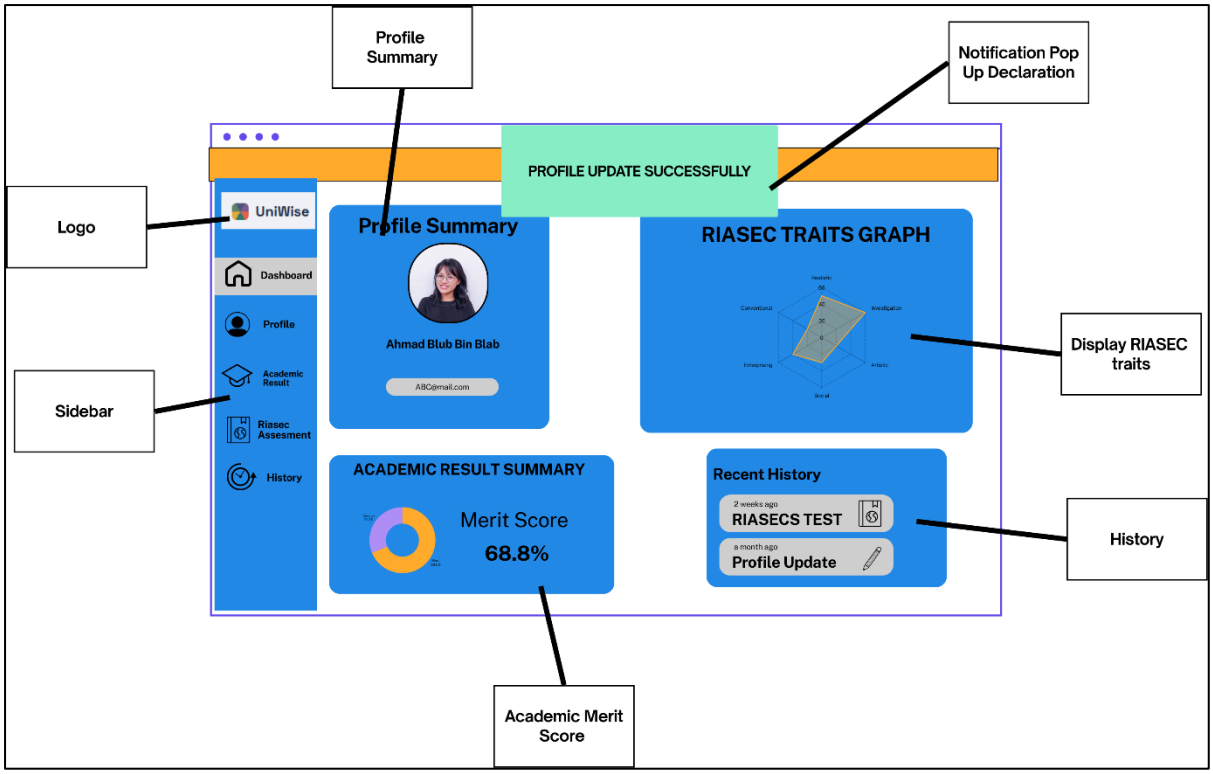


Figure 3.2.1.2.4: User Edit Profile Success Description

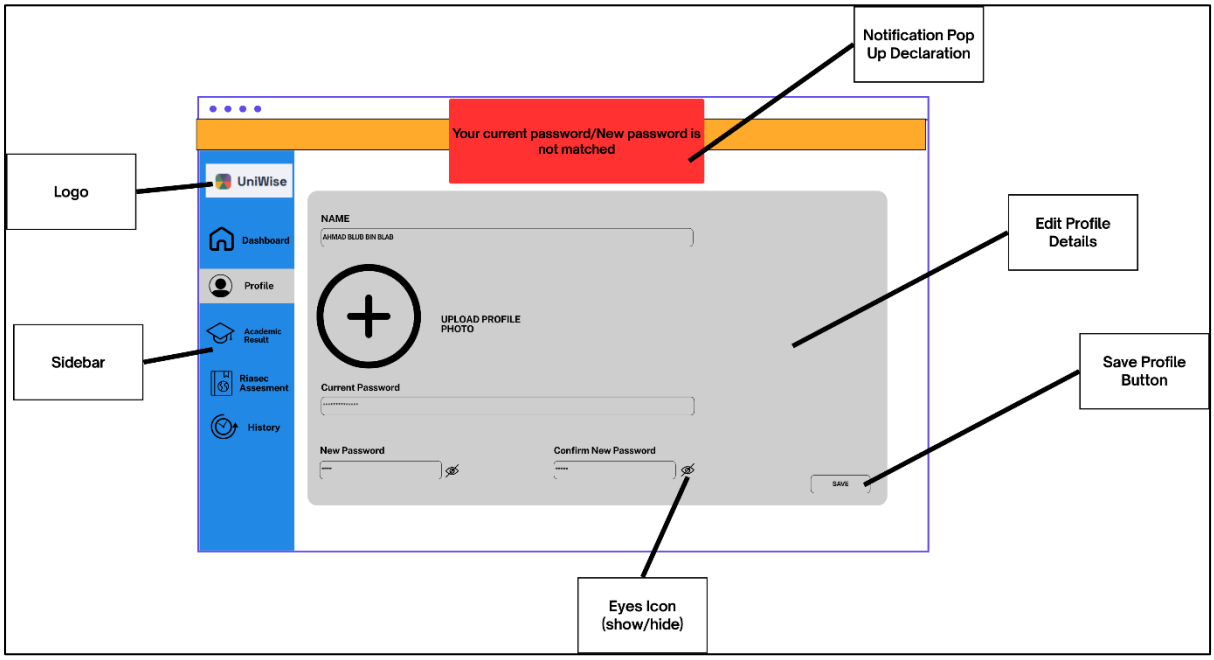


Figure 3.2.1.2.5: User Edit Profile Unsuccessful Description

3.2.1.3 User Academic

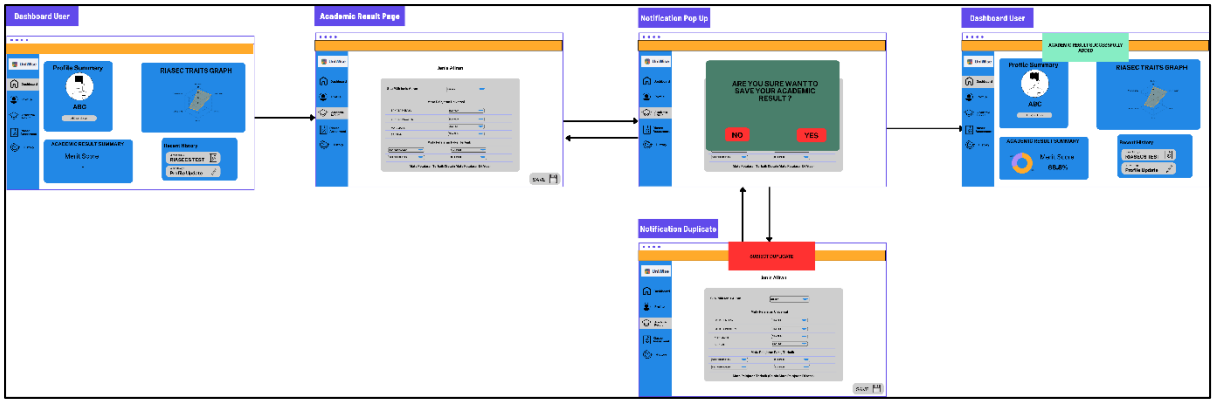


Figure 3.2.1.3.1: User Academic Screenflow

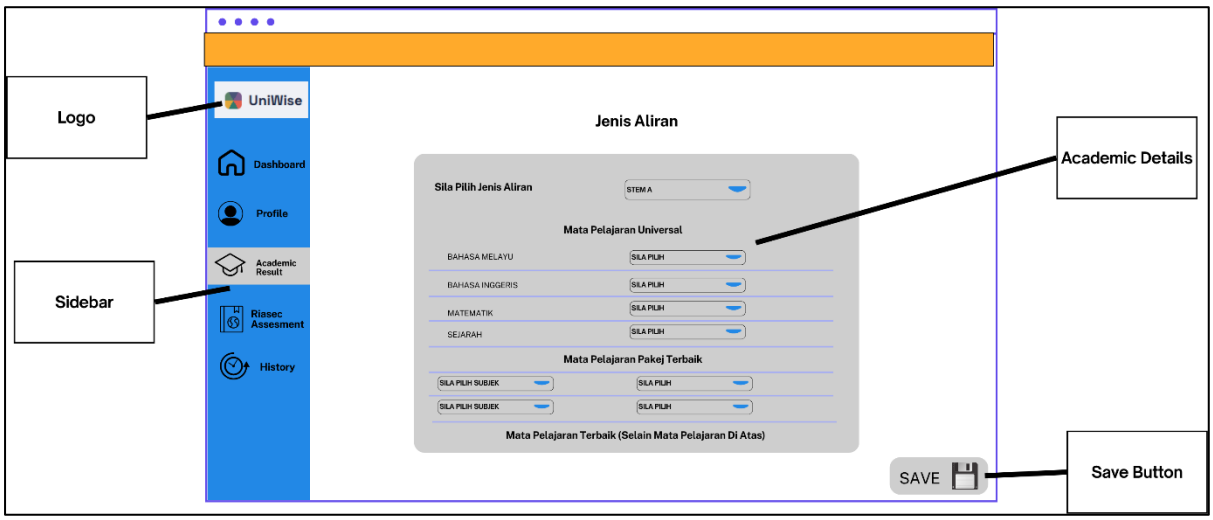


Figure 3.2.1.3.2: User Academic Detail Form Description

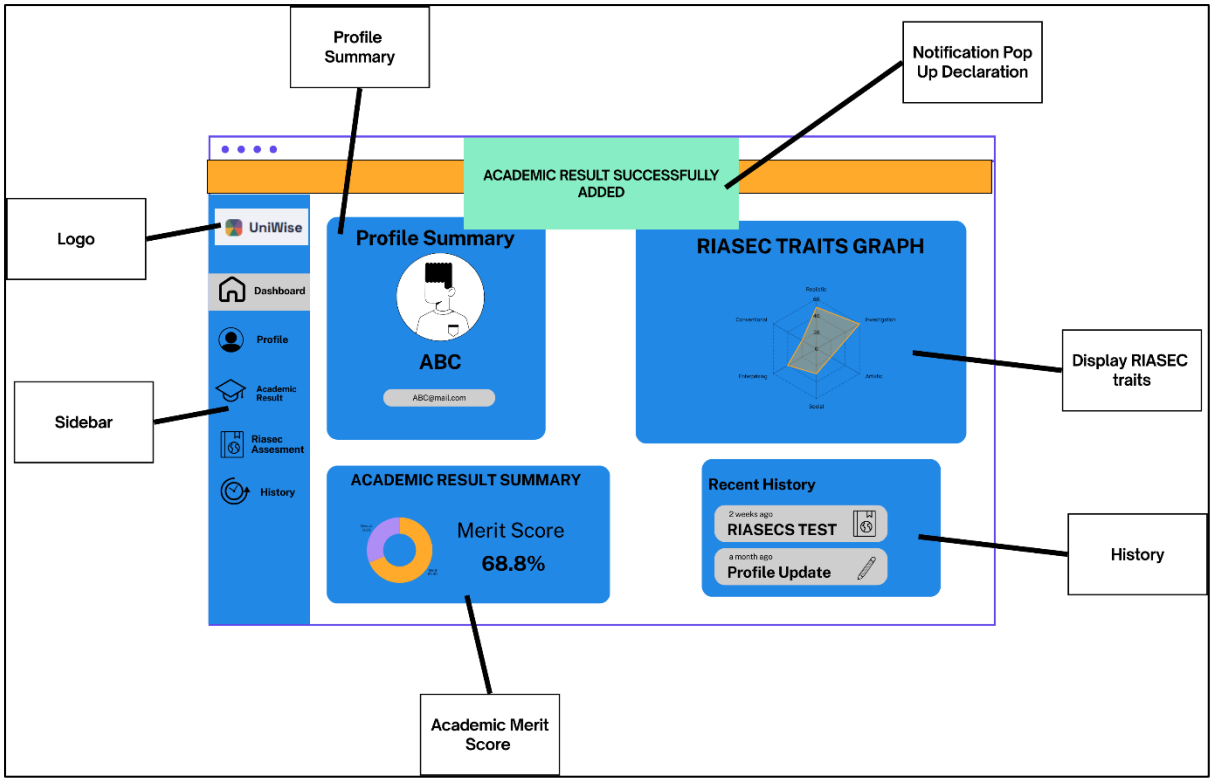


Figure 3.2.1.3.3: User Academic Detail Success Description

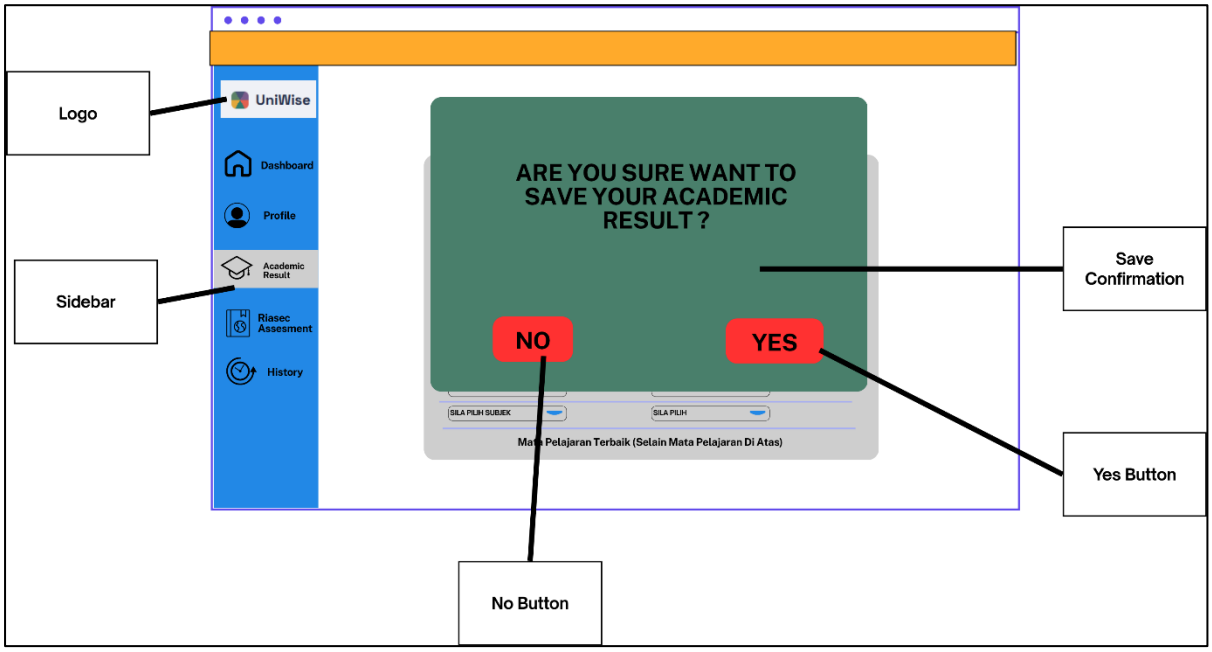


Figure 3.2.1.3.4: User Academic Detail Confirmation Description

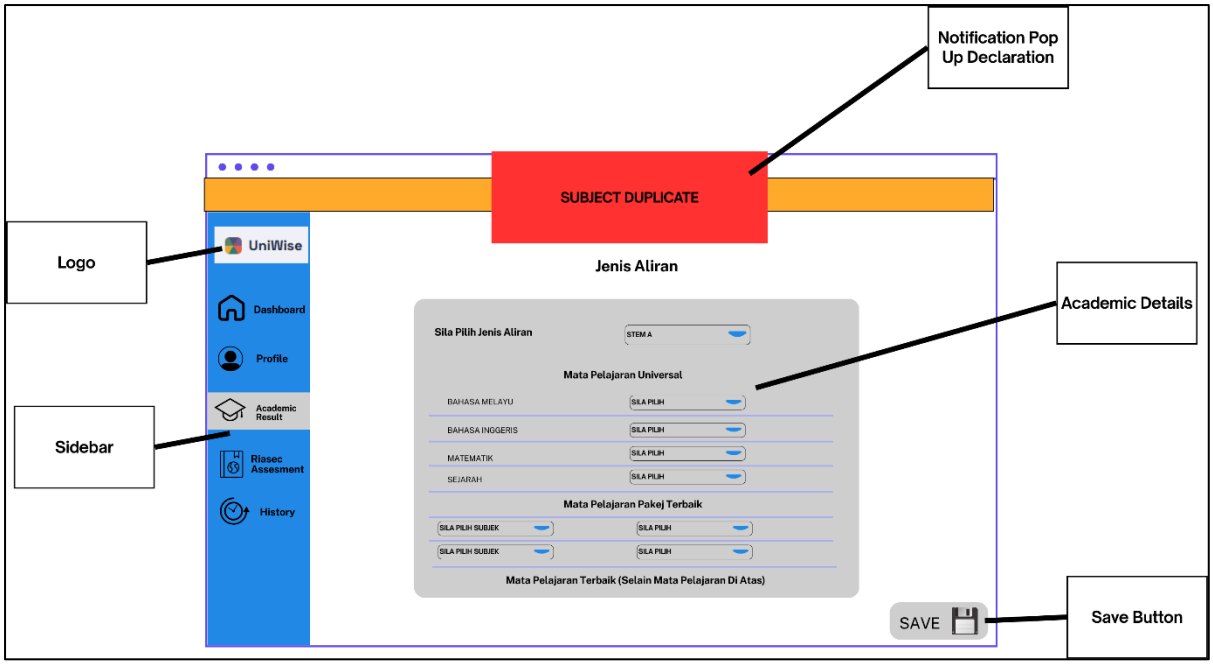


Figure 3.2.1.3.5: User Academic Detail Unsuccessful Description

3.2.1.4 User RIASEC Assessment

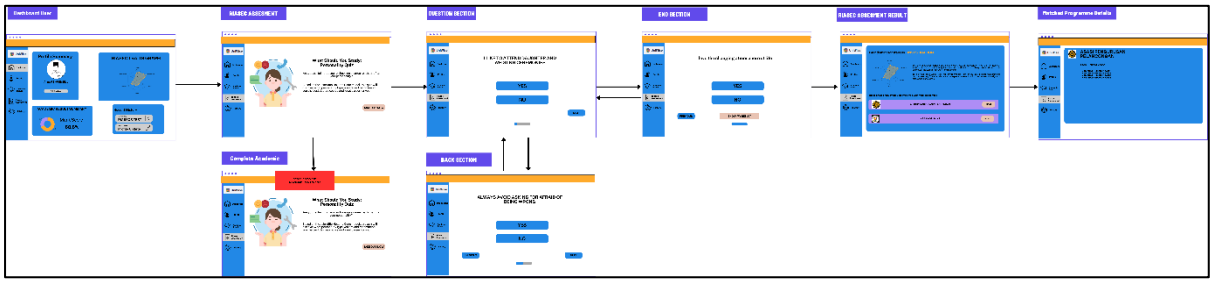


Figure 3.2.1.4.1: User RIASEC Assessment Screenflow

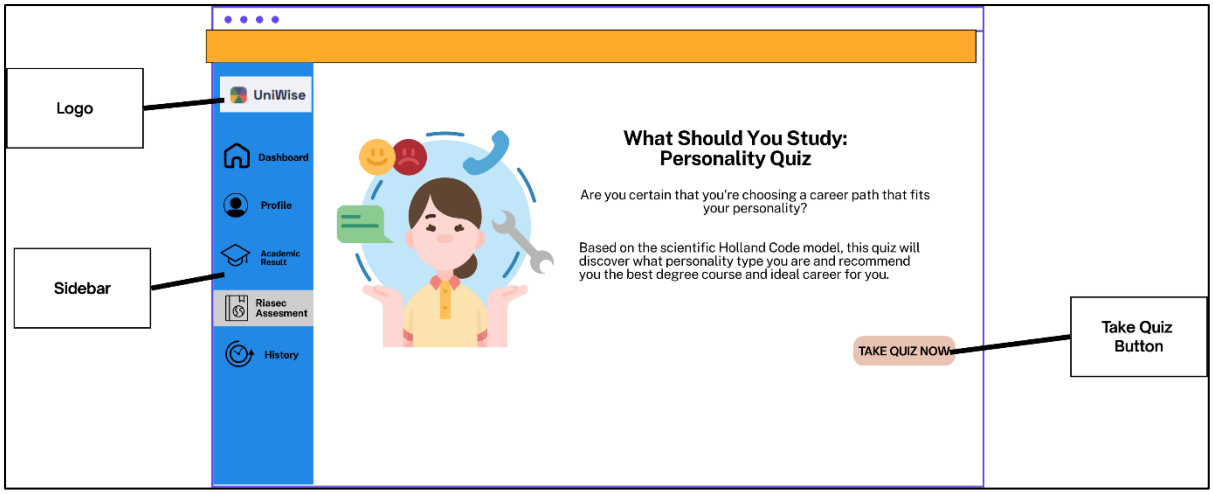


Figure 3.2.1.4.2: User RIASEC Assessment Description

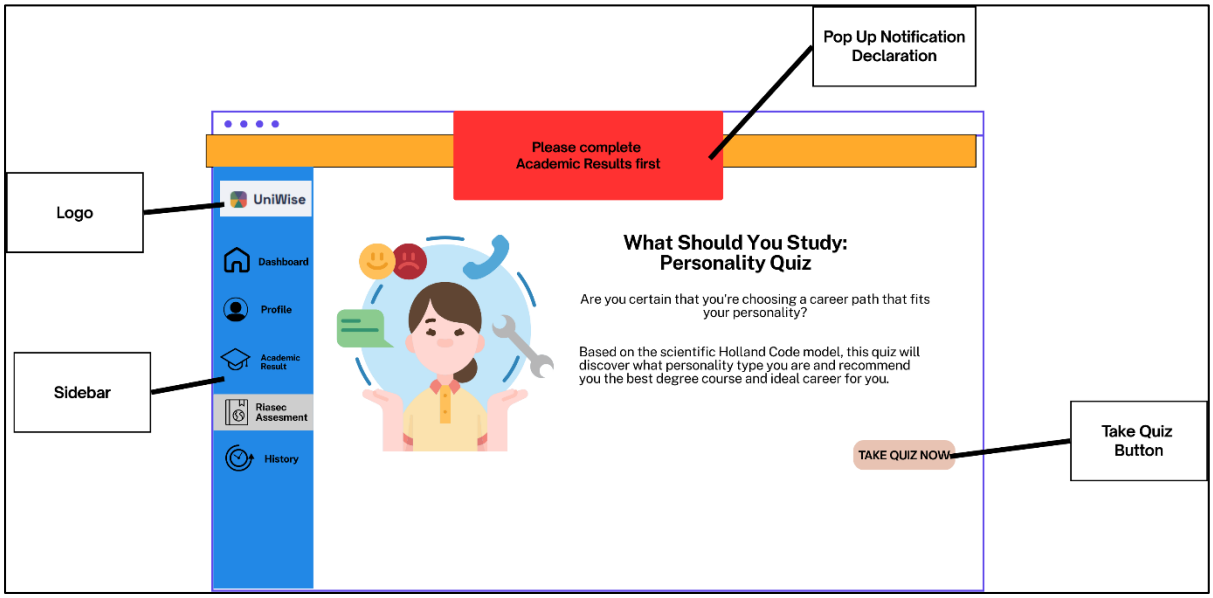


Figure 3.2.1.4.3: User RIASEC Assessment Fail Description

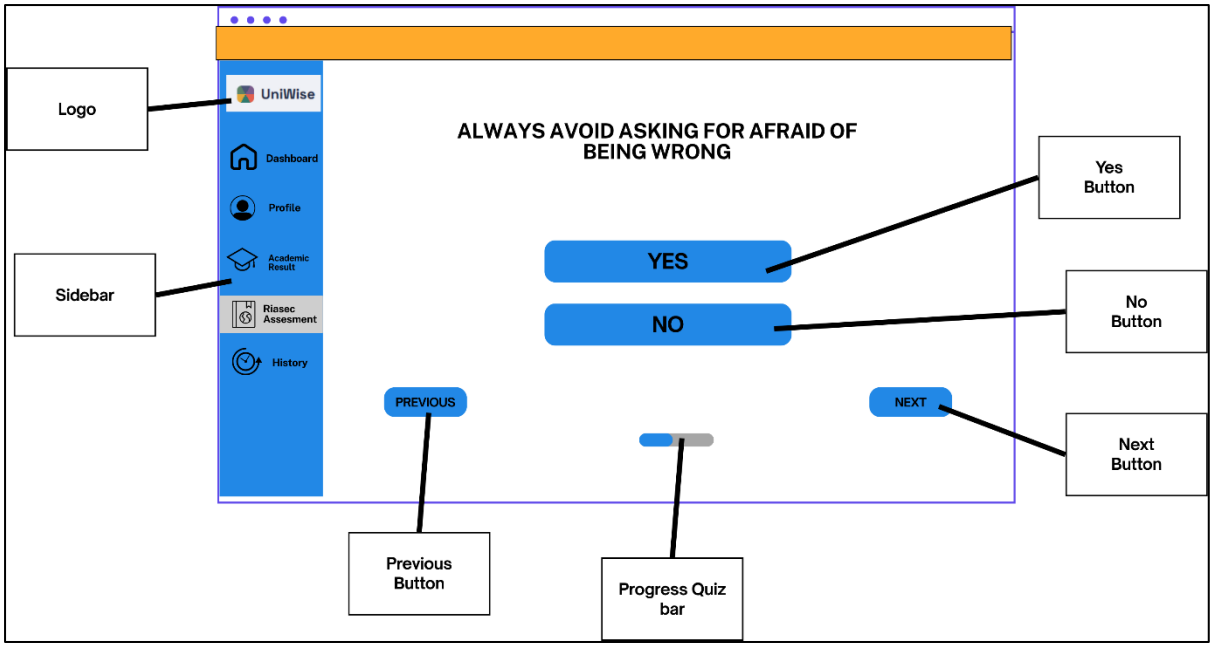


Figure 3.2.1.4.4: User Take RIASEC Assessment Description

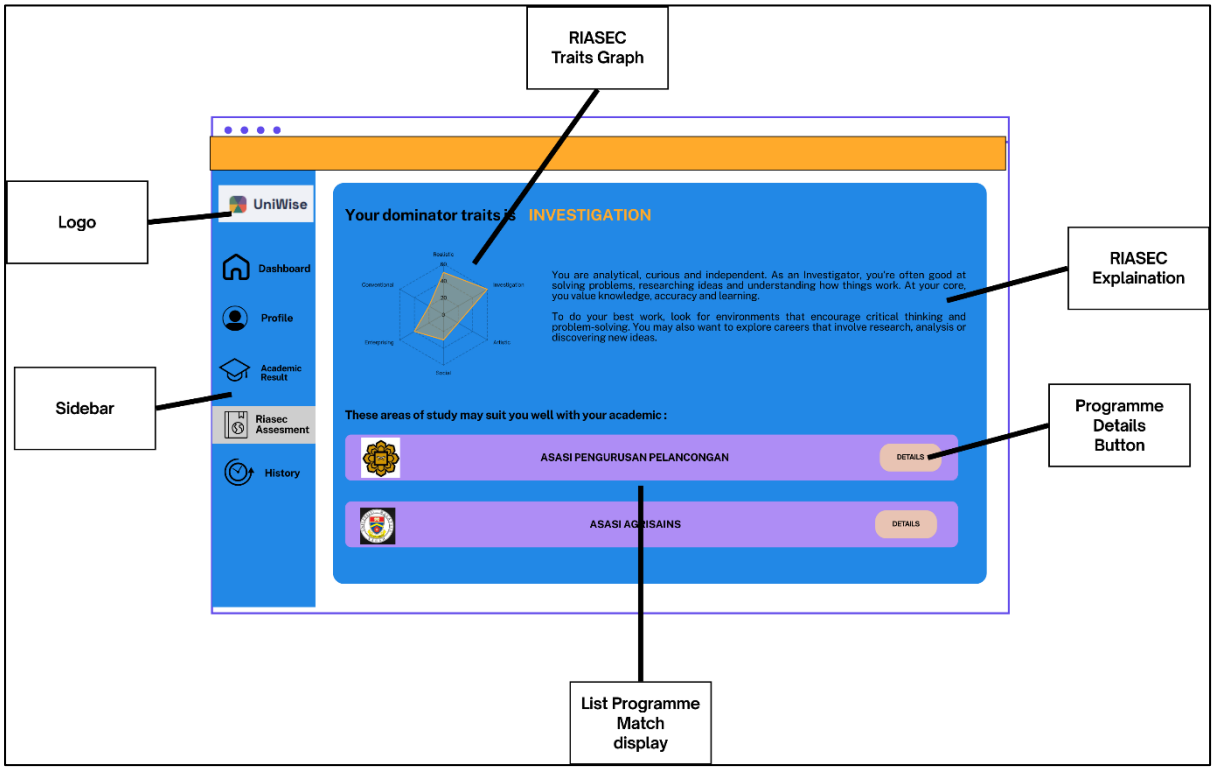


Figure 3.2.1.4.5: RIASEC Assessment Result Description

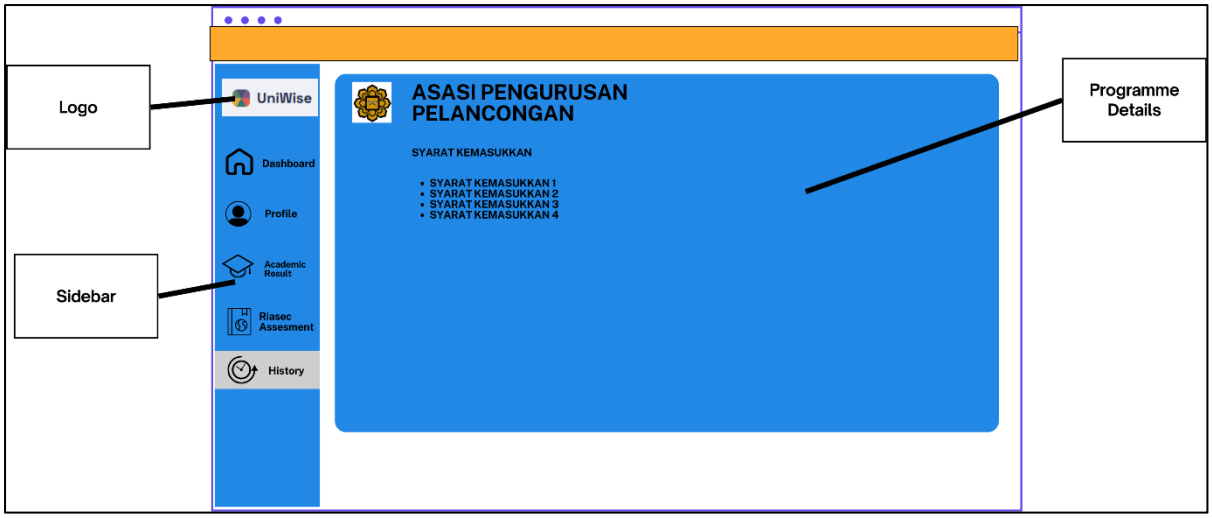


Figure 3.2.1.4.6: Programme Details Description

3.2.1.5 User History

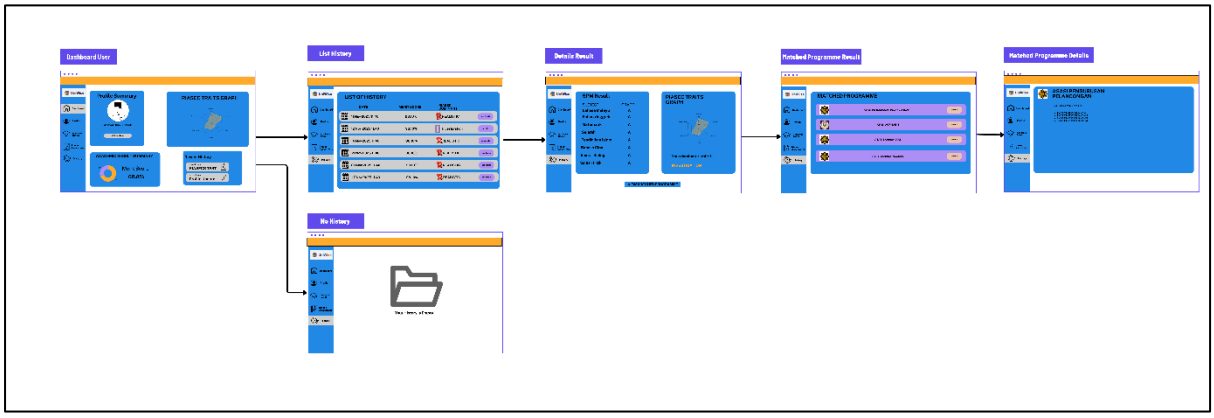


Figure 3.2.1.5.1: User History Screenflow

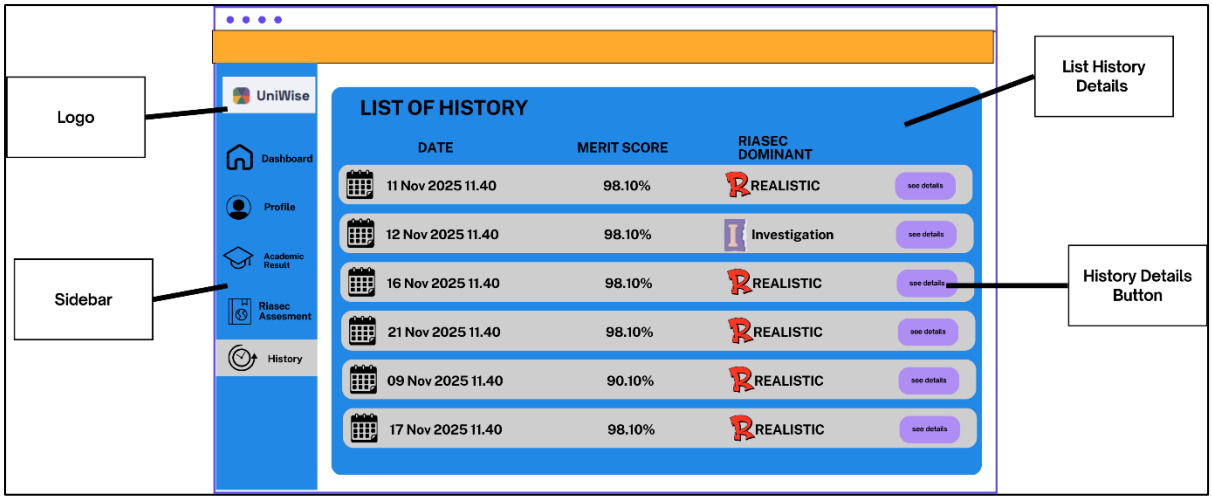


Figure 3.2.1.5.2: List User History Description

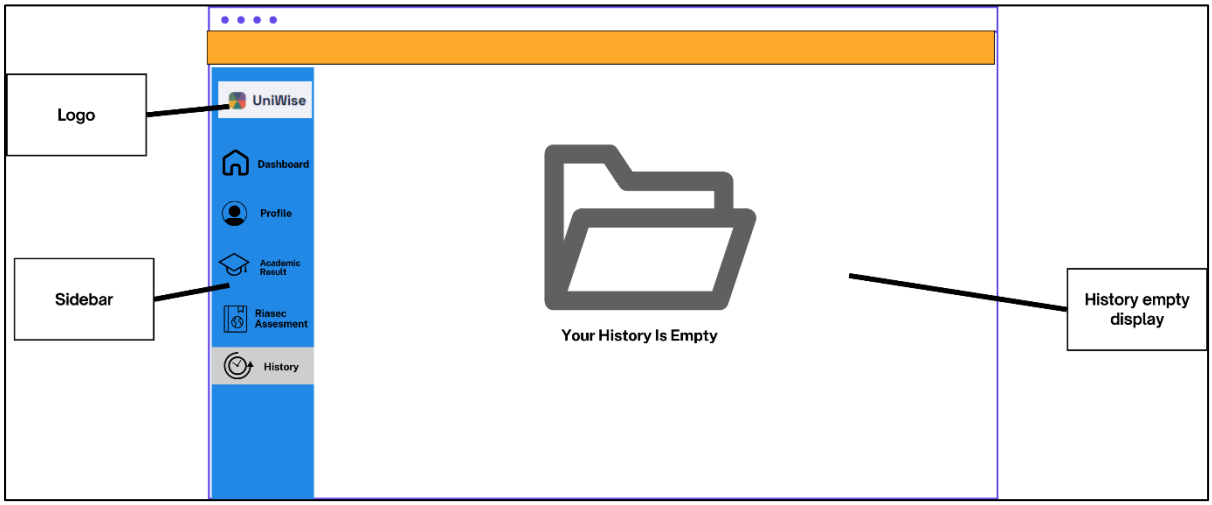


Figure 3.2.1.5.3: No List User History Description

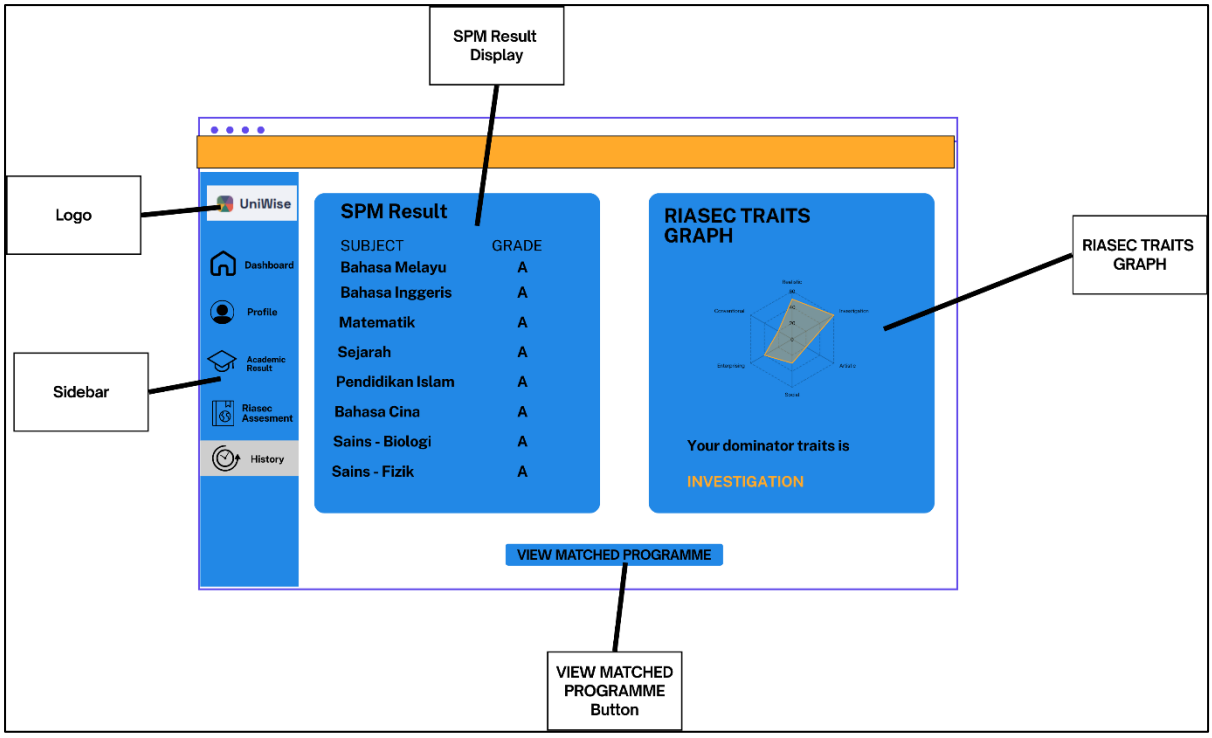


Figure 3.2.1.5.4: User History Detail Description

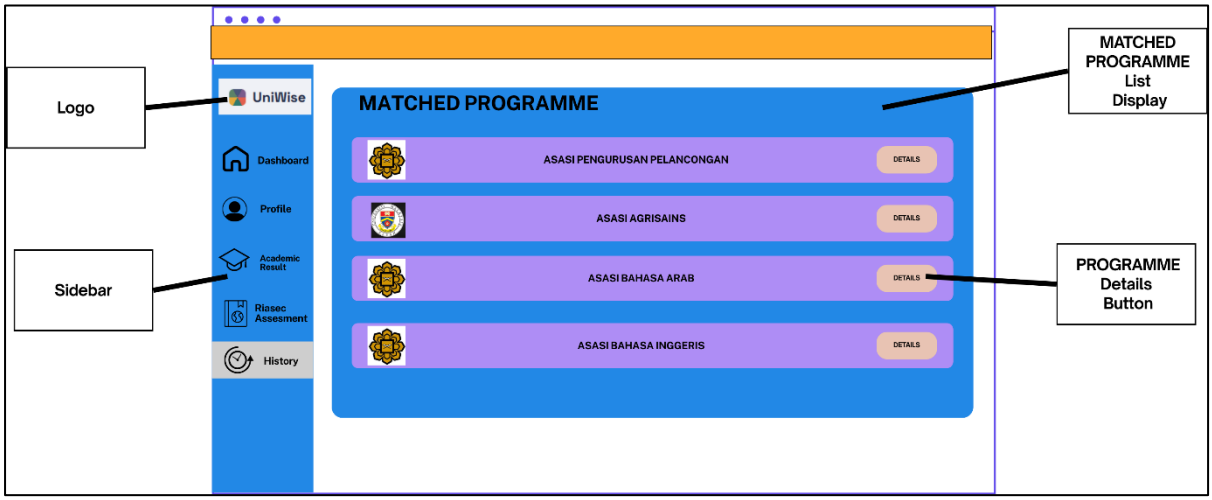


Figure 3.2.1.5.5: List User History Programme Description

3.2.1.6 Admin Manage User

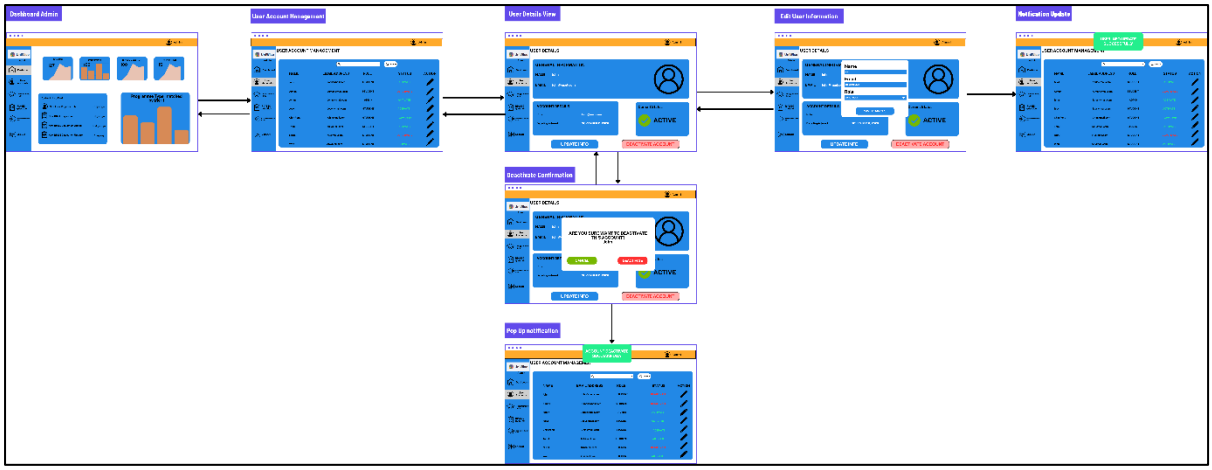


Figure 3.2.1.6.1: User History Screenflow

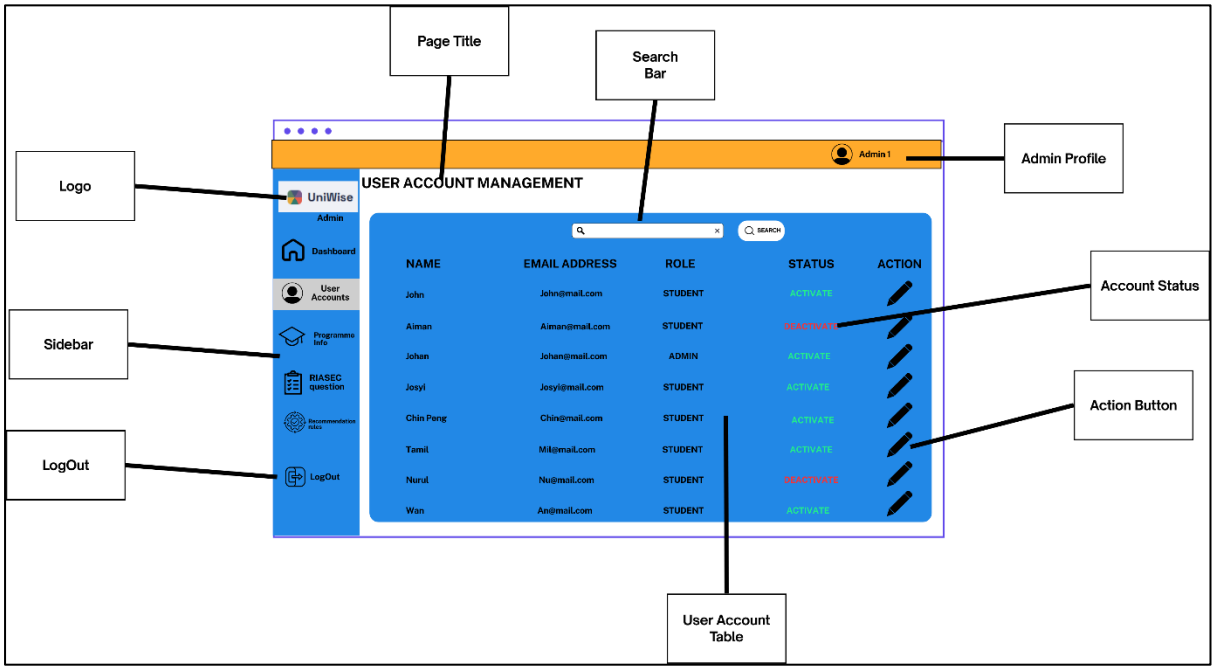


Figure 3.2.1.6.2: User Account Management Page Description

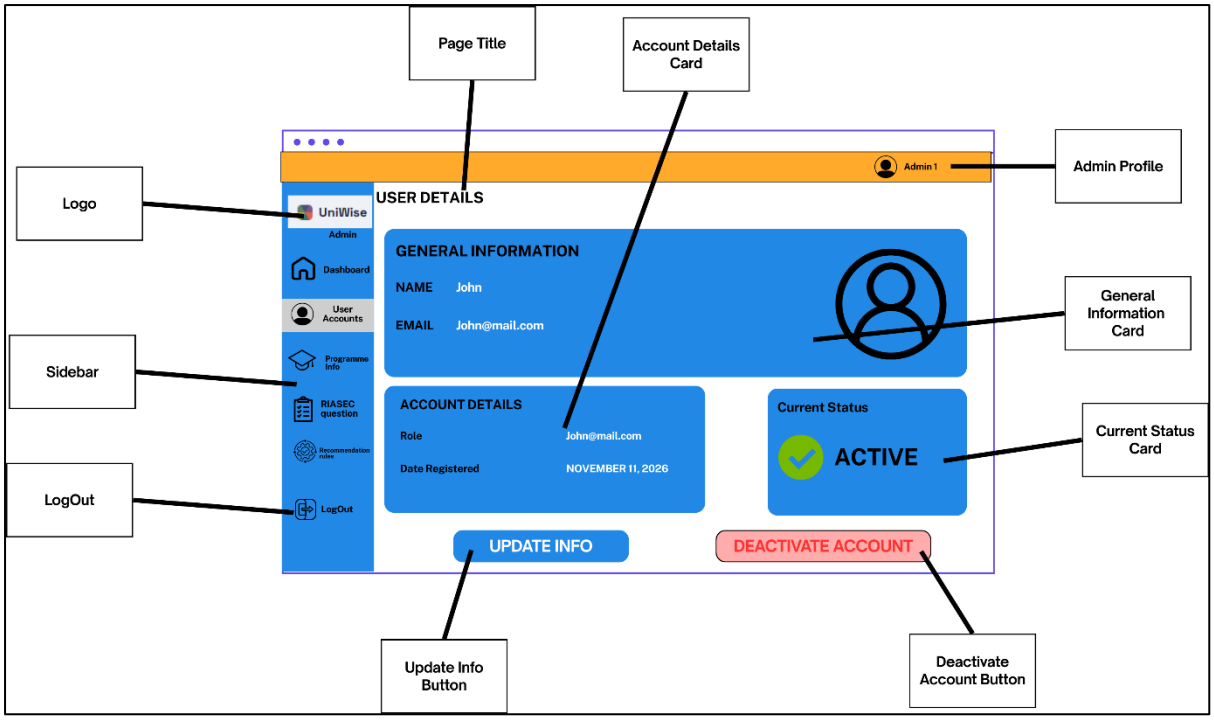


Figure 3.2.1.6.3: User Detail Page Description

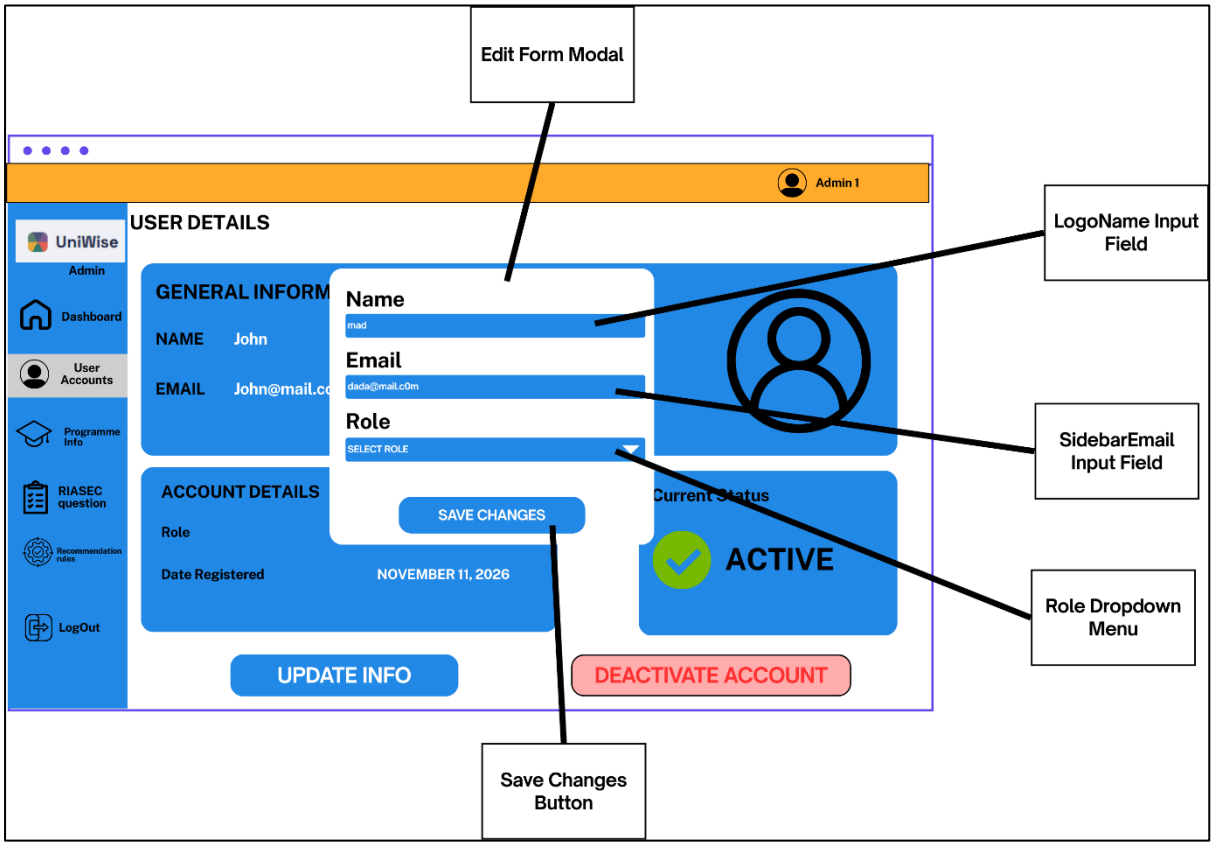


Figure 3.2.1.6.4: Admin Edit Detail User Description

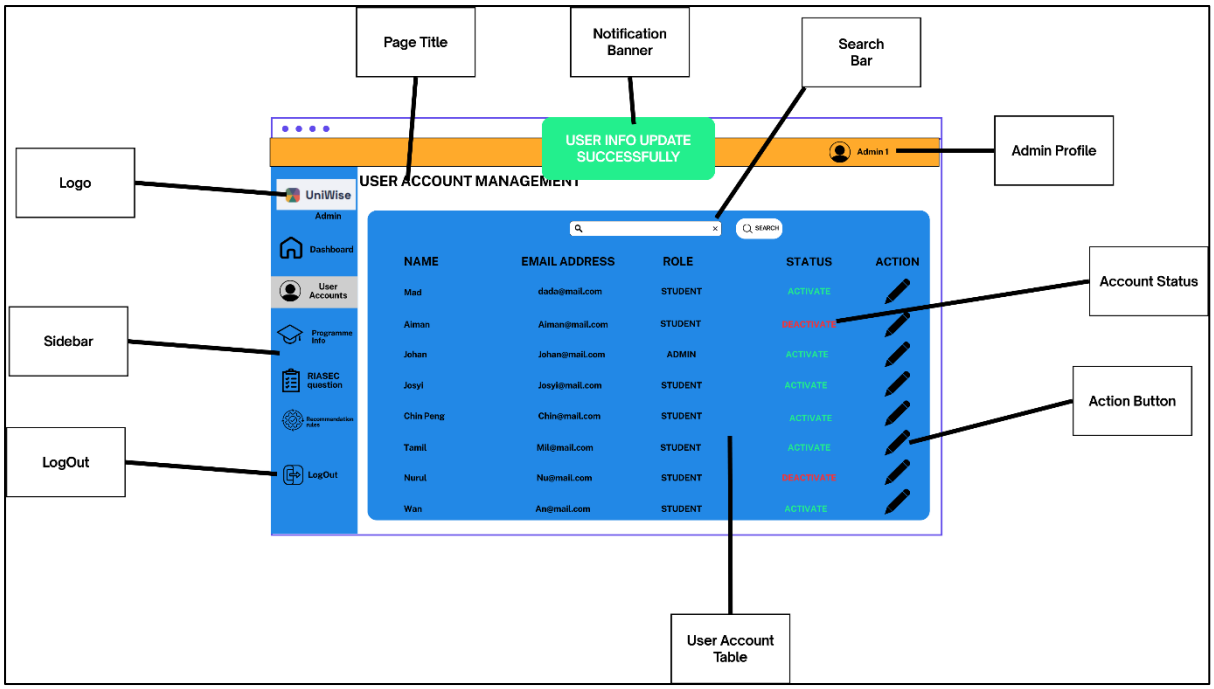


Figure 3.2.1.6.5: Edit Successful Description

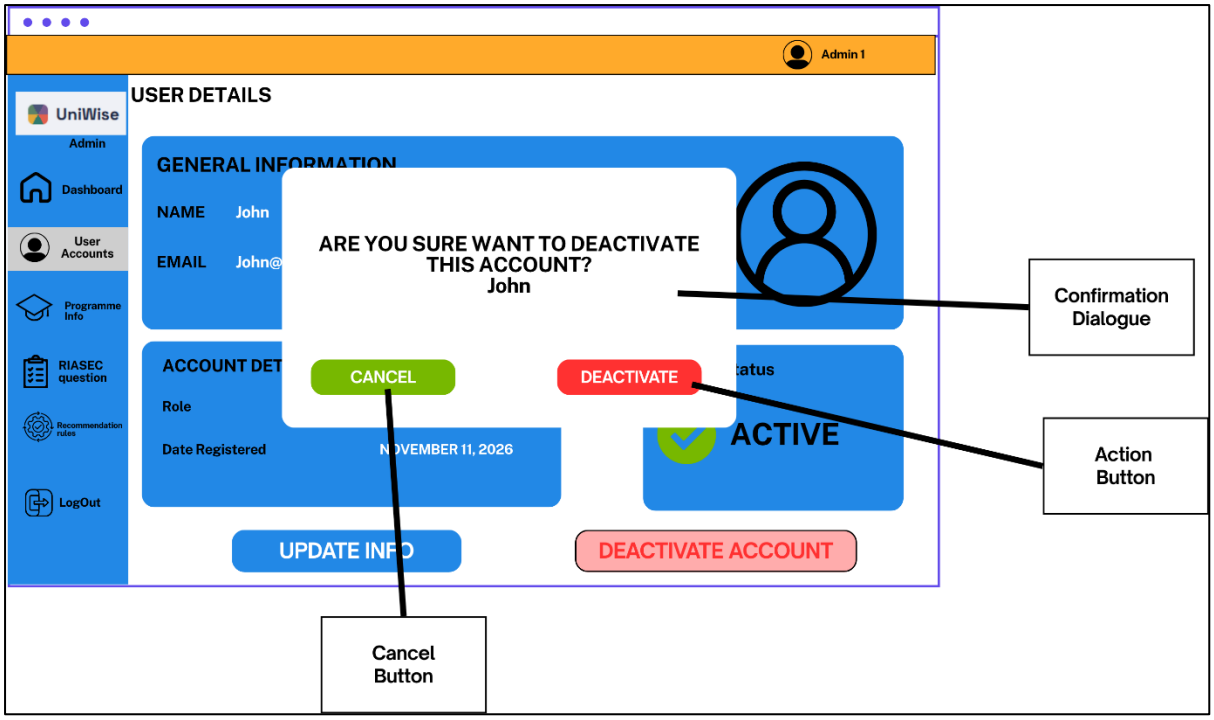


Figure 3.2.1.6.6: Deactivate User Description

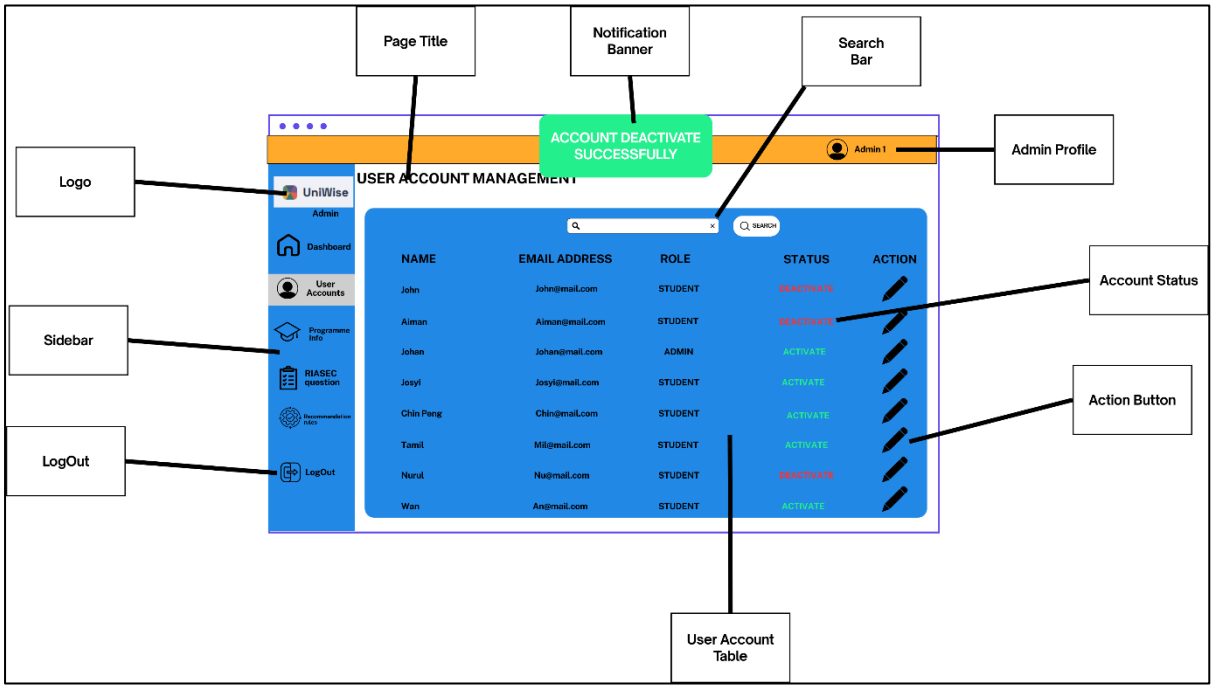


Figure 3.2.1.6.7: Deactivate User Successful Description

	DOCUMENT IDENTIFICATION	
	APPLICATION NAME	VERSION
	UNIWISE	1.0

3.2.1.7 Admin Manage Programme

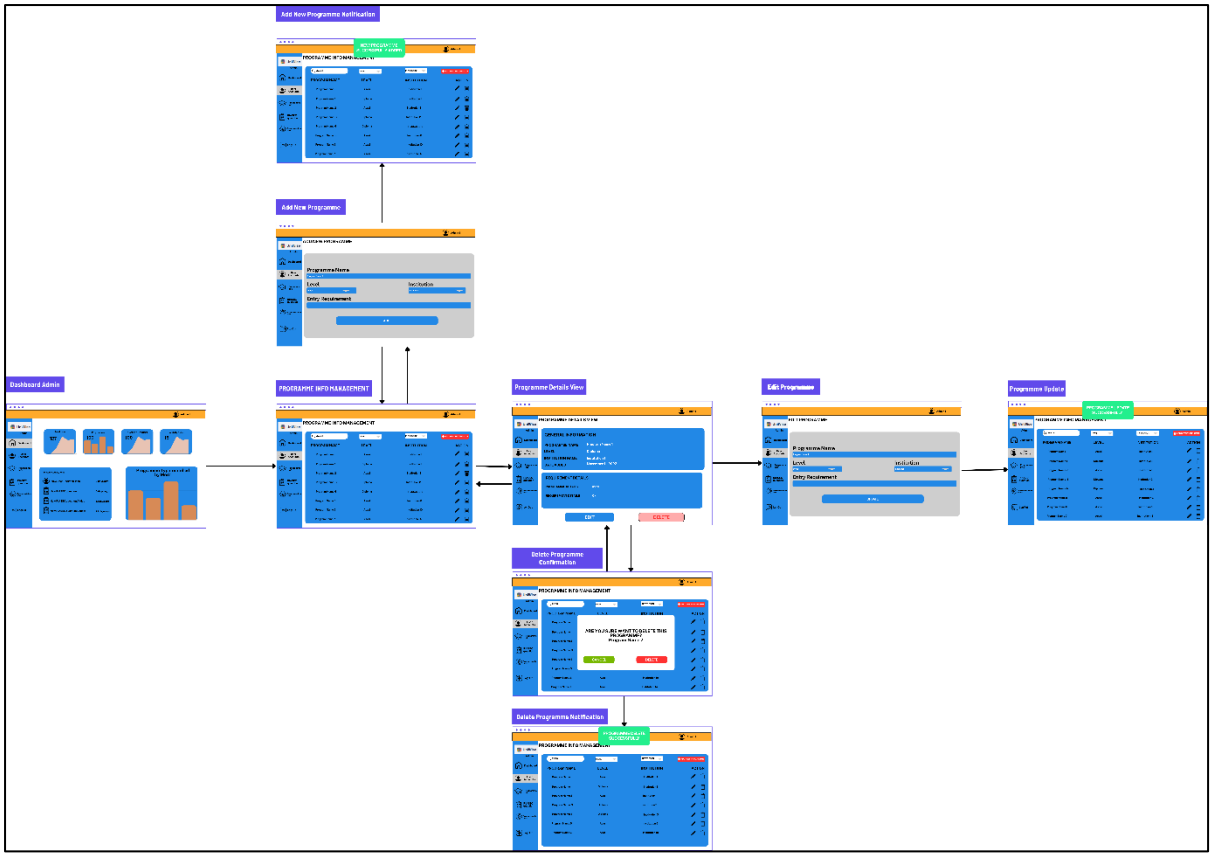


Figure 3.2.1.7.1: Admin Manage Programme Screenflow

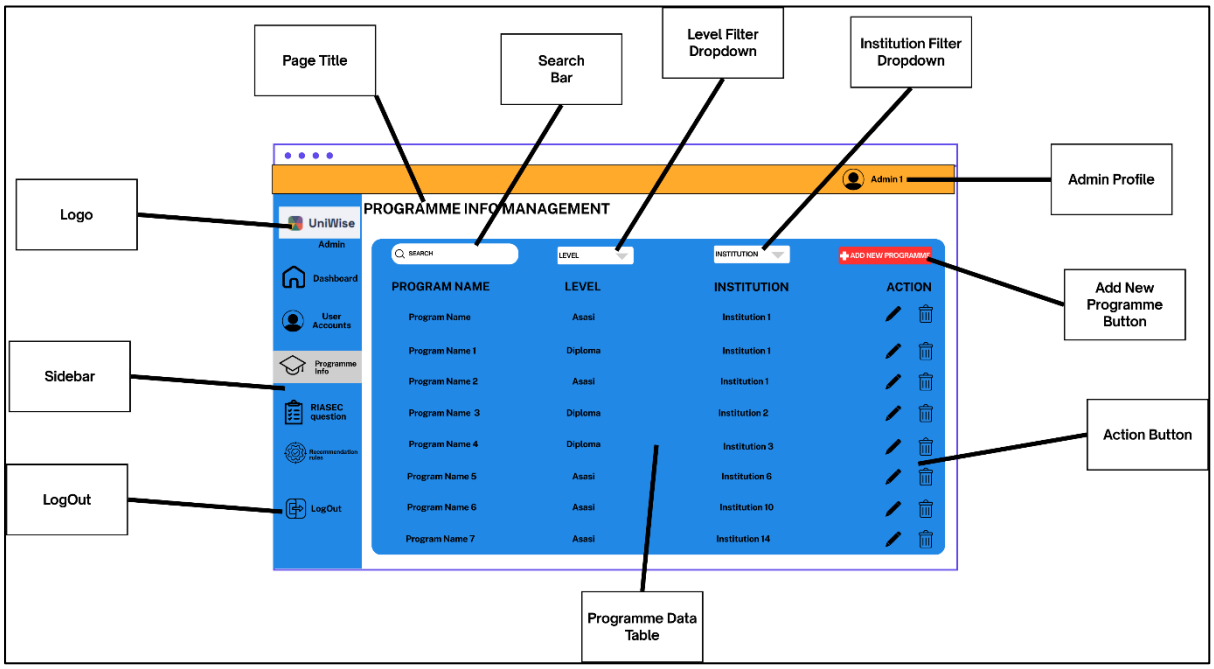


Figure 3.2.1.7.2: Programme Management Page Description

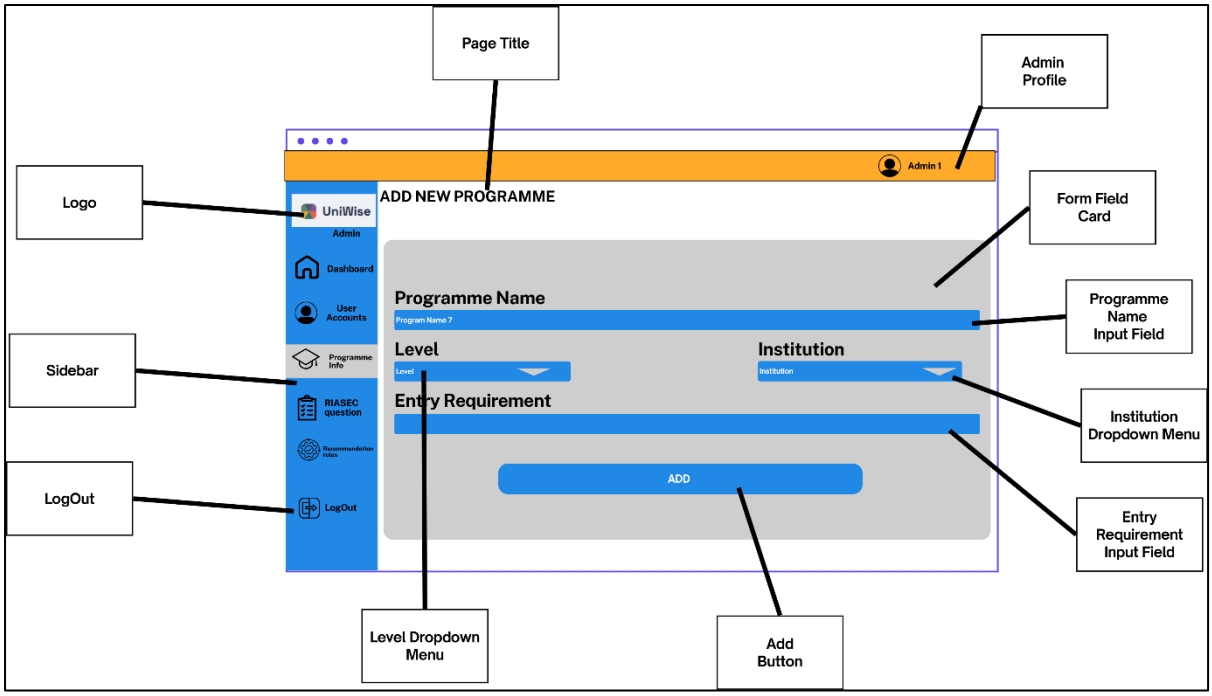


Figure 3.2.1.7.3: Add New Programme Description

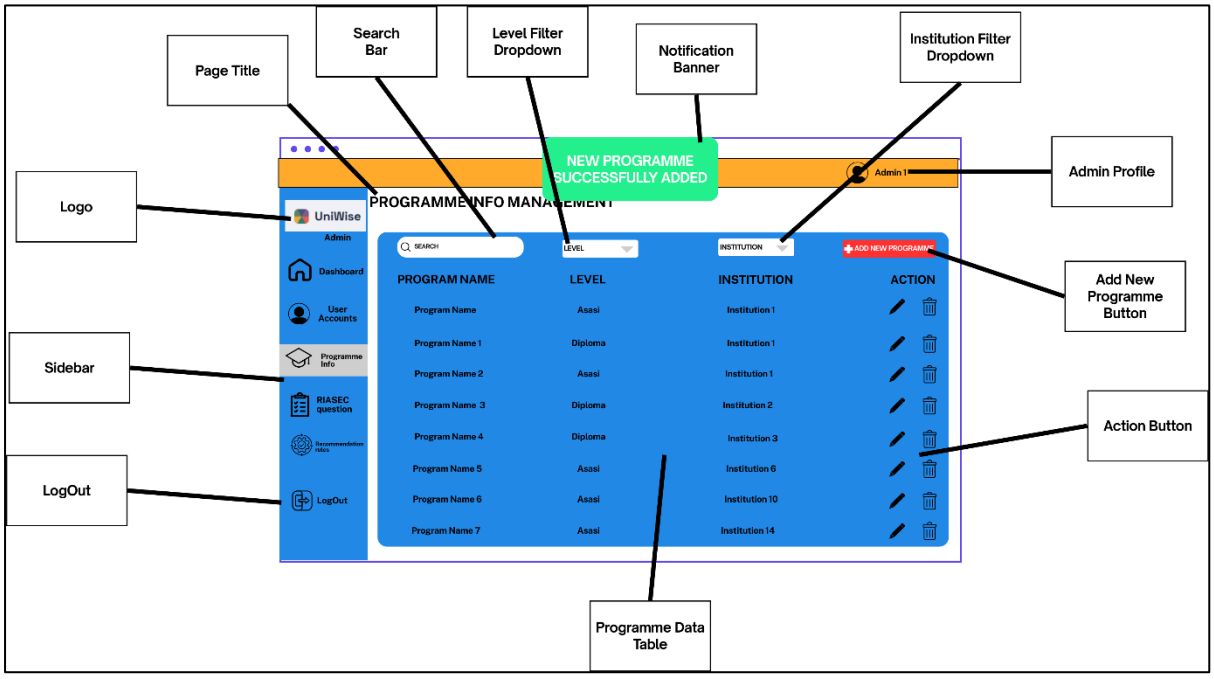


Figure 3.2.1.7.4: Add New Programme Notification Description

	DOCUMENT IDENTIFICATION	
	APPLICATION NAME	VERSION
	UNIWISE	1.0

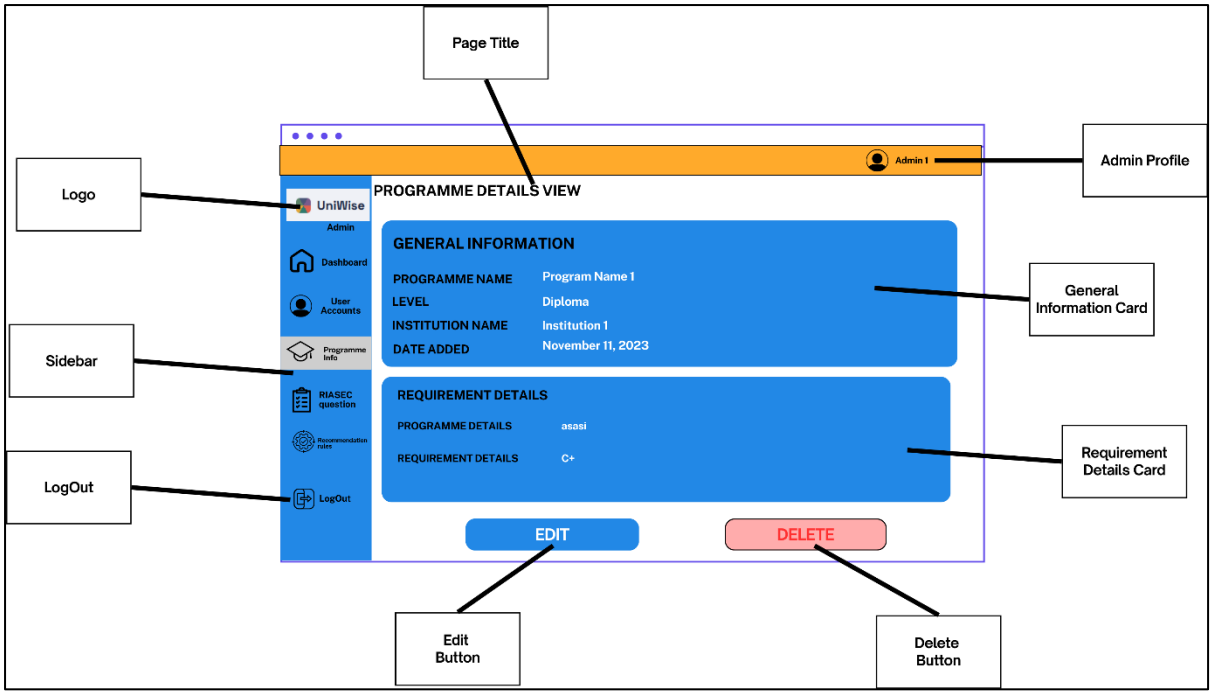


Figure 3.2.1.7.5: Programme Details Page Description

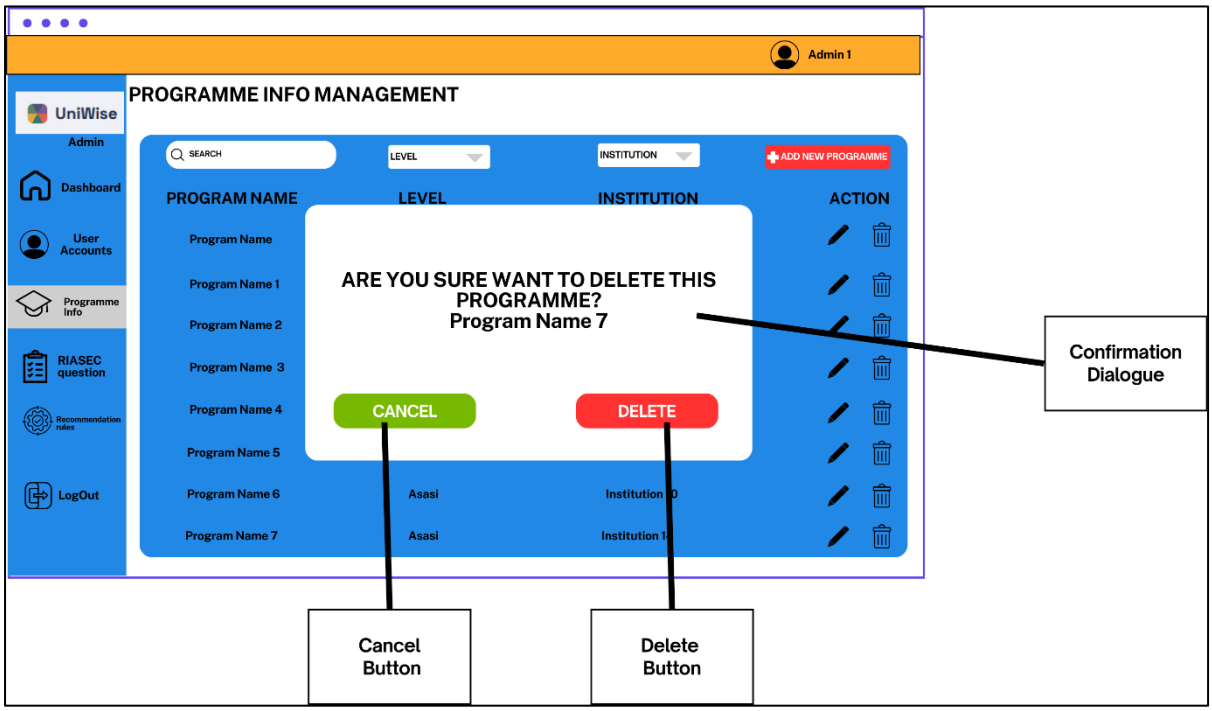


Figure 3.2.1.7.6: Delete Programme Description

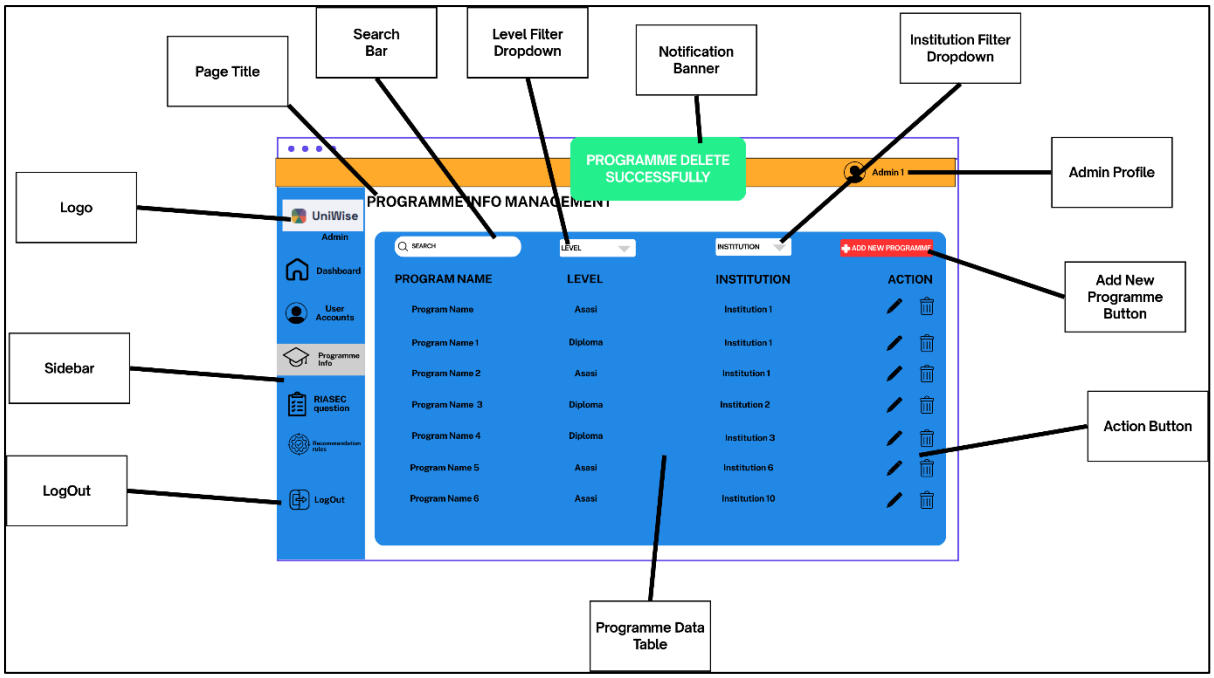


Figure 3.2.1.7.7: Delete Programme Notification Description

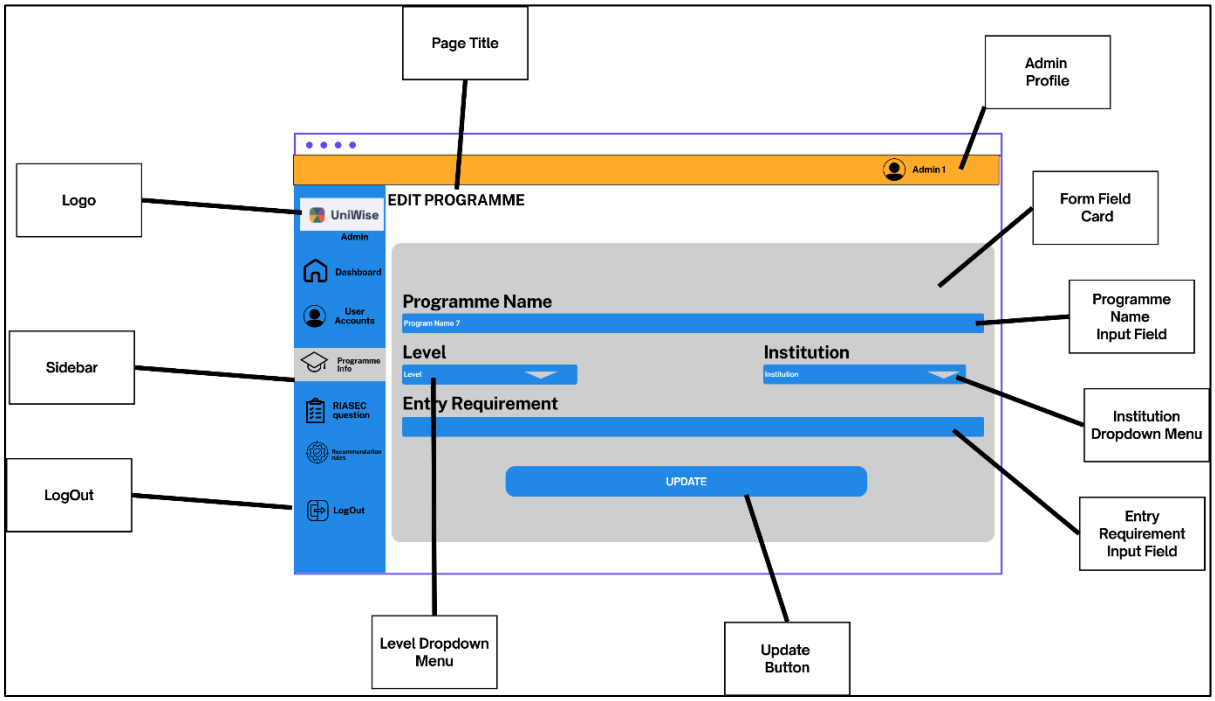


Figure 3.2.1.7.8: Edit Programme Description

	DOCUMENT IDENTIFICATION	
	APPLICATION NAME	VERSION
	UNIWISE	1.0

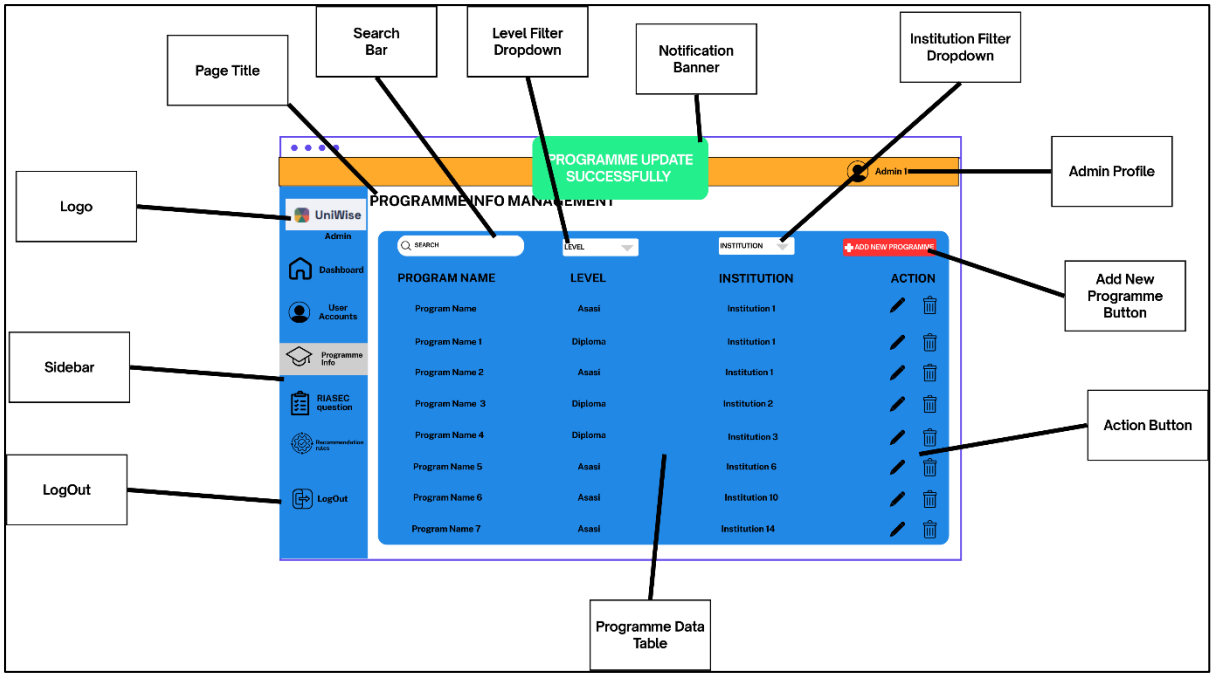


Figure 3.2.1.7.9: Edit Programme Notification Description

3.2.1.8 Admin Manage RIASEC Question

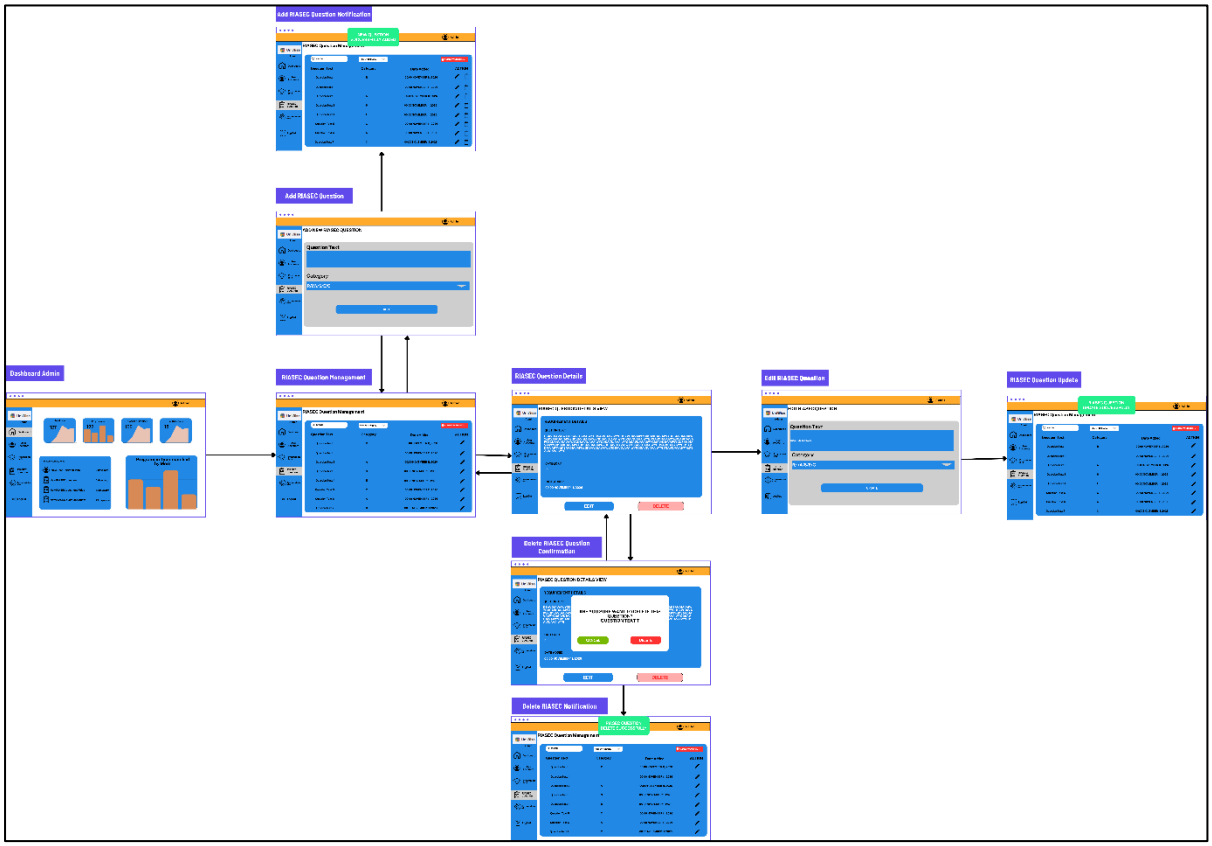


Figure 3.2.1.8.1: Admin Manage RIASEC Question Screenflow

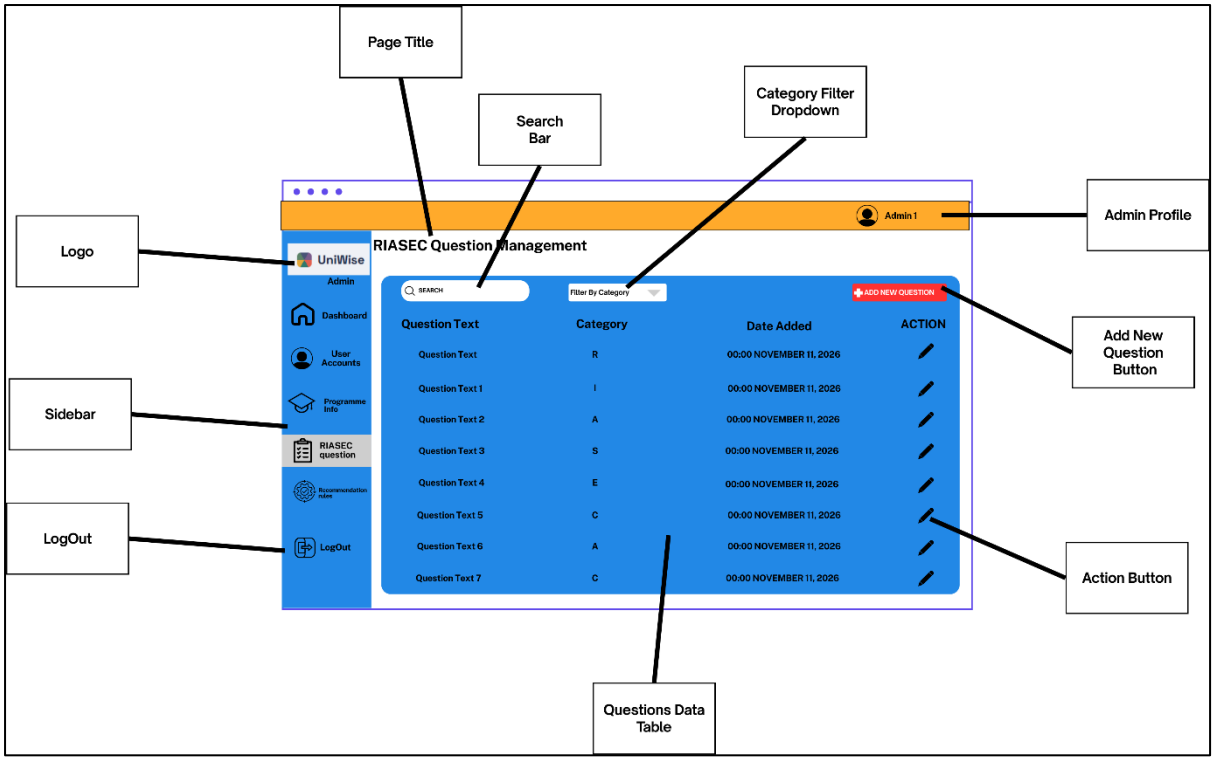


Figure 3.2.1.8.2: Admin Manage RIASEC Question Page Description

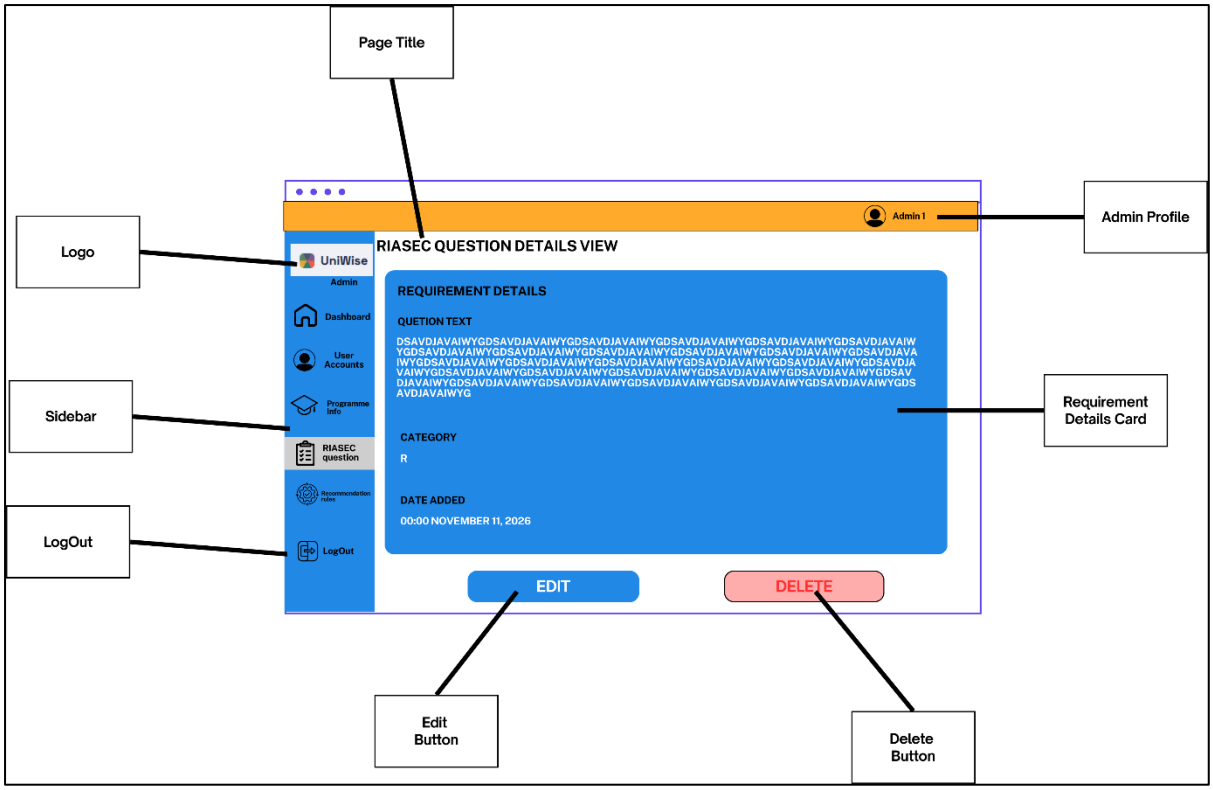


Figure 3.2.1.8.3: RIASEC Question Details Page Description

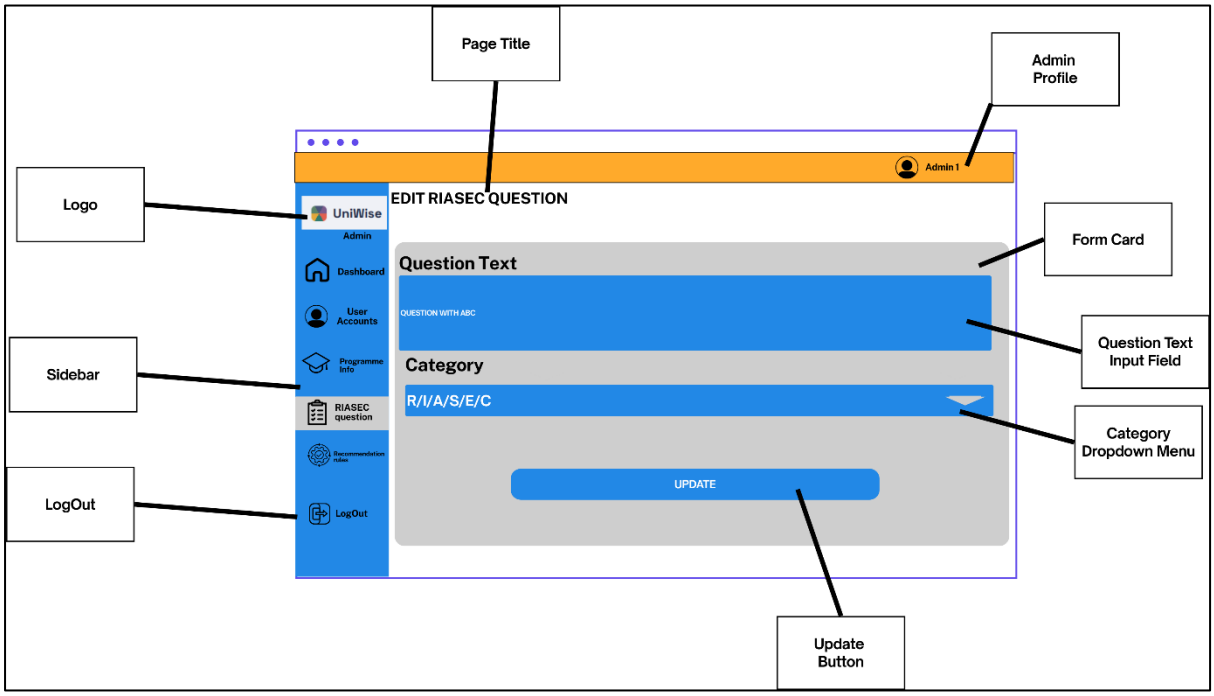


Figure 3.2.1.8.4: Edit RIASEC Question Details Description

	DOCUMENT IDENTIFICATION	
	APPLICATION NAME	VERSION
	UNIWISE	1.0

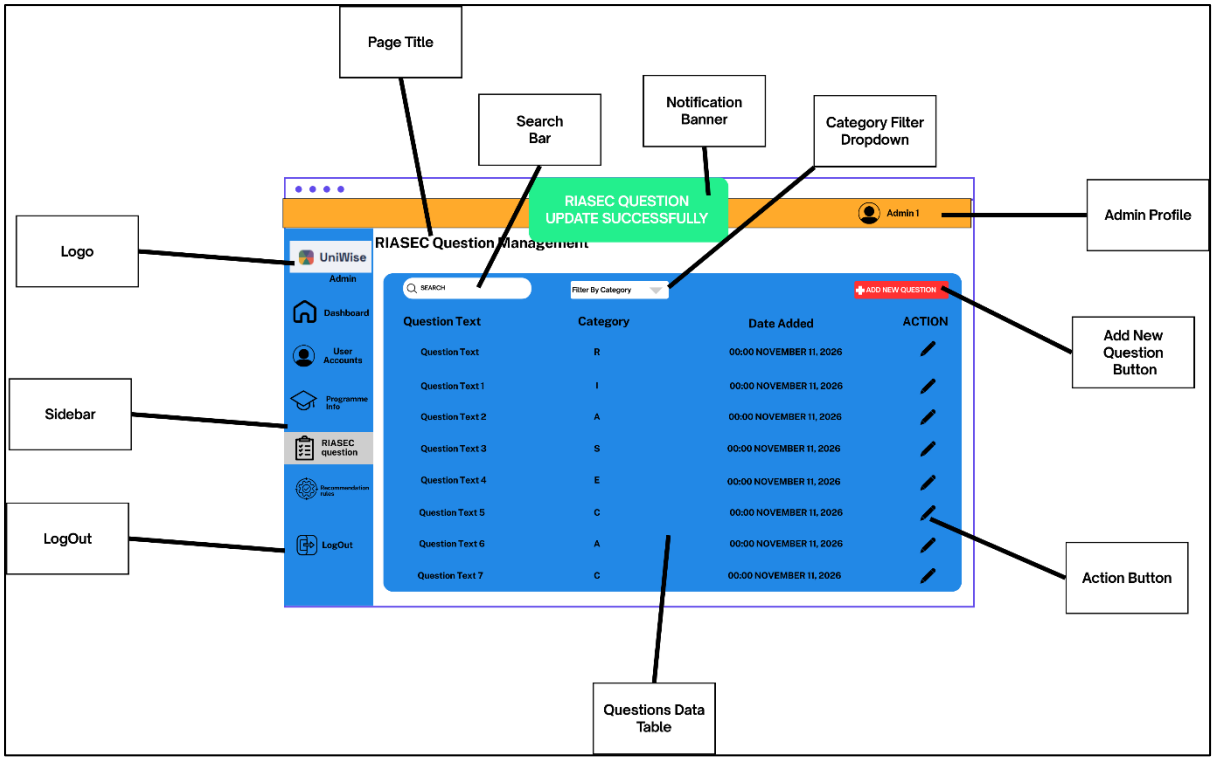


Figure 3.2.1.8.5: Edit RIASEC Question Details Notification Description

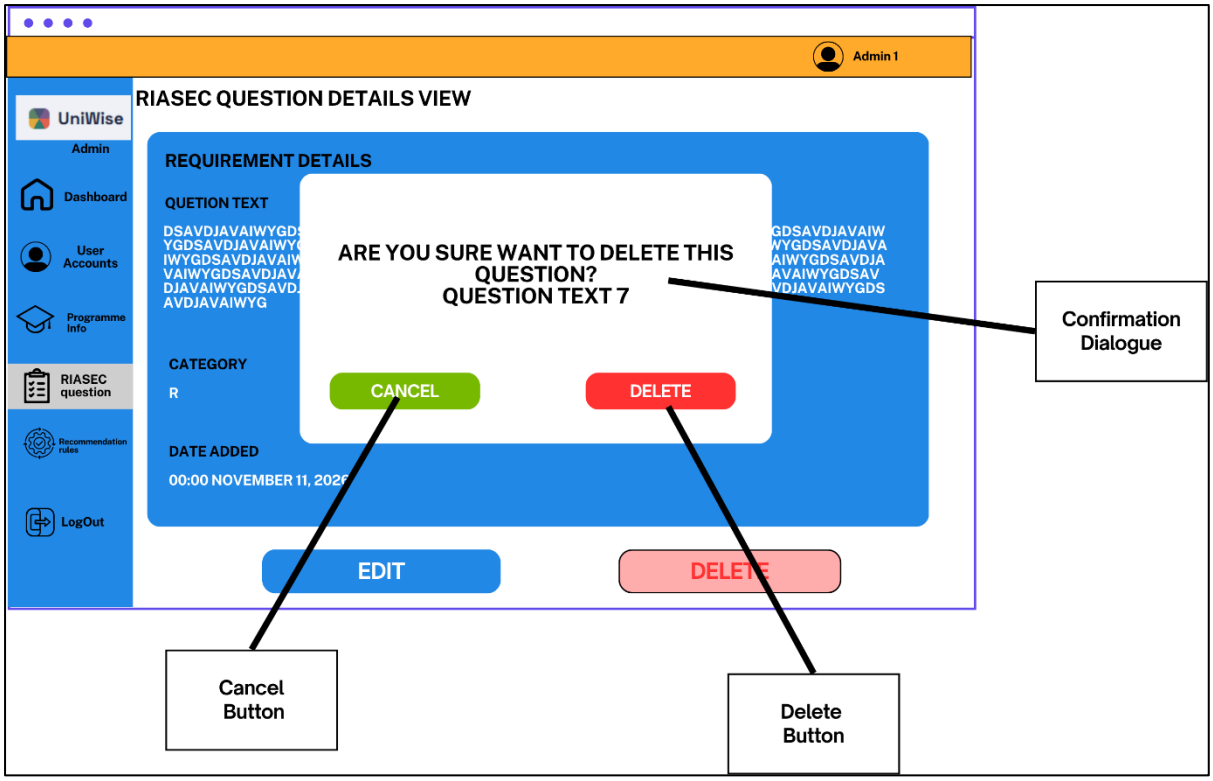


Figure 3.2.1.8.6: Delete RIASEC Question Confirmation Description

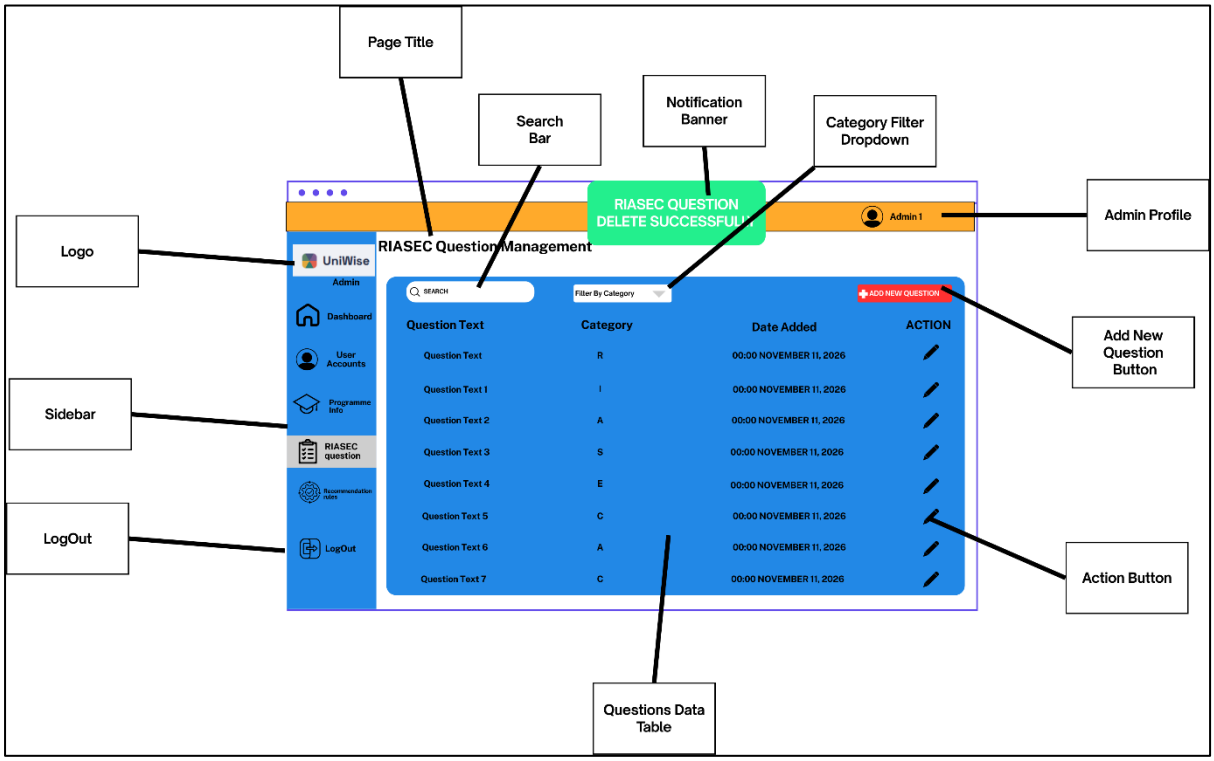


Figure 3.2.1.8.7: Delete RIASEC Question Successful Description

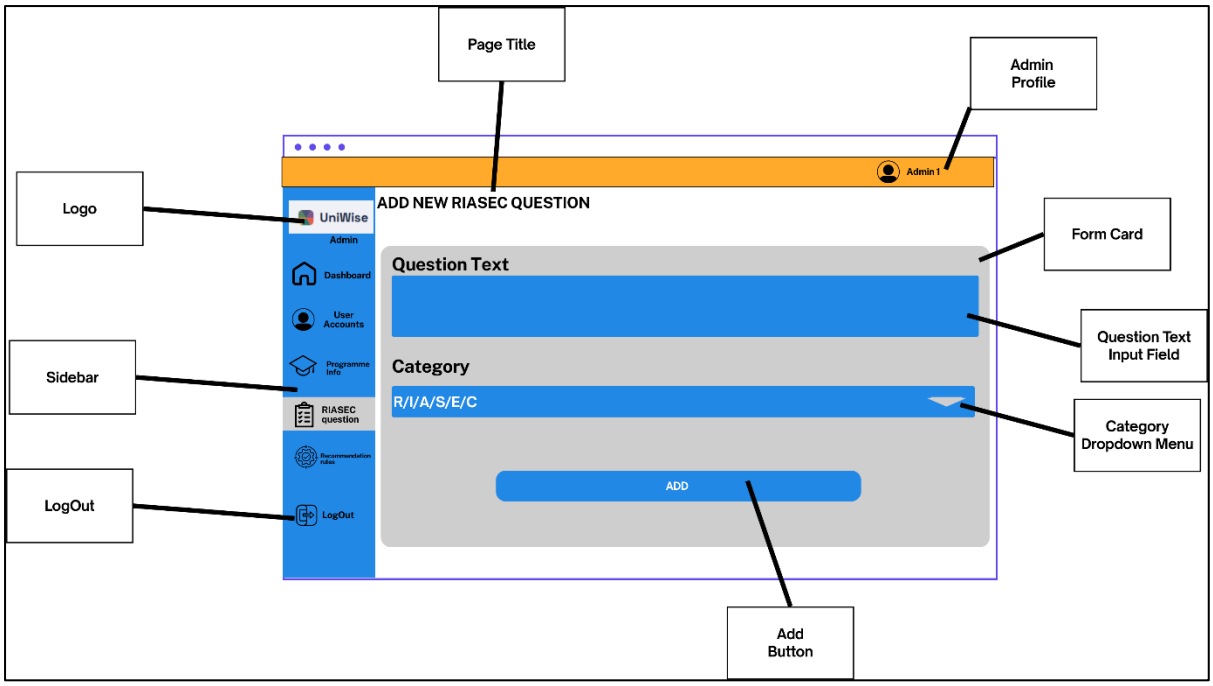


Figure 3.2.1.8.8: Add RIASEC Question Description

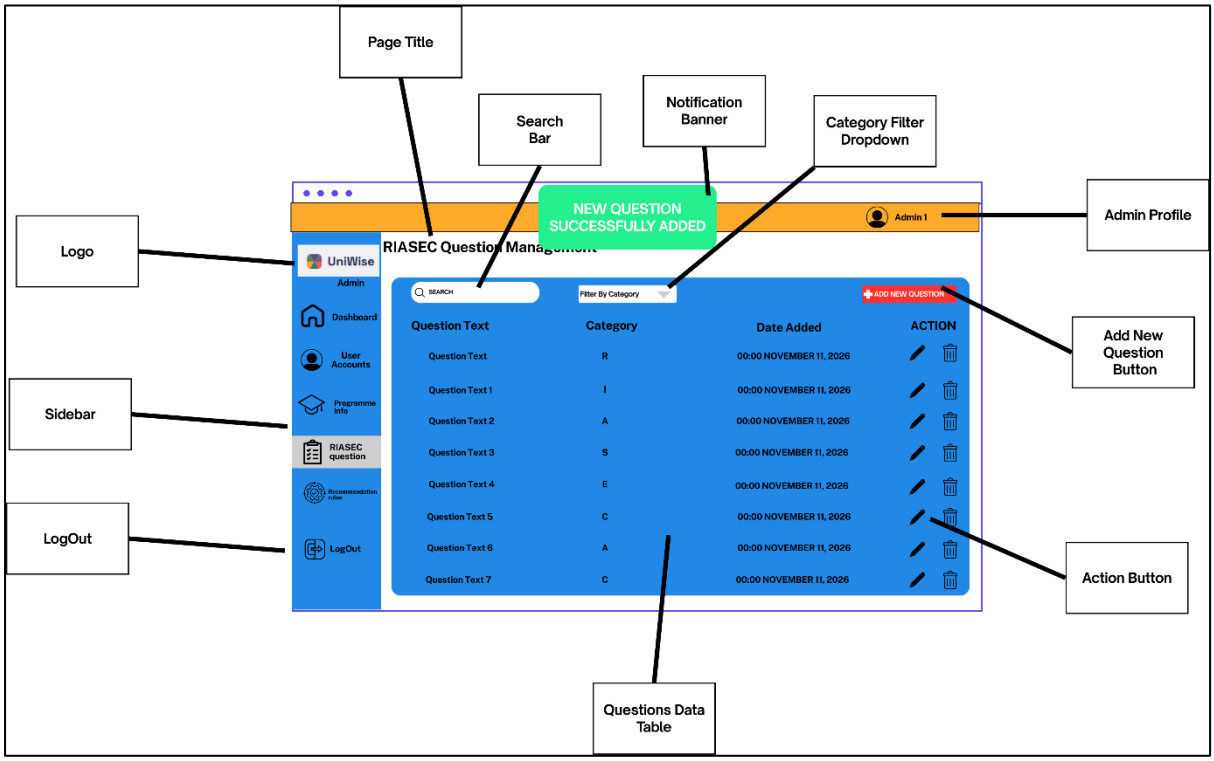


Figure 3.2.1.8.9: Add RIASEC Question Successful Description

3.2.1.9 Admin Manage Recommendation Rule

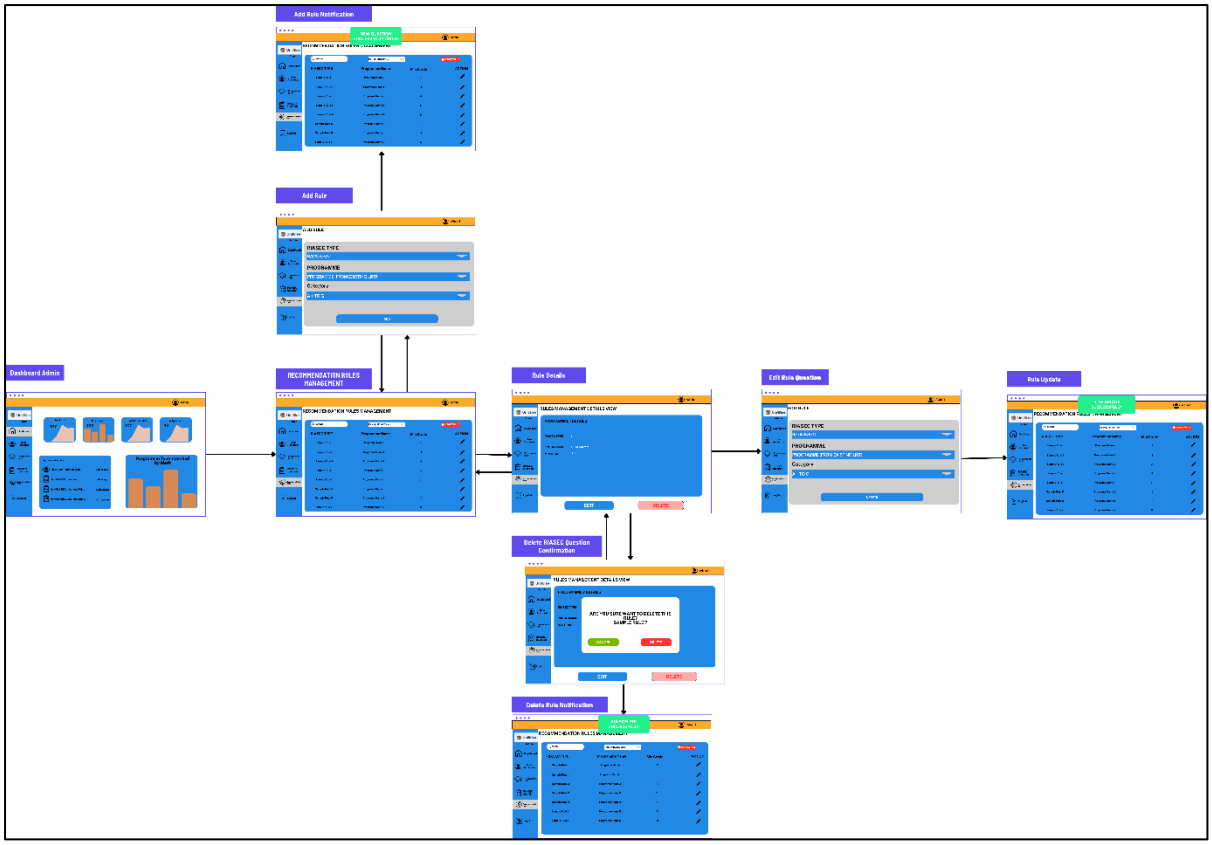


Figure 3.2.1.9.1: Admin Manage Recommendation Rule Screenflow

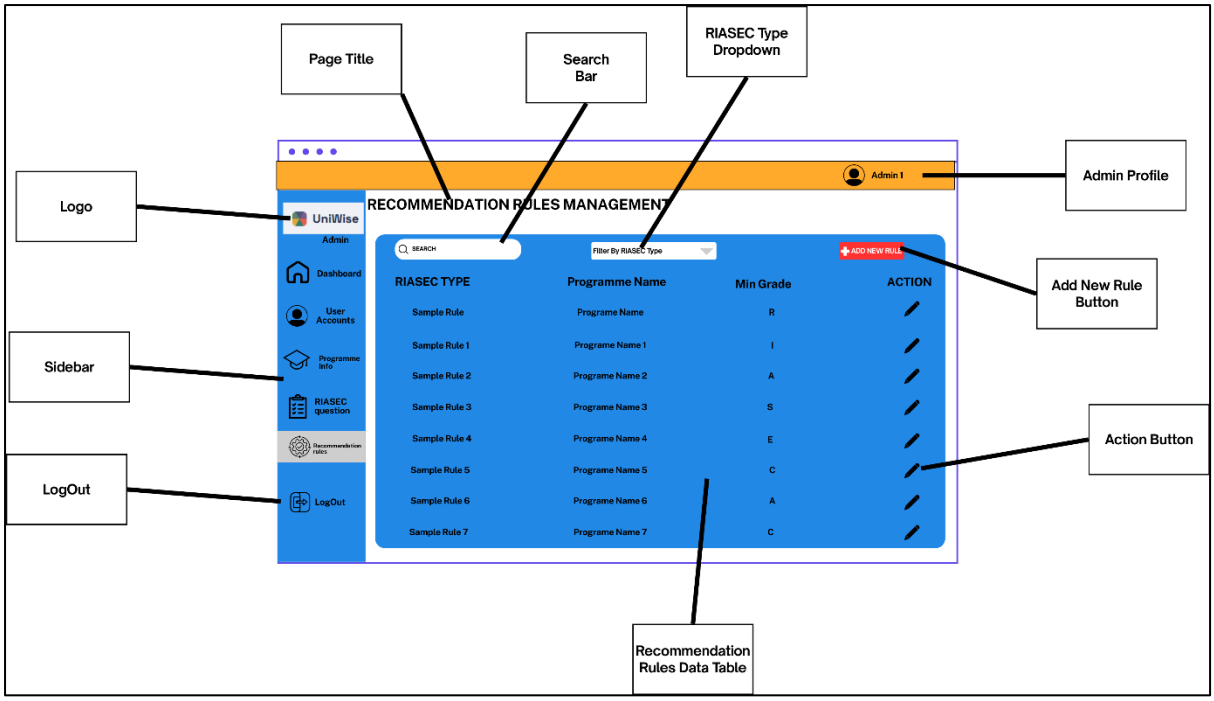


Figure 3.2.1.9.2: Admin Manage Recommendation Rule Page Description

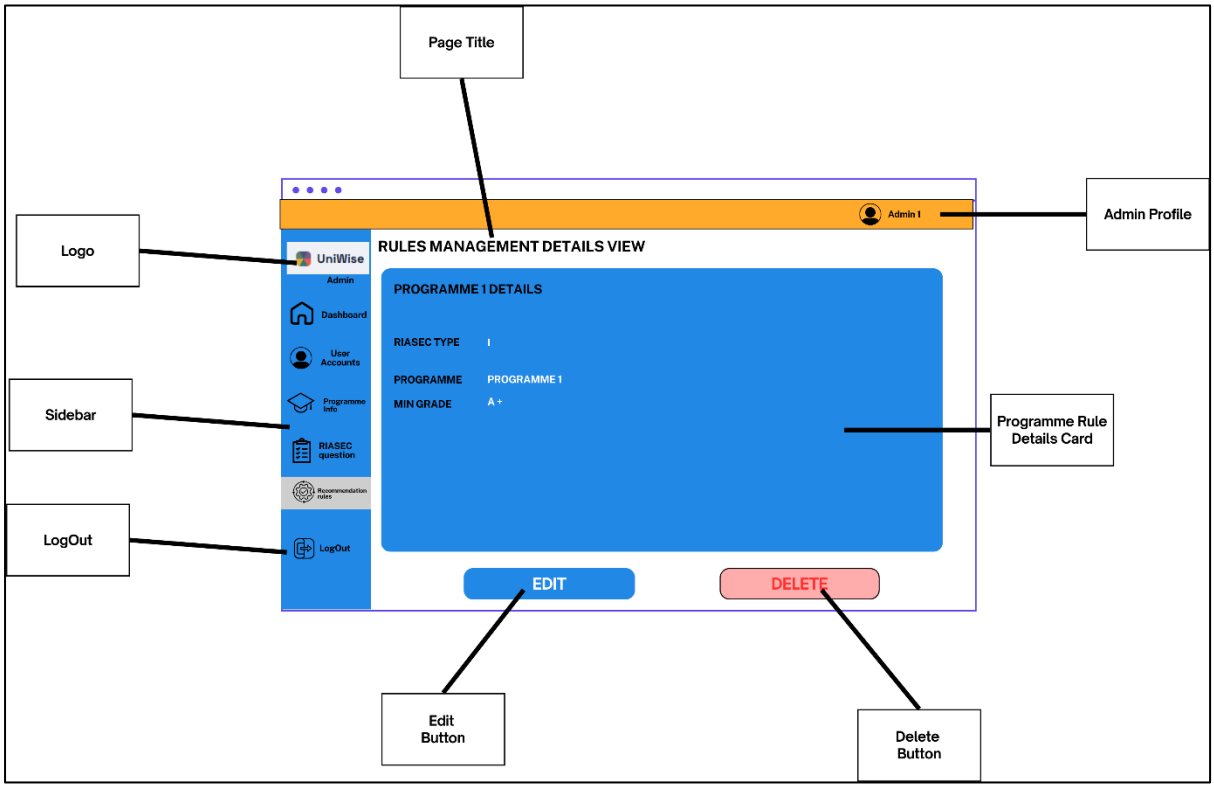


Figure 3.2.1.9.3: Rule Recommendation Details Description

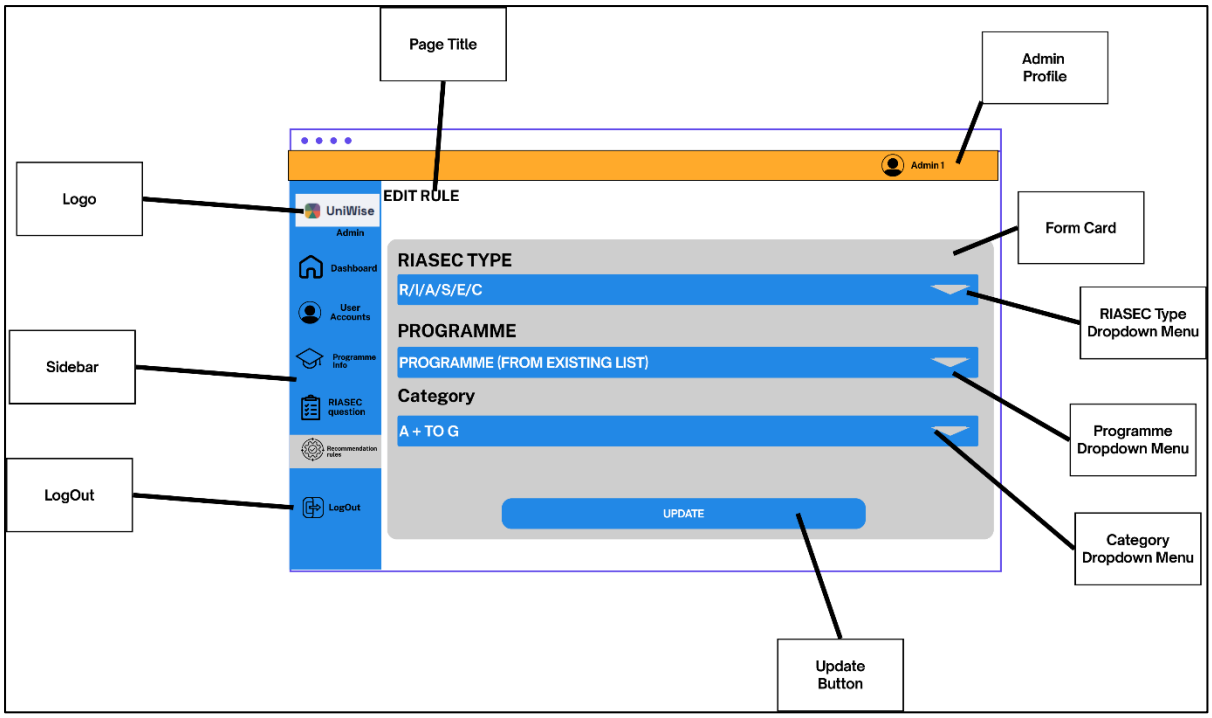


Figure 3.2.1.9.4: Edit Rule Recommendation Details Description

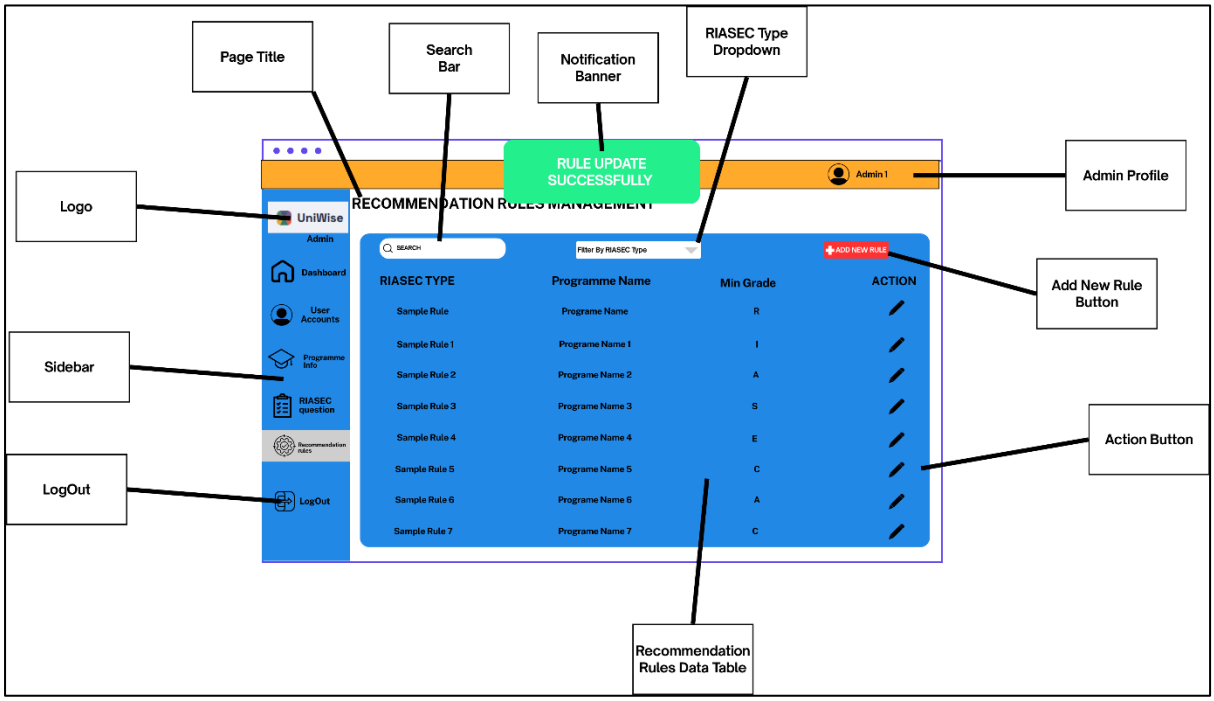


Figure 3.2.1.9.5: Rule Recommendation Update Notification Description

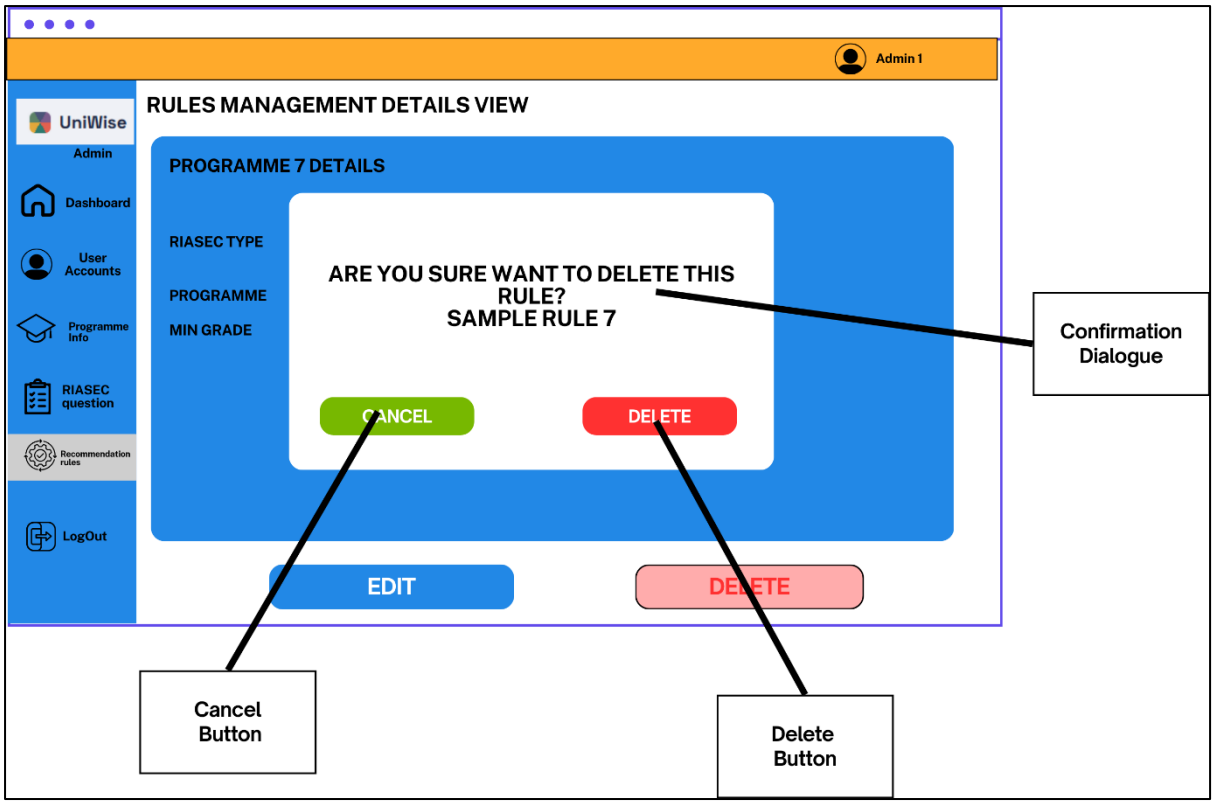


Figure 3.2.1.9.6: Delete Rule Recommendation Description

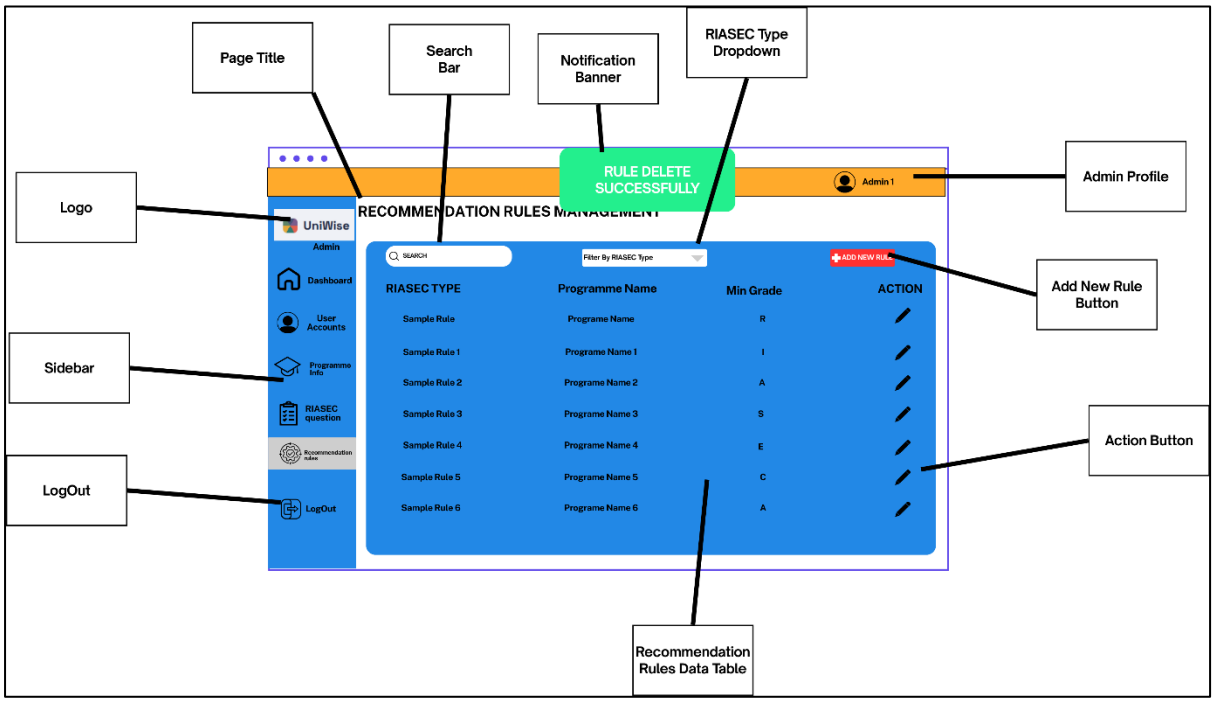


Figure 3.2.1.9.7: Delete Rule Recommendation Successful Description

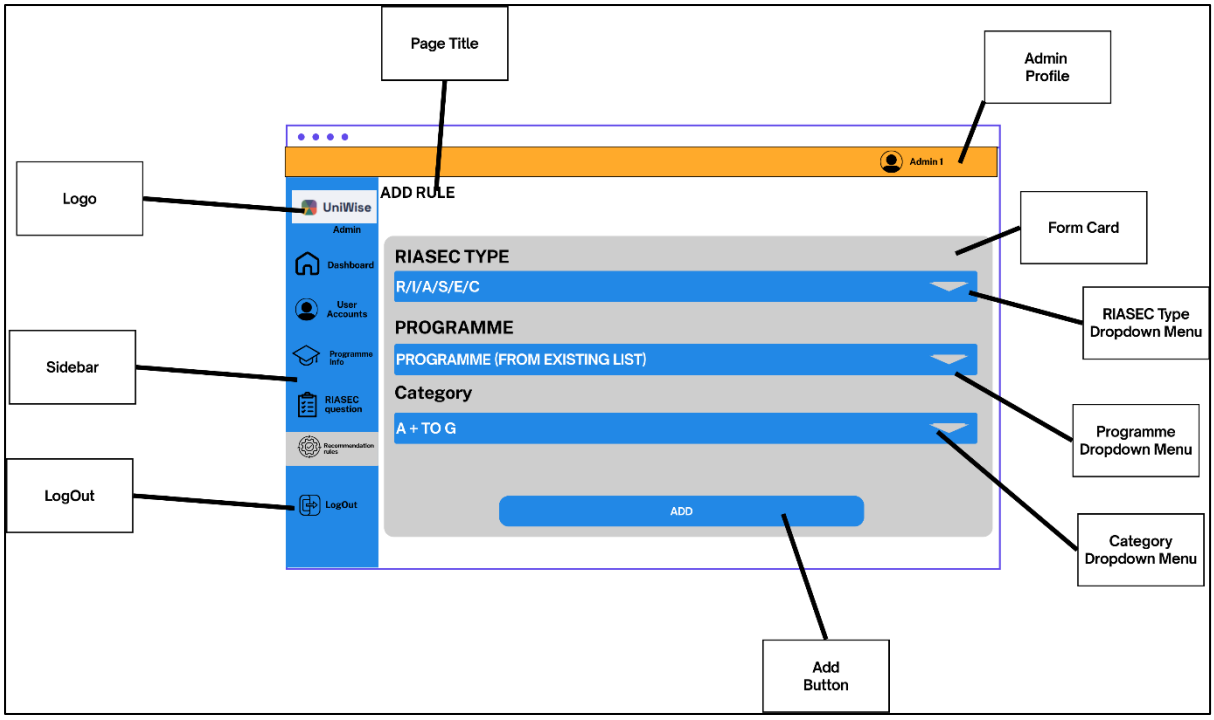


Figure 3.2.1.9.8: Add Rule Recommendation Description

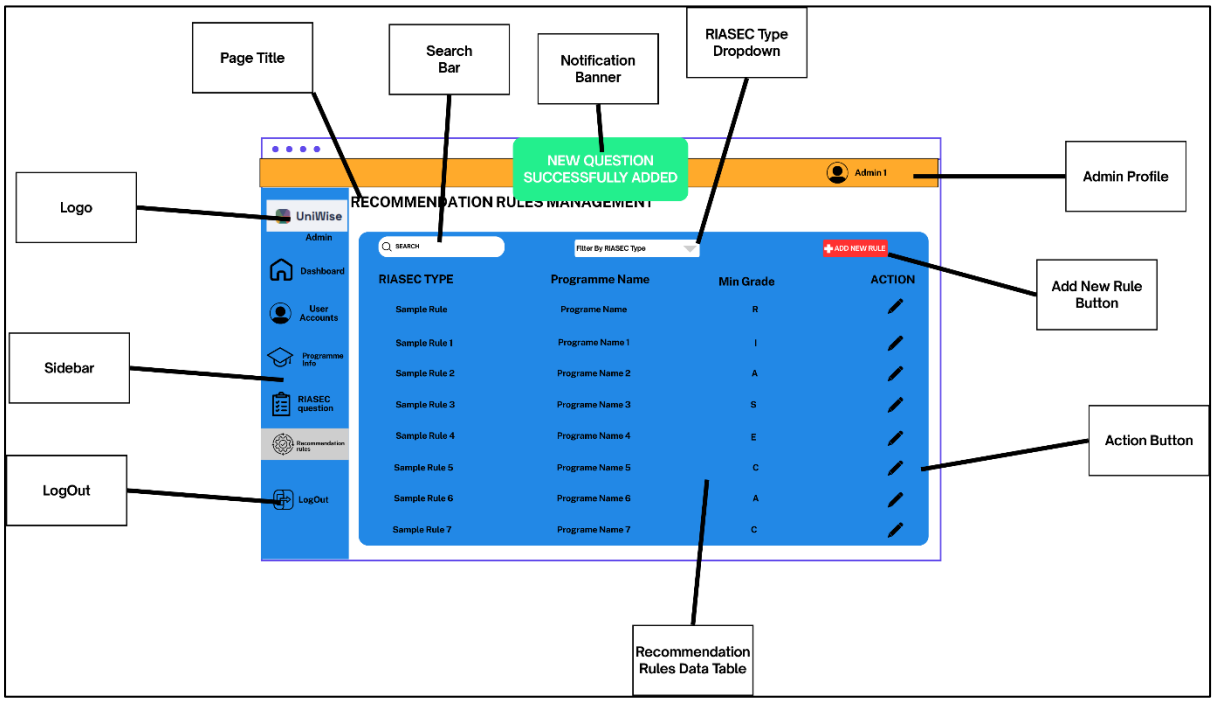


Figure 3.2.1.9.9: Add Rule Recommendation Successful Description

3.3 Functional Requirements

This section lists the functional requirements of the UniWise system. Each requirement describes a specific behaviour that the system must provide, together with its priority level.

FR ID	Description
FR001	Register Account
FR002	User Login
FR003	User Logout
FR004	Update Profile
FR005	Enter SPM Results
FR006	Edit SPM Results
FR007	Validate SPM Results
FR008	Take RIASEC Assessment
FR009	Record RIASEC Responses
FR010	Calculate Personality Type
FR011	View RIASEC Result

FR012	Generate Recommendations
FR013	View Recommended Programmes
FR014	View Programme Details
FR015	Save Recommendation History
FR016	View Recommendation History
FR017	Manage Programme Information
FR018	Manage RIASEC Questions
FR019	Manage Recommendation Rules
FR020	Manage User Accounts

3.4 Use Cases

This section presents the use case diagram and detailed use case descriptions that model the interactions between the system's actors Guest, User, and Administrator and the UniWise system.

3.4.1 Use Case Diagram (Main)

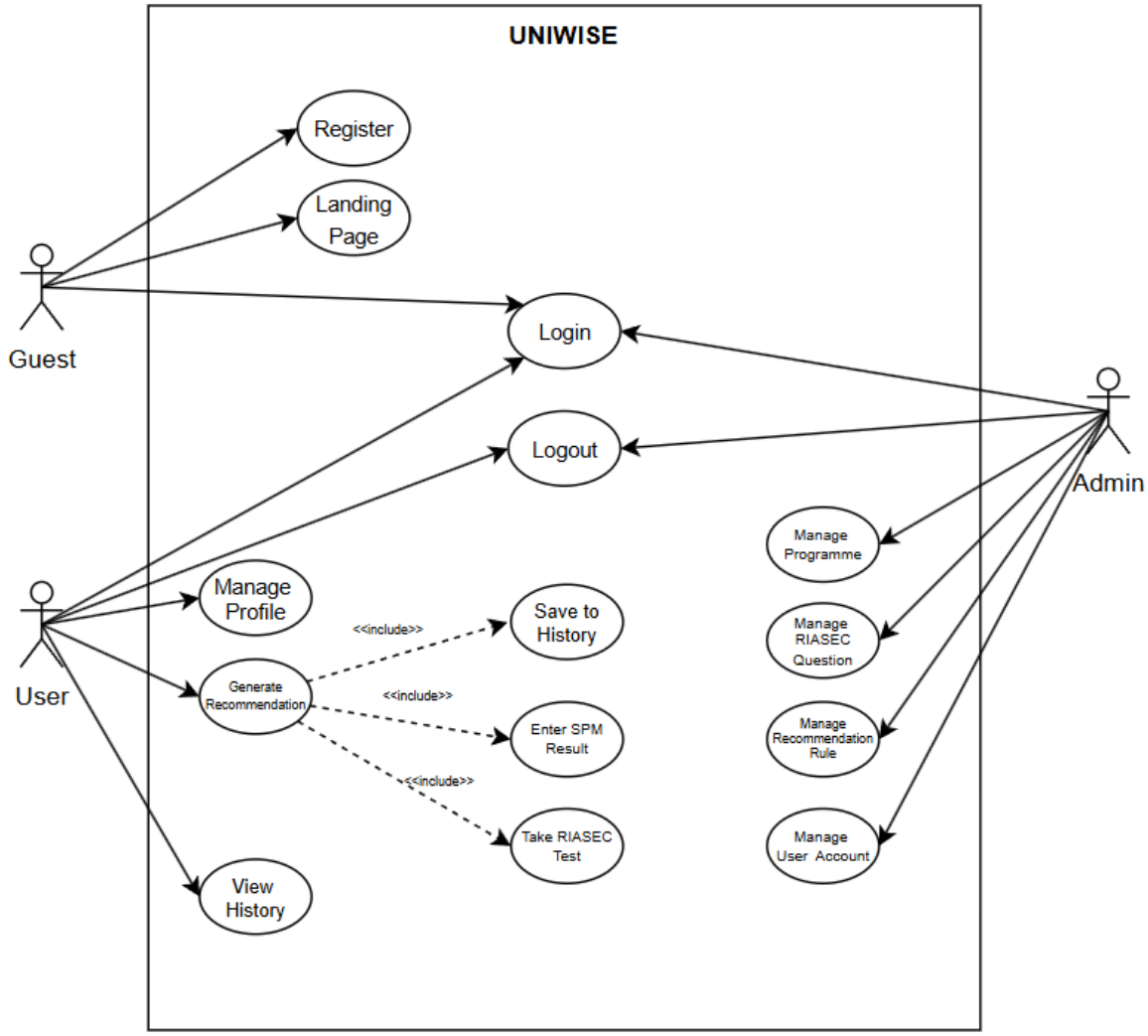


Figure 3.4.1: UniWise Use Case Diagram (Main)

The use case diagram identifies three actors. A Guest can register an account and view general system information. A User, once authenticated, can manage their profile, enter SPM results, complete the RIASEC assessment, generate and view programme recommendations, and review recommendation history. An Administrator manages programme information, RIASEC questions, recommendation rules, and user accounts.

3.4.1.1 Use Case Diagram for Register Account & Authentication

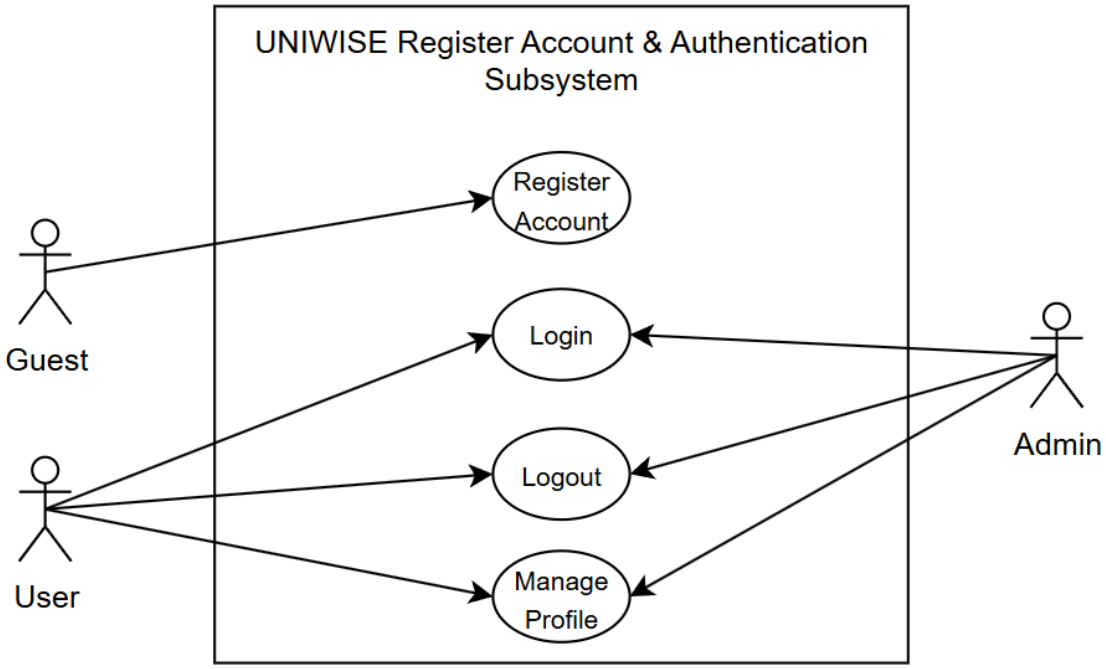


Figure 3.4.1.1: UniWise Use Case Diagram for Register Account & Authentication

3.4.1.2 Use Case Diagram for Manage Profile

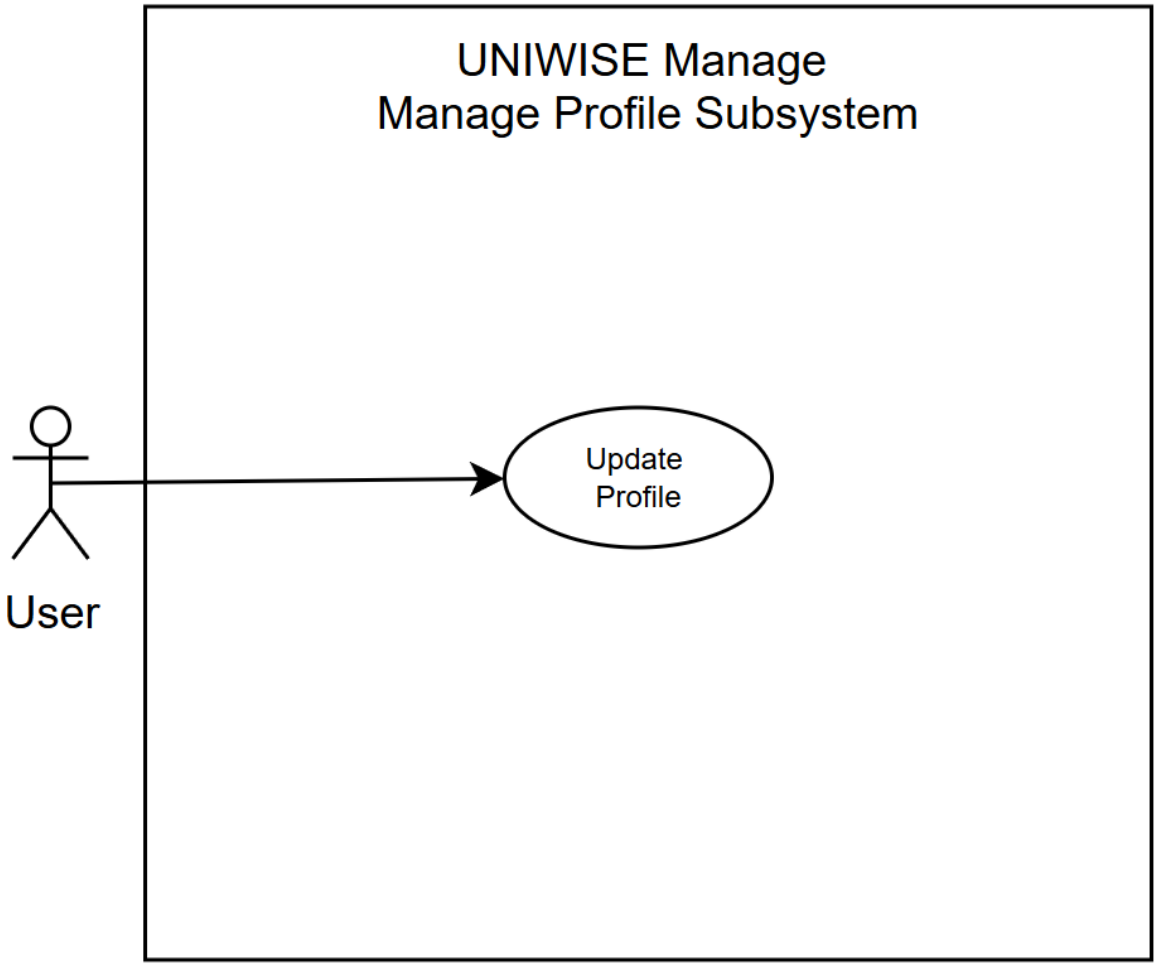


Figure 3.4.1.2: UniWise Use Case Diagram for Manage Profile

3.4.1.3 Use Case Diagram for ManageRIASEC

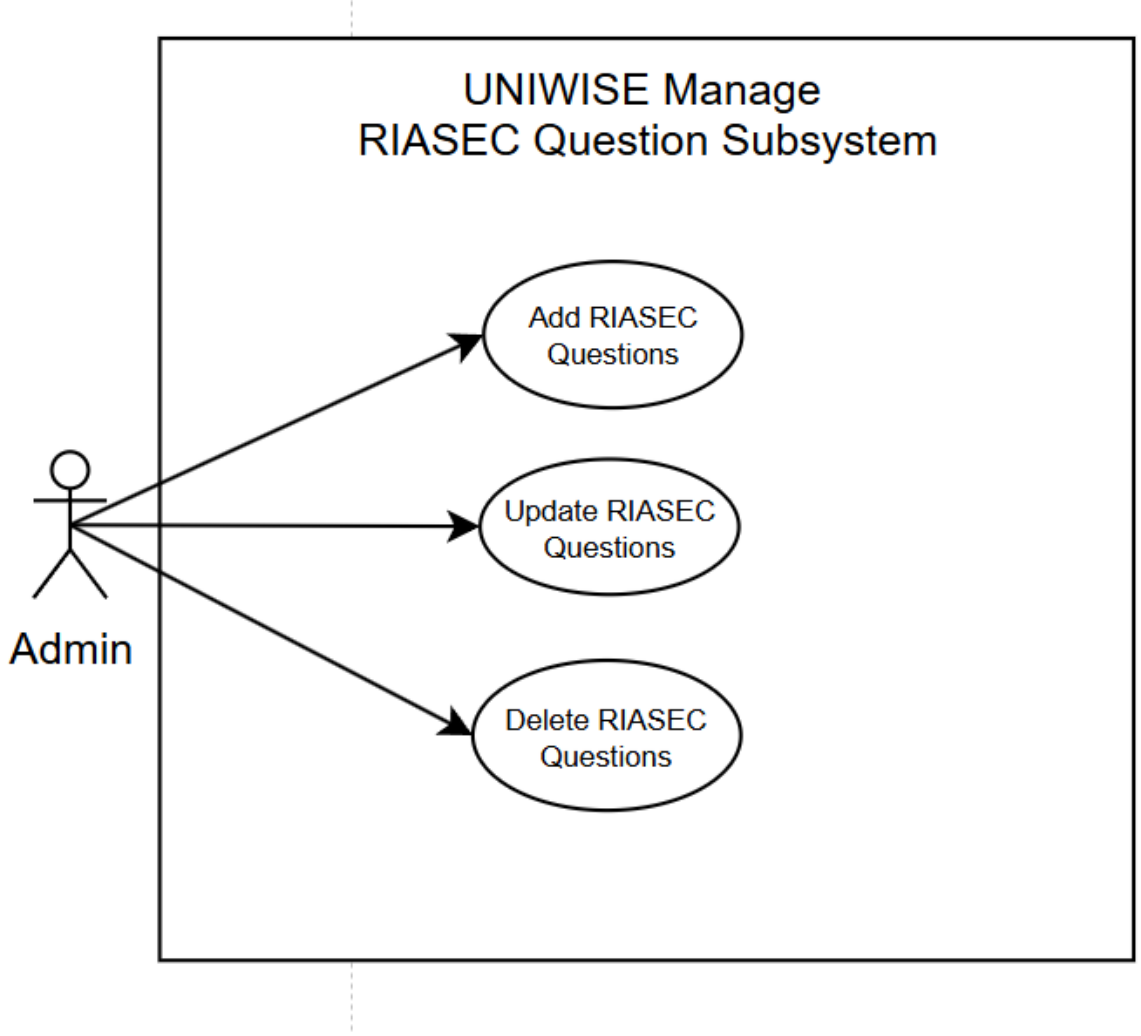


Figure 3.4.1.3: UniWise Use Case Diagram for Manage RIASEC

3.4.1.4 Use Case Diagram for Recommendation History

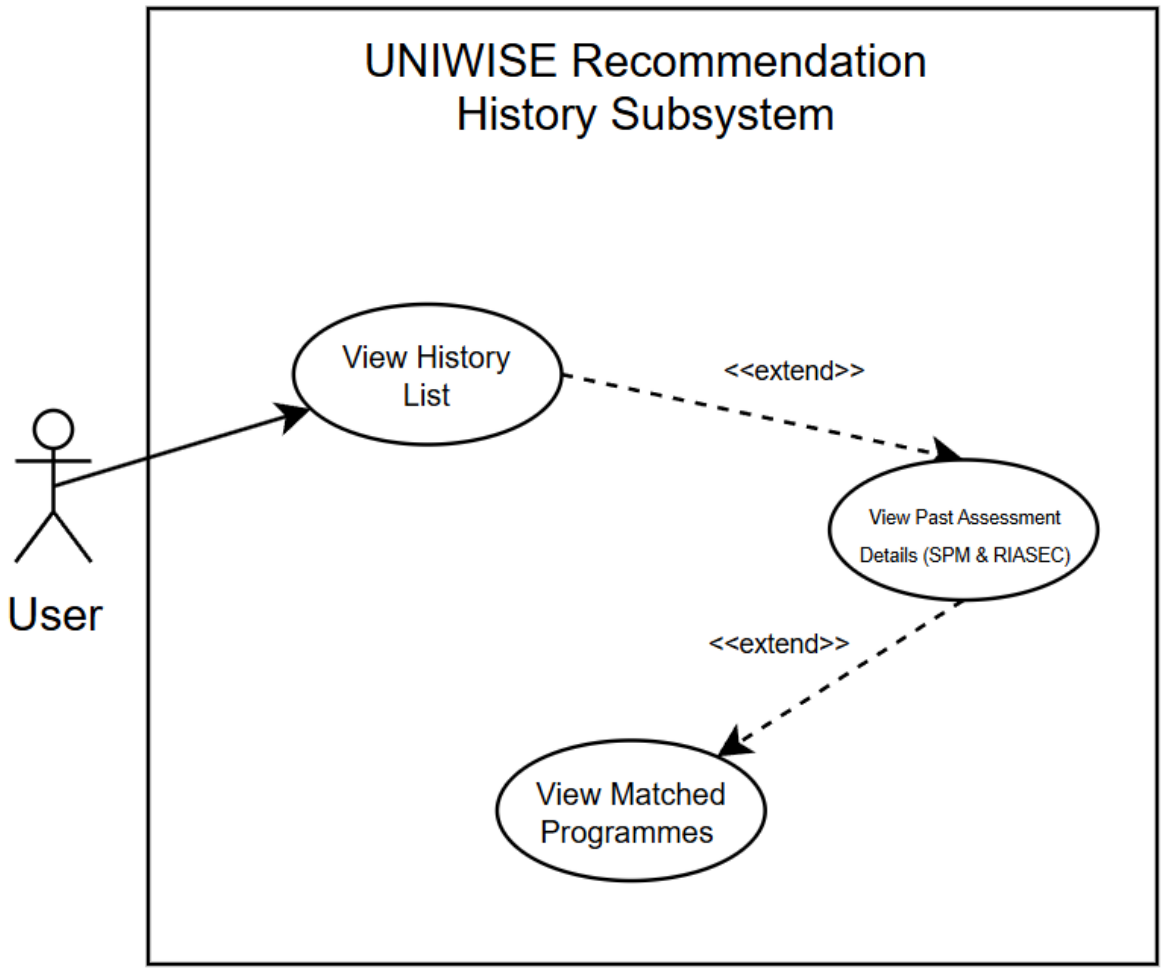


Figure 3.4.1.4: UniWise Use Case Diagram for Recommendation History

3.4.1.5 Use Case Diagram for Recommendation Rule

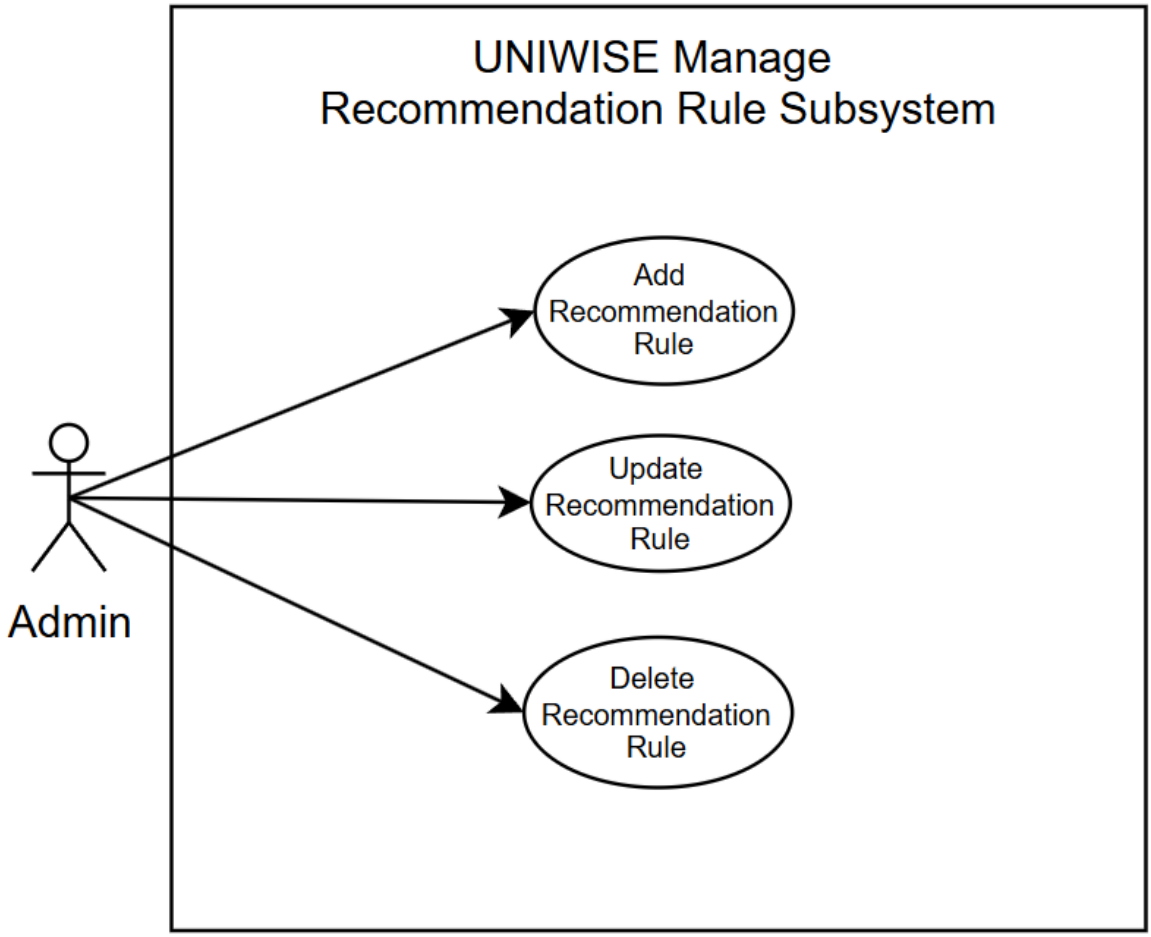


Figure 3.4.1.5: UniWise Use Case Diagram for Recommendation Rule

3.4.1.6 Use Case Diagram for Recommendation Programme

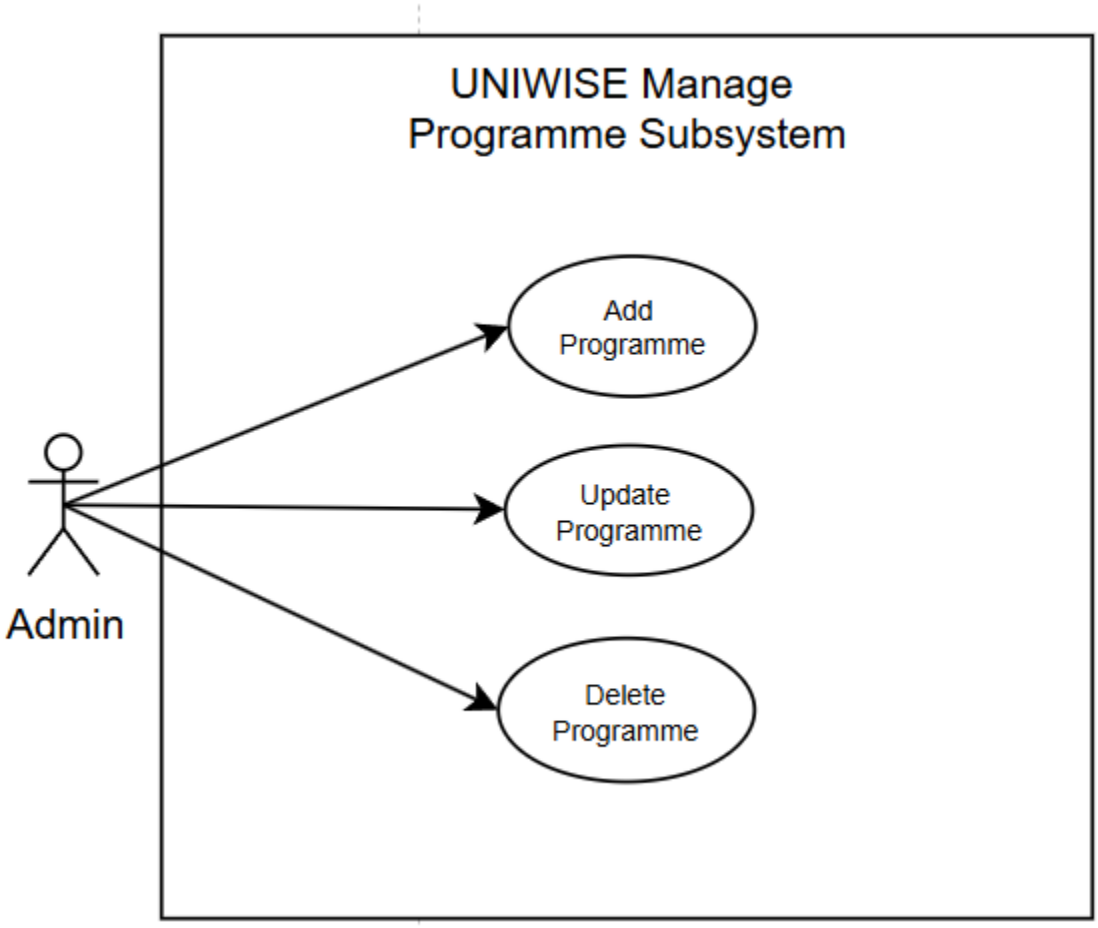


Figure 3.4.1.6: UniWise Use Case Diagram for Recommendation Programme

3.4.2 Use Case Description

3.4.2.1 Use Case Description Register Account

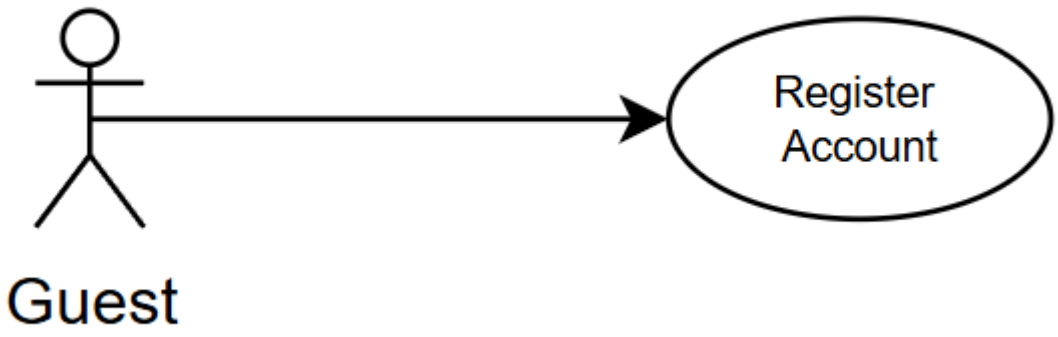


Figure 3.4.2.1 Use Case Description for Register Account

Table 3.4.2.1 Use Case Description for Register Account

Use Case ID	UC001	**Author**	Rosli M. A. H.
Use Case Name	Register Account	**Written Date**	21/07/2026
Actor	Guest		
Description	Allows a new SPM graduate (Guest) to create a user account in the UniWise system using Firebase Authentication.		
Pre-condition	Guest is on the Registration page and has internet connectivity.		
Steps	**Basic Flow:** 1. Guest clicks on the 'Register' button from the Landing Page. 2. System displays the Registration Form requiring Name, Email, and Password. 3. Guest enters required registration details and submits the form. 4. System sends registration details to Firebase Authentication for user creation. 5. Firebase returns successful registration confirmation. 6. System creates corresponding user record in MySQL database and redirects Guest to the Login Page with a success notification. **Alternate / Exception Flow:** 3a. Invalid input format (e.g., weak password, invalid email): 1. System displays validation error message. 2. Guest corrects details and resubmits.		

	4a. Email already registered: 1. System displays 'Email already registered' notification. 2. Guest is prompted to log in instead.
Post-Condition	• User credentials record is created in Firebase Authentication. • Corresponding profile entry is saved in MySQL Users table with role set to 'STUDENT'.
Exception	• Backend or Firebase Authentication service unavailable (Error 500/503). • Network timeout during registration request.
Business Rules	• Email address must be valid and unique across the system. • Password must follow minimum security requirements (minimum 6 characters).

3.4.2.2 Use Case Description Login

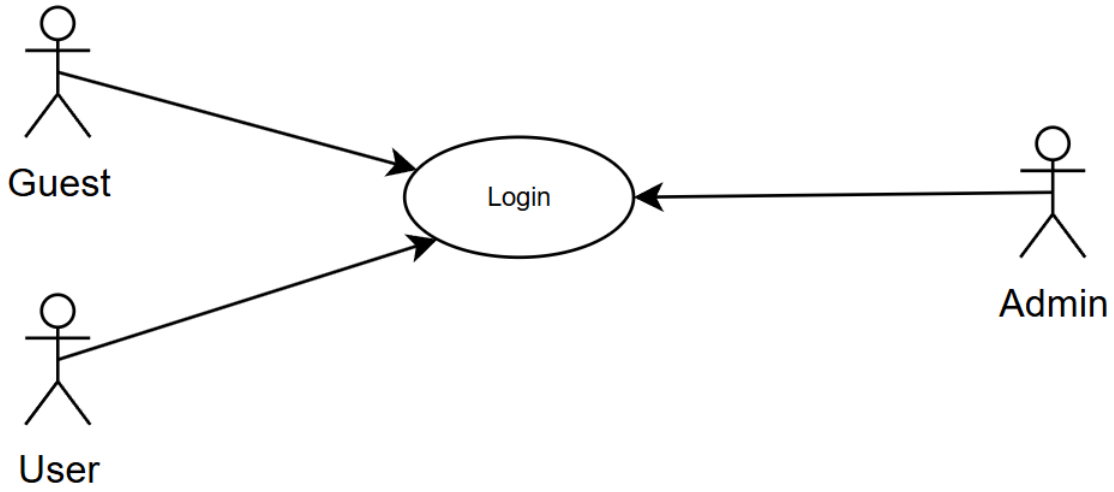


Figure 3.4.2.2 Use Case Description for Login

Table 3.4.2.2 Use Case Description for Login

Use Case ID	UC002	**Author**	Rosli M. A. H.
Use Case Name	Login	**Written Date**	21/07/2026
Actor	Guest, User, Admin		
Description	Authenticates registered users or administrators to grant access to restricted UniWise system features.		
Pre-condition	Actor has a registered account and is on the Login page.		

Steps	**Basic Flow:** 1. Actor enters Email and Password on the Login page. 2. Actor clicks the 'LOG IN' button. 3. System verifies credentials with Firebase Authentication. 4. Backend verifies Firebase ID Token and checks user role in MySQL database. 5. System redirects User to User Dashboard or Admin to Admin Dashboard based on role. **Alternate / Exception Flow:** 3a. Invalid email or password: 1. System displays authentication error notification. 2. Actor remains on Login page to retry. 4a. Account deactivated by Admin: 1. System displays 'Account Deactivated' notification. 2. Login process is terminated.
Post-Condition	• Firebase ID Token issued and session token verified on Laravel backend. • User redirected to User Dashboard or Admin Dashboard according to role.
Exception	• Authentication service connection failure. • User attempts login with a disabled/deactivated account.
Business Rules	• Account status must be ACTIVATED. • Access to administrative functions restricted strictly to authenticated ADMIN accounts.

3.4.2.3 Use Case Description Logout

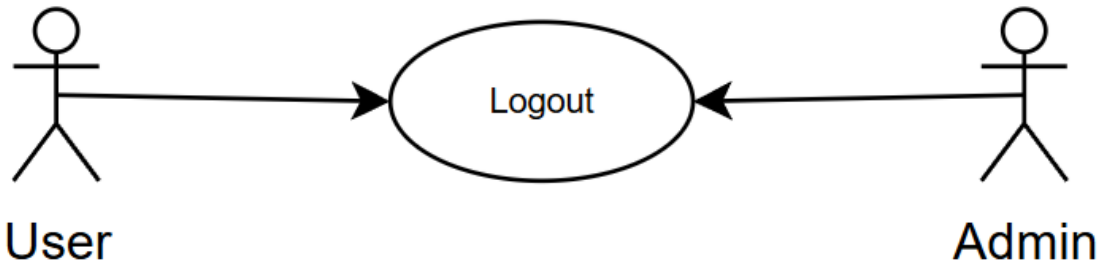


Figure 3.4.2.3 Use Case Description for Logout

Table 3.4.2.3 Use Case Description for Logout

Use Case ID	UC003	**Author**	Rosli M. A. H.
Use Case Name	User Logout	**Written Date**	21/07/2026
Actor	User, Admin		
Description	Logs out the authenticated user or admin and terminates active session.		
Pre-condition	User or Admin is logged into the UniWise system.		
Steps	**Basic Flow:** 1. Actor clicks the 'LogOut' button in the sidebar menu. 2. System clears user session, token, and local authentication state. 3. System redirects the user to the Landing Page / Login Page. **Alternate / Exception Flow:** None.		
Post-Condition	• Client authentication tokens and local storage session states are completely cleared. • User is safely returned to the landing page.		
Exception	• Unexpected client-side session state error during logout execution.		
Business Rules	• User must be logged in before executing session termination.		

3.4.2.4 Use Case Description Update Profile

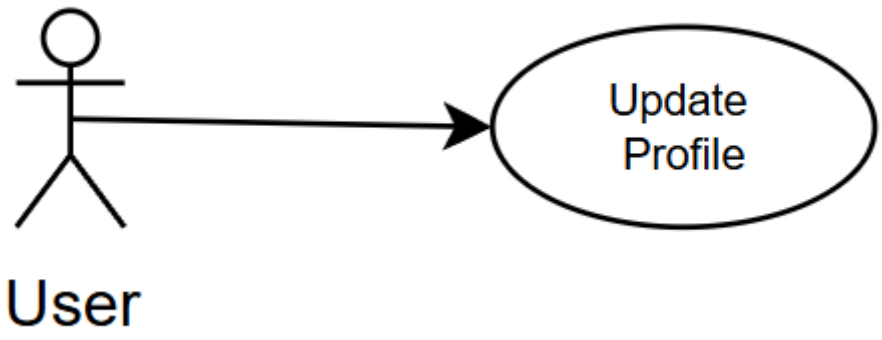


Figure 3.4.2.4 Use Case Description for Update Profile

Table 3.4.2.4 Use Case Description for Update Profile

Use Case ID	UC004	**Author**	Rosli M. A. H.
Use Case Name	Update Profile	**Written Date**	21/07/2026
Actor	User		
Description	Allows a registered user to view and update their profile details (Name, Password, Profile Photo).		
Pre-condition	User is logged in and navigates to the Profile section.		
Steps	**Basic Flow:** 1. User clicks on 'Profile' in the sidebar menu. 2. System displays Profile Details Page. 3. User clicks 'EDIT PROFILE' button. 4. System displays Edit Profile form. 5. User updates Name, uploads new profile picture, or enters new password details. 6. User clicks 'SAVE' button. 7. System validates input, updates profile details in database/Firebase, and displays 'PROFILE UPDATE SUCCESSFULLY' popup notification. **Alternate / Exception Flow:** 7a. Current password mismatch when trying to change password: 1. System displays 'Your current password/New password is not matched' alert. 2. Changes are not saved; user stays on Edit Profile form.		
Post-Condition	• Updated profile information (Name, Password, or Profile Picture) stored in database/Firebase. • Updated user profile details immediately reflected on the user dashboard.		
Exception	• Image file upload fails due to format mismatch or exceeding file size limit. • Database update error during profile details saves.		
Business Rules	• When changing passwords, current password entry must match existing records. • Uploaded profile picture must be in valid image formats (JPEG, PNG) within max size limits.		

3.4.2.5 Use Case Description Enter SPM Results

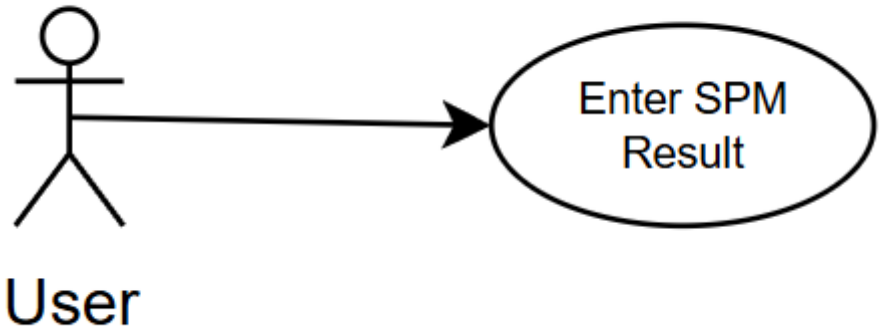


Figure 3.4.2.5 Use Case Description for Enter SPM Results

Table 3.4.2.5 Use Case Description for Enter SPM Results

Use Case ID	UC005	**Author**	Rosli M. A. H.
Use Case Name	Enter SPM Results	**Written Date**	21/07/2026
Actor	User		
Description	Allows a user to input SPM stream and examination grades into the system.		
Pre-condition	User is logged in and navigated to Academic Result page.		
Steps	**Basic Flow:** 1. User selects 'Academic Result' from sidebar menu. 2. System displays Academic Result form showing stream selection (e.g., STEM A) and core/elective subject dropdowns. 3. User selects stream type, universal subjects (Bahasa Melayu, English, Math, History), and package/elective subjects along with earned grades. 4. User clicks 'SAVE' button. 5. System displays confirmation modal ('ARE YOU SURE WANT TO SAVE YOUR ACADEMIC RESULT?'). 6. User confirms by clicking 'YES'. 7. System calculates merit score, stores SPM results in SPM_Results table, and shows success notification ('ACADEMIC RESULT SUCCESSFULLY ADDED'). **Alternate / Exception Flow:** 6a. User clicks 'NO' on confirmation modal: 1. Confirmation modal closes, returning user to edit form without saving. 7a. Duplicate subject selection detected:		

	1. System displays red banner 'SUBJECT DUPLICATE'. 2. Result is not saved until duplicate is resolved.
Post-Condition	• SPM subjects and grades recorded into the SPM_Results table. • Overall Merit Score calculated and updated on User Dashboard summary.
Exception	• Unselected mandatory subject fields during submission. • Database transaction failure when saving subject grades.
Business Rules	• Must select mandatory Universal Subjects (Bahasa Melayu, Bahasa Inggeris, Matematik, Sejarah). • Subject duplication is strictly disallowed. • Grades must be valid SPM letter grades (A+, A, A-, B+, B, C+, C, D, E, G).

3.4.2.6 Use Case Description Edit SPM Results

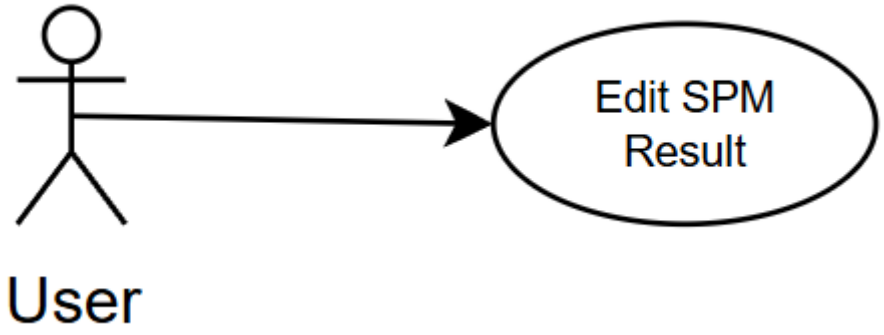


Figure 3.4.2.6 Use Case Description for Edit SPM Results

Table 3.4.2.6 Use Case Description for Edit SPM Results

Use Case ID	UC006	**Author**	Rosli M. A. H.
Use Case Name	Edit SPM Results	**Written Date**	21/07/2026

Actor	User
Description	Allows a user to update previously entered SPM subjects and grades.
Pre-condition	User has previously saved SPM results.
Steps	**Basic Flow:** 1. User navigates to Academic Result page. 2. System populates form with user's previously stored SPM subjects and grades. 3. User modifies subject selections or grades. 4. User clicks 'SAVE' button and confirms on pop-up dialog. 5. System updates SPM_Results table, recalculates merit score, and updates user dashboard. **Alternate / Exception Flow:** Same as UC005 alternate flows.
Post-Condition	• SPM_Results table updated with revised subject selections and grades. • User's Merit Score automatically recalculated.
Exception	• Data sync error when retrieving previously saved SPM subjects.
Business Rules	• Same subject duplicate checks and grade validation rules as UC005 apply.

3.4.2.7 Use Case Description Validate SPM Results

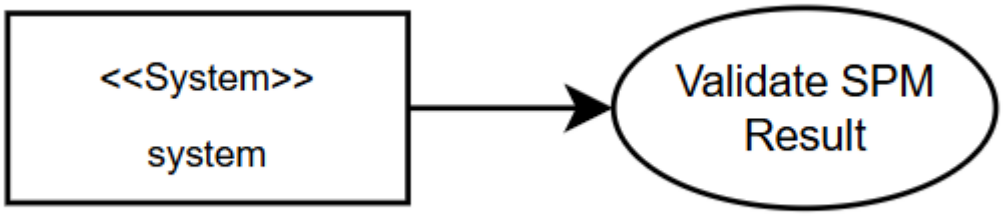


Figure 3.4.2.7 Use Case Description for Validate SPM Results

Table 3.4.2.7 Use Case Description for Validate SPM Results

Use Case ID	UC007	**Author**	Rosli M. A. H.
Use Case Name	Validate SPM Results	**Written Date**	21/07/2026

Actor	System (Automated)
Description	Automated system process to validate SPM subject entries, ensure required core subjects are filled, check grade validity, and compute overall merit score.
Pre-condition	User submits SPM results form.
Steps	**Basic Flow:** 1. System receives SPM subject-grade data from frontend request. 2. System checks that all mandatory universal subjects (BM, BI, Math, History) are selected. 3. System checks for duplicate subjects in user selection. 4. System converts letter grades (A+, A, A-, B+, etc.) to numerical values and computes overall Merit Score percentage. 5. System passes validated data for storage and recommendation evaluation. **Alternate / Exception Flow:** 2a/3a. Validation failure (missing required core subject or duplicate subject): 1. System returns validation error response to frontend. 2. Frontend displays appropriate error banner to user.
Post-Condition	• Validation status flag returned as successful. • Normalized numerical values prepared for Merit Score computation.
Exception	• Malformed payload or invalid grade value sent to validation endpoint.
Business Rules	• All core universal subjects must be present before proceeding to merit score calculation.

3.4.2.8 Use Case Description Take RIASEC Assessment

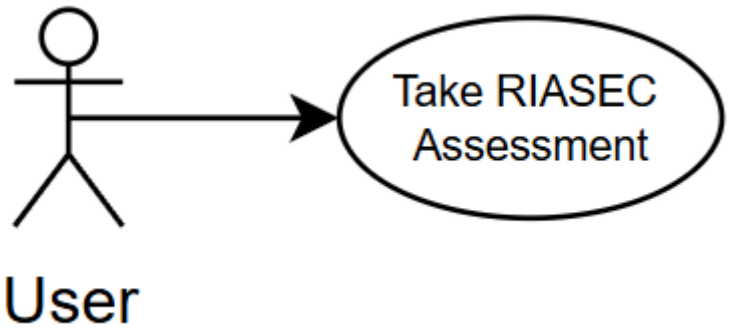


Figure 3.4.2.8 Use Case Description for Take RIASEC Assessment

Table 3.4.2.8 Use Case Description for Take RIASEC Assessment

Use Case ID	UC008	**Author**	Rosli M. A. H.
Use Case Name	Take RIASEC Assessment	**Written Date**	21/07/2026
Actor	User		
Description	Presents RIASEC personality questionnaire to user and records answers to evaluate Holland personality traits.		
Pre-condition	User is logged in and has completed SPM results entry.		
Steps	**Basic Flow:** 1. User selects 'RIASEC Assessment' from sidebar. 2. System displays RIASEC test intro page ('What Should You Study: Personality Quiz'). 3. User clicks 'TAKE QUIZ NOW' button. 4. System presents assessment questions sequentially with 'YES' / 'NO' response options and progress bar. 5. User answers each question and navigates through questions using 'NEXT' and 'PREVIOUS' buttons. 6. User reaches final question and submits completed assessment. **Alternate / Exception Flow:** 3a. User has not completed SPM Academic Results entry yet: 1. System blocks quiz initiation and displays red pop-up notification 'Please complete Academic Results first'. 2. User is prompted to fill Academic Results first.		
Post-Condition	• User responses to all questionnaire items successfully captured. • System proceeds to calculate the dominant personality trait.		
Exception	• Assessment session interrupted due to network loss.		
Business Rules	• Prerequisite Rule: User MUST complete SPM Academic Results entry before taking the RIASEC assessment.		

3.4.2.9 Use Case Description Record RIASEC Responses

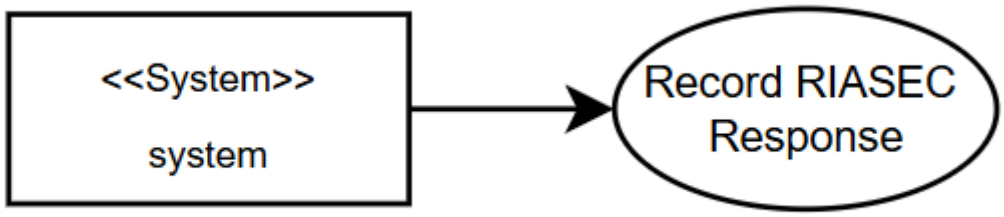


Figure 3.4.2.9 Use Case Description for Record RIASEC Responses

Table 3.4.2.9 Use Case Description for Record RIASEC Responses

Use Case ID	UC009	**Author**	Rosli M. A. H.
Use Case Name	Record RIASEC Responses	**Written Date**	21/07/2026
Actor	System (Automated)		
Description	Stores individual user answers for each RIASEC question into database for trait calculation.		
Pre-condition	User answers RIASEC quiz questions.		
Steps	**Basic Flow:** 1. System receives individual question answers ('YES'/'NO' / values 1 or 0) from frontend. 2. System maps answers to respective question IDs and RIASEC personality categories (R, I, A, S, E, C). 3. System saves response records into RIASEC_Responses table in MySQL database. **Alternate / Exception Flow:** None.		
Post-Condition	• Individual user choices inserted into RIASEC_Responses database table.		
Exception	• Database write failure during batch insert of question responses.		
Business Rules	• Each response must map directly to a valid question_id and category (R, I, A, S, E, C).		

3.4.2.10 Use Case Description Calculate Personality Type

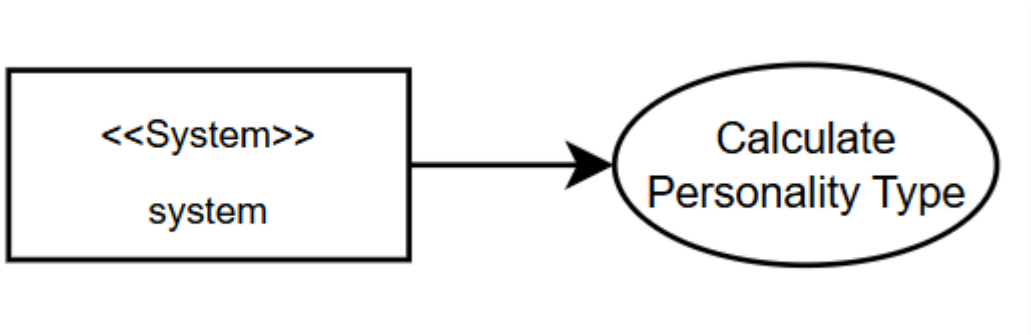


Figure 3.4.2.10 Use Case Description for Calculate Personality Type

Table 3.4.2.10 Use Case Description for Calculate Personality Type

Use Case ID	UC010	**Author**	Rosli M. A. H.
Use Case Name	Calculate Personality Type	**Written Date**	21/07/2026
Actor	System (Automated)		
Description	Calculates category totals from RIASEC responses to determine user's dominant RIASEC personality trait and trait scores.		
Pre-condition	User completes all RIASEC assessment questions.		
Steps	**Basic Flow:** 1. System aggregates scores across the 6 personality categories (Realistic, Investigative, Artistic, Social, Enterprising, Conventional). 2. System identifies the category with highest score as user's Dominant Trait (e.g., INVESTIGATION). 3. System generates RIASEC trait score vector for radar chart visualization. **Alternate / Exception Flow:** None.		
Post-Condition	• Total category scores aggregated and dominant trait code stored. • Radar chart coordinates prepared for display.		
Exception	• Incomplete assessment response set encountered during calculation.		
Business Rules	• In case of a tie between traits, predefined secondary priority rules establish the dominant trait.		

3.4.2.11 Use Case Description View RIASEC Result

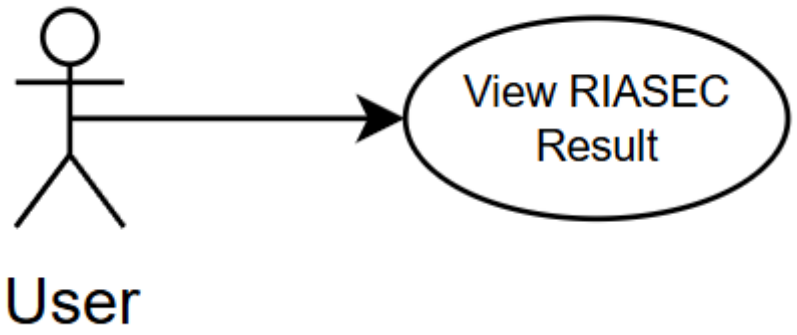


Figure 3.4.2.11 Use Case Description for View RIASEC Result

Table 3.4.2.11 Use Case Description for View RIASEC Result

Use Case ID	UC011	**Author**	Rosli M. A. H.
Use Case Name	View RIASEC Result	**Written Date**	21/07/2026
Actor	User		
Description	Displays user's RIASEC personality profile including dominant trait, detailed explanation, and radar chart graph.		
Pre-condition	User completed RIASEC assessment.		
Steps	**Basic Flow:** 1. Upon completing quiz, system redirects User to RIASEC Assessment Result screen. 2. System displays dominant trait title (e.g., 'Your dominant trait is INVESTIGATION'). 3. System displays RIASEC radar traits graph and explanation paragraph. 4. System displays list of matched tertiary programmes fitting personality and academic eligibility. **Alternate / Exception Flow:** None.		
Post-Condition	• Personality traits radar graph, trait description, and initial matching programmes displayed.		
Exception	• Chart rendering failure on frontend browser.		
Business Rules	• Results displayed must strictly reflect the latest calculated test session records.		

3.4.2.12 Use Case Description Generate Recommendations

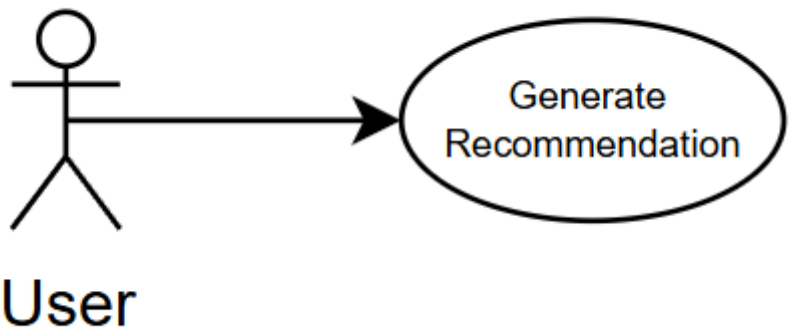


Figure 3.4.2.12 Use Case Description for Generate Recommendations

Table 3.4.2.12 Use Case Description for Generate Recommendations

Use Case ID	UC012	**Author**	Rosli M. A. H.
Use Case Name	Generate Recommendations	**Written Date**	21/07/2026
Actor	User / System		
Description	Rule-based recommendation engine matching user's SPM grades and RIASEC trait against IPTA programme requirements.		
Pre-condition	User SPM results and RIASEC personality results are available.		
Steps	**Basic Flow:** 1. System fetches user's calculated Merit Score / SPM subject grades and dominant RIASEC personality code. 2. System queries Recommendation_Rules table matching RIASEC type and minimum academic requirements. 3. System filters eligible IPTA programmes (Diploma, Foundation, Certificate). 4. System generates tailored programme recommendation list. 5. System automatically logs session to Recommendations history table. **Alternate / Exception Flow:** 3a. No matching programmes found for user eligibility/personality: 1. System displays notification 'No matching programmes found based on current criteria'. 2. Suggests user update SPM results or retake quiz.		
Post-Condition	• Matching IPTA higher education programmes generated and saved. • Recommendation session logged into history records.		

Exception	• No matching criteria found within Recommendation_Rules database table.
Business Rules	• Recommendation engine targets ONLY public IPTA institutions. • Restricted to Diploma, Foundation (Asasi), and Certificate levels. • Programmes match only if academic minimum grades and RIASEC trait rules are satisfied.

3.4.2.13 Use Case Description View Recommended Programmes

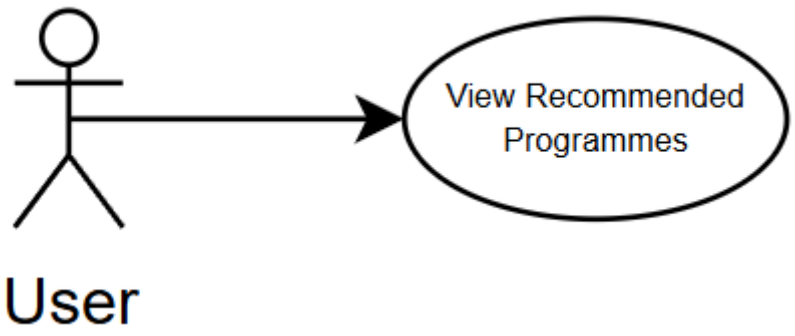


Figure 3.4.2.13 Use Case Description for View Recommended Programmes

Table 3.4.2.13 Use Case Description for View Recommended Programmes

Use Case ID	UC013	**Author**	Rosli M. A. H.
Use Case Name	View Recommended Programmes	**Written Date**	21/07/2026
Actor	User		
Description	Displays list of IPTA programmes recommended to user based on recommendation rules.		
Pre-condition	Recommendations have been generated.		

Steps	**Basic Flow:** 1. System displays list card elements showing recommended IPTA programmes (e.g., ASASI PENGURUSAN PELANCONGAN, ASASI AGRISAINS). 2. Each programme card displays institution logo, programme title, and 'DETAILS' button. 3. User can click 'DETAILS' button on any programme card to view complete entry requirements. **Alternate / Exception Flow:** None.
Post-Condition	• Targeted list of eligible programmes presented with institution logo and details buttons.
Exception	• Failed loading programme list UI components.
Business Rules	• Only programmes for which the student meets minimum entry criteria are displayed.

3.4.2.14 Use Case Description View Programme Details

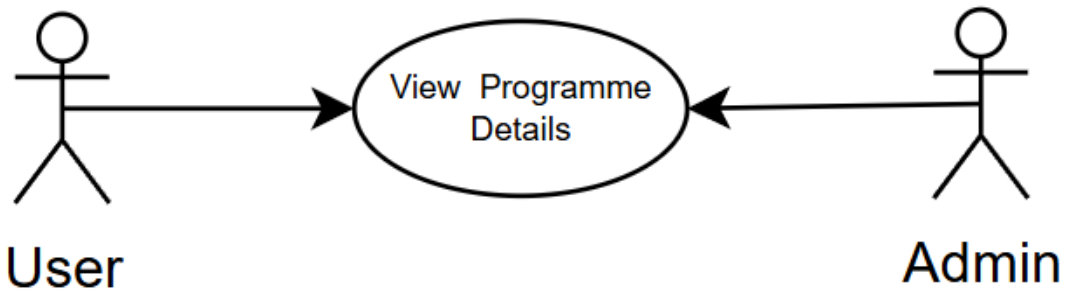


Figure 3.4.2.14 Use Case Description for View Programme Details

Table 3.4.2.14 Use Case Description for View Programme Details

Use Case ID	UC014	**Author**	Rosli M. A. H.
Use Case Name	View Programme Details	**Written Date**	21/07/2026
Actor	User, Admin		
Description	Displays detailed information and specific entry requirements (Syarat Kemasukan) for a selected higher education programme.		
Pre-condition	Programme list is displayed.		

Steps	**Basic Flow:** 1. Actor clicks 'DETAILS' button next to a programme name. 2. System retrieves programme record from Programmes table. 3. System displays Programme Details view showing Institution Name, Programme Level, and itemized entry requirements (Syarat Kemasukan). **Alternate / Exception Flow:** None.
Post-Condition	• Detailed view showing specific entry requirements (Syarat Kemasukan) displayed.
Exception	• Selected programme_id not found in database.
Business Rules	• Accessible to both regular Users and Administrators.

3.4.2.15 Use Case Description Save Recommendation History

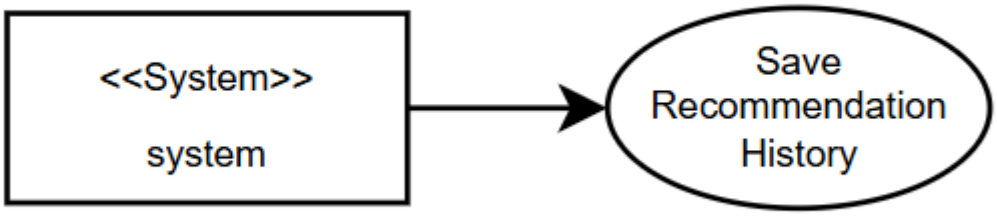


Figure 3.4.2.15 Use Case Description for Save Recommendation History

Table 3.4.2.15 Use Case Description for Save Recommendation History

Use Case ID	UC015	**Author**	Rosli M. A. H.
Use Case Name	Save Recommendation History	**Written Date**	21/07/2026
Actor	System (Automated)		
Description	Stores snapshot of generated recommendation session (date, merit score, dominant trait, matched programmes) into Recommendations table.		
Pre-condition	Programme recommendation generation completes.		

Steps	**Basic Flow:** 1. System captures timestamp, user_id, calculated merit score, dominant RIASEC trait, and matched programme IDs. 2. System inserts snapshot records into Recommendations history database table. **Alternate / Exception Flow:** None.
Post-Condition	• Snapshot containing timestamp, merit score, dominant RIASEC trait, and matched programmes logged in Recommendations table.
Exception	• Database transaction rollback during history snapshot saving.
Business Rules	• Every completed recommendation generation event must create an immutable history record.

3.4.2.16 Use Case Description View Recommendation History

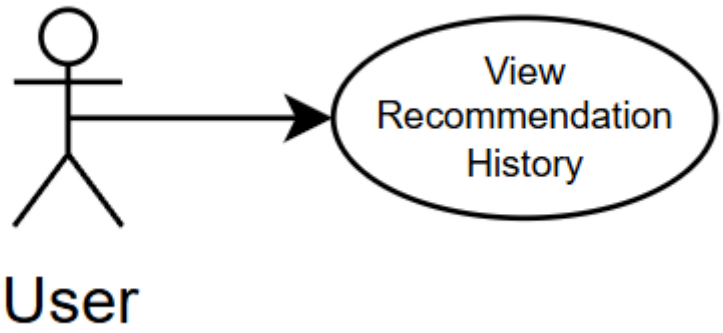


Figure 3.4.2.16 Use Case Description for View Recommendation History

Table 3.4.2.16 Use Case Description for View Recommendation History

Use Case ID	UC016	**Author**	Rosli M. A. H.
Use Case Name	View Recommendation History	**Written Date**	21/07/2026
Actor	User		
Description	Allows user to view past recommendation results and historical SPM/RIASEC test logs.		

Pre-condition	User is logged in.
Steps	**Basic Flow:** 1. User clicks 'History' in sidebar menu. 2. System checks history records for user. 3. System displays 'LIST OF HISTORY' table listing date, merit score, RIASEC dominant trait icon, and 'see details' button. 4. User clicks 'see details' button on a historical record. 5. System displays historical SPM results summary, RIASEC radar graph, and list of matched programmes from that session. **Alternate / Exception Flow:** 2a. User has no prior assessment history: 1. System displays empty state screen showing folder graphic and message 'Your History Is Empty'.
Post-Condition	• History table rendered or empty state banner displayed if no records exist.
Exception	• Corrupted or missing historical session data.
Business Rules	• Users can access only their own recommendation history entries.

3.4.2.17 Use Case Description Manage Programme Information

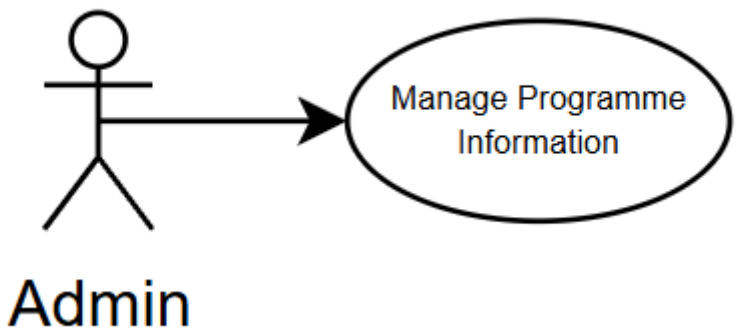


Figure 3.4.2.17 Use Case Description for Manage Programme Information

Table 3.4.2.17 Use Case Description for Manage Programme Information

Use Case ID	UC017	**Author**	Rosli M. A. H.

Use Case Name	Manage Programme Information	**Written Date**	21/07/2026
Actor	Admin		
Description	Allows Administrator to view, add, edit, and delete higher education programme details in the system.		
Pre-condition	Admin is logged into Admin Dashboard.		
Steps	**Basic Flow:** 1. Admin selects 'Programme Info' from sidebar menu. 2. System displays Programme Info Management table with search bar and Level/Institution filter dropdowns. 3. Admin can perform management tasks: a. Add: Clicks '+ ADD NEW PROGRAMME', enters Name, Level, Institution, Entry Requirements, clicks 'ADD'. b. Edit: Clicks Edit icon, modifies details, clicks 'UPDATE'. c. Delete: Clicks Delete icon, confirms on dialog ('ARE YOU SURE WANT TO DELETE THIS PROGRAMME?'). 4. System updates MySQL database and displays success notification banner (e.g., 'NEW PROGRAMME SUCCESSFULLY ADDED', 'PROGRAMME UPDATE SUCCESSFULLY', 'PROGRAMME DELETE SUCCESSFULLY'). **Alternate / Exception Flow:** 3c-1. Admin cancels deletion on dialog: 1. Dialog closes without deleting record.		
Post-Condition	• Programme record added, modified, or soft-deleted in Programmes table.		
Exception	• Database integrity violation when attempting to delete a programme linked to active recommendation rules.		
Business Rules	• Only authenticated ADMIN accounts can perform CRUD operations on programmes. • Required fields: Programme Name, Institution, Level, and Entry Requirements.		

3.4.2.18 Use Case Description Manage RIASEC Questions

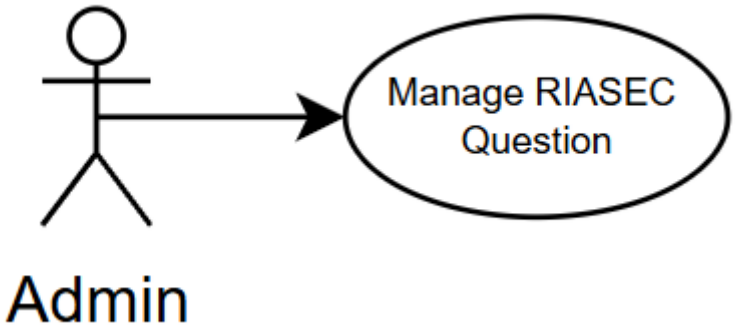


Figure 3.4.2.18 Use Case Description for Manage RIASEC Questions

Table 3.4.2.18 Use Case Description for Manage RIASEC Questions

Use Case ID	UC018	**Author**	Rosli M. A. H.
Use Case Name	Manage RIASEC Questions	**Written Date**	21/07/2026
Actor	Admin		
Description	Allows Administrator to view, add, edit, and delete RIASEC personality assessment questions.		
Pre-condition	Admin is logged into Admin Dashboard.		
Steps	**Basic Flow:** 1. Admin selects 'RIASEC question' from sidebar menu. 2. System displays RIASEC Question Management table with search and category filter dropdown. 3. Admin can perform management tasks: a. Add: Clicks '+ ADD NEW QUESTION', fills Question Text, selects Category (R/I/A/S/E/C), clicks 'ADD'. b. Edit: Clicks Edit icon, updates question text/category, clicks 'UPDATE'. c. Delete: Clicks Delete icon on details view, confirms on dialog ('ARE YOU SURE WANT TO DELETE THIS QUESTION?'). 4. System updates RIASEC_Questions table and shows success notification banner. **Alternate / Exception Flow:** 3c-1. Admin cancels deletion dialog: 1. Dialog closes without deleting question.		

Post-Condition	• Questions added, updated, or removed in RIASEC_Questions table.
Exception	• Database failure during question record updates.
Business Rules	• Question category must strictly belong to one of the 6 RIASEC types (R, I, A, S, E, C). • Question text cannot be left blank.

3.4.2.19 Use Case Description Manage Recommendation Rules

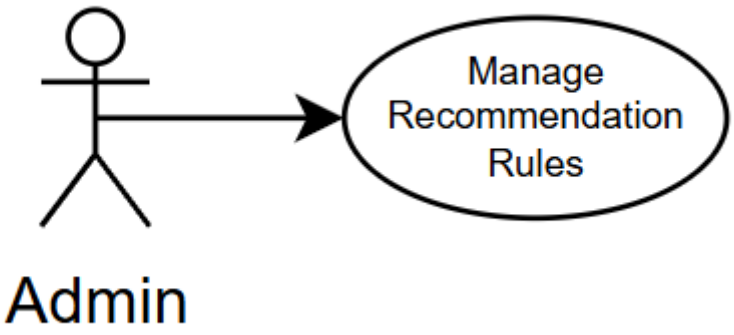


Figure 3.4.2.19 Use Case Description for Manage Recommendation Rules

Table 3.4.2.19 Use Case Description for Manage Recommendation Rules

Use Case ID	UC019	**Author**	Rosli M. A. H.
Use Case Name	Manage Recommendation Rules	**Written Date**	21/07/2026
Actor	Admin		
Description	Allows Administrator to create, update, and remove recommendation rules mapping RIASEC types and minimum grades to programmes.		
Pre-condition	Admin is logged into Admin Dashboard.		
Steps	**Basic Flow:** 1. Admin selects 'Recommendation rules' from sidebar menu. 2. System displays Recommendation Rules Management table. 3. Admin can perform management tasks: a. Add: Clicks '+ ADD NEW RULE', selects RIASEC Type, Programme Name from existing list, Minimum Grade, clicks 'ADD'. b. Edit: Clicks Edit icon, modifies rule parameters, clicks 'UPDATE'.		

	c. Delete: Clicks Delete icon, confirms on dialog ('ARE YOU SURE WANT TO DELETE THIS RULE?'). 4. System updates Recommendation_Rules table and displays success notification banner. **Alternate / Exception Flow:** 3c-1. Admin cancels rule deletion: 1. Dialog closes without removing rule.
Post-Condition	• Rule record added, modified, or removed in Recommendation_Rules table.
Exception	• Conflict or duplication in recommendation rule mapping.
Business Rules	• Mapped programme must exist in the Programmes table. • Mapped RIASEC code must be valid (R, I, A, S, E, C).

3.4.2.20 Use Case Description Manage User Accounts

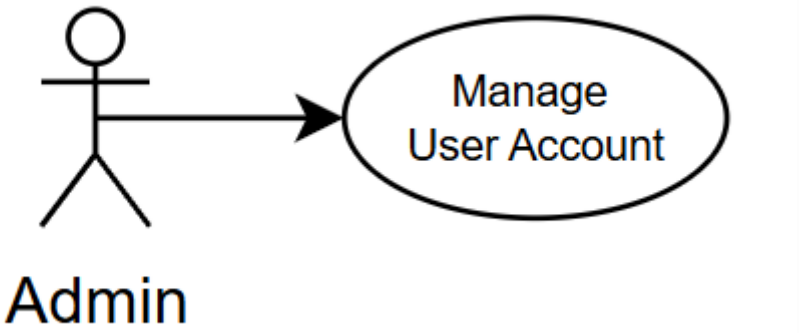


Figure 3.4.2.20 Use Case Description for Manage User Accounts

Table 3.4.2.20 Use Case Description for Manage User Accounts

Use Case ID	UC020	**Author**	Rosli M. A. H.
Use Case Name	Manage User Accounts	**Written Date**	21/07/2026
Actor	Admin		
Description	Allows Administrator to view registered user accounts, search/filter users, edit user role/details, and activate/deactivate user accounts.		
Pre-condition	Admin is logged into Admin Dashboard.		

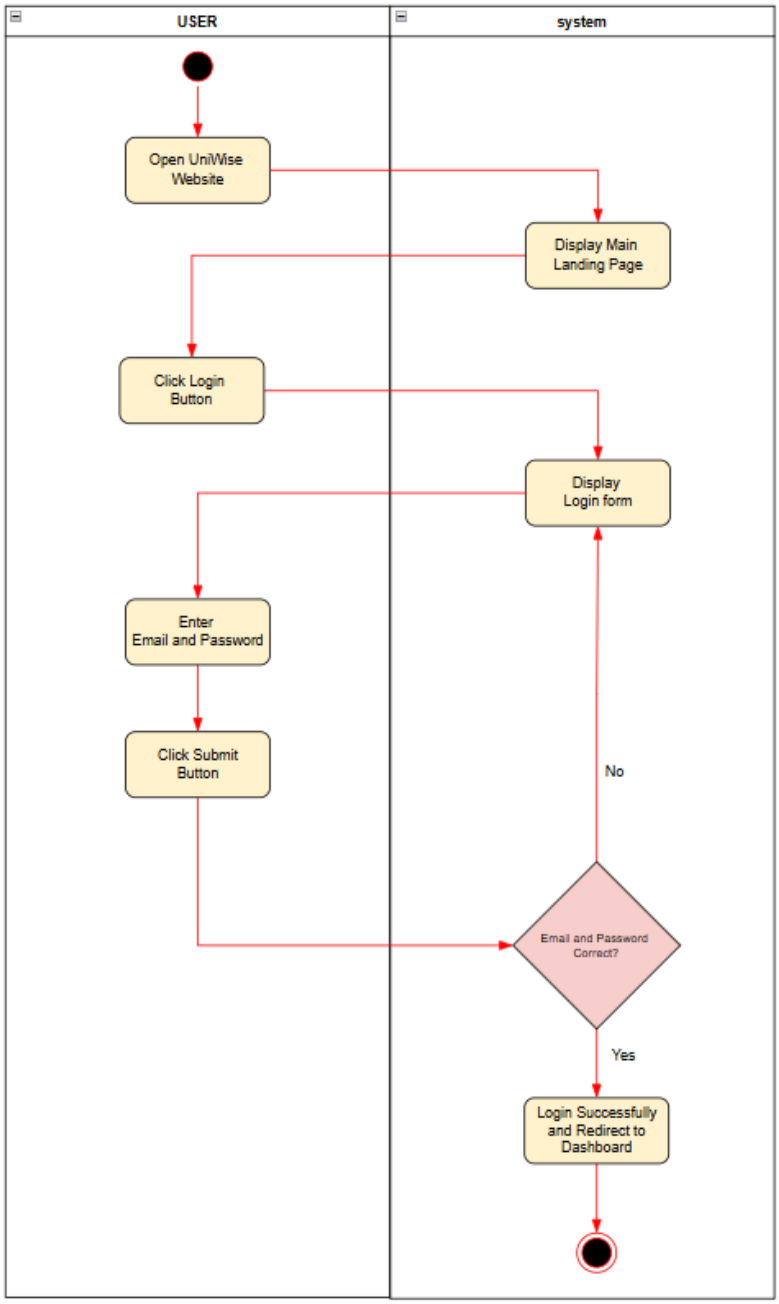
Steps	**Basic Flow:** 1. Admin selects 'User Accounts' from sidebar menu. 2. System displays User Account Management table listing Name, Email Address, Role (STUDENT/ADMIN), and Status (ACTIVATED/DEACTIVATED). 3. Admin clicks Action icon for a specific user to open User Details view. 4. Admin can perform actions: a. Update Info: Clicks 'UPDATE INFO', changes Name, Email, or Role dropdown, clicks 'SAVE CHANGES'. b. Deactivate/Activate: Clicks 'DEACTIVATE ACCOUNT' button, confirms on popup dialog ('ARE YOU SURE WANT TO DEACTIVATE THIS ACCOUNT?'). 5. System updates user state in database and shows success banner ('USER INFO UPDATE SUCCESSFULLY' / 'ACCOUNT DEACTIVATE SUCCESSFULLY'). **Alternate / Exception Flow:** 4b-1. Admin cancels deactivation dialog: 1. System closes confirmation popup without changing account status.
Post-Condition	• Target user details updated or account status toggled between ACTIVATED and DEACTIVATED.
Exception	• Admin attempts to deactivate their own active administrator account.
Business Rules	• Only authenticated ADMIN accounts can manage user account statuses. • Deactivated users are blocked from logging into the system.

3.5 Data and Behavioral Models

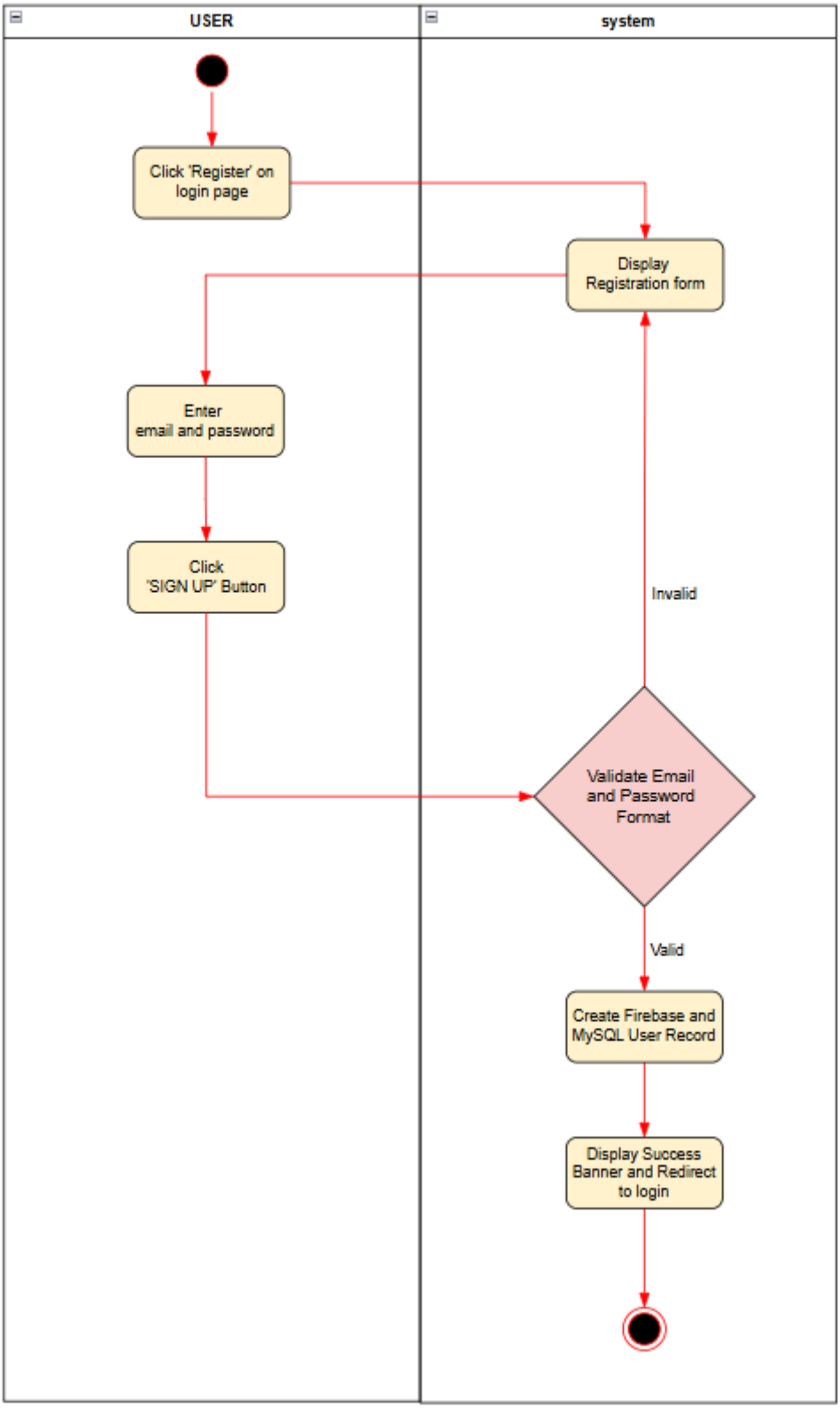
This section presents the structural, behavioral, and data persistence models of the UniWise system. The system behavior is specified using Activity Diagrams with swimlane partitions, while the system structure and database schema are represented via a Class Diagram and a Logical Database Table.

3.5.1 Activity Diagrams

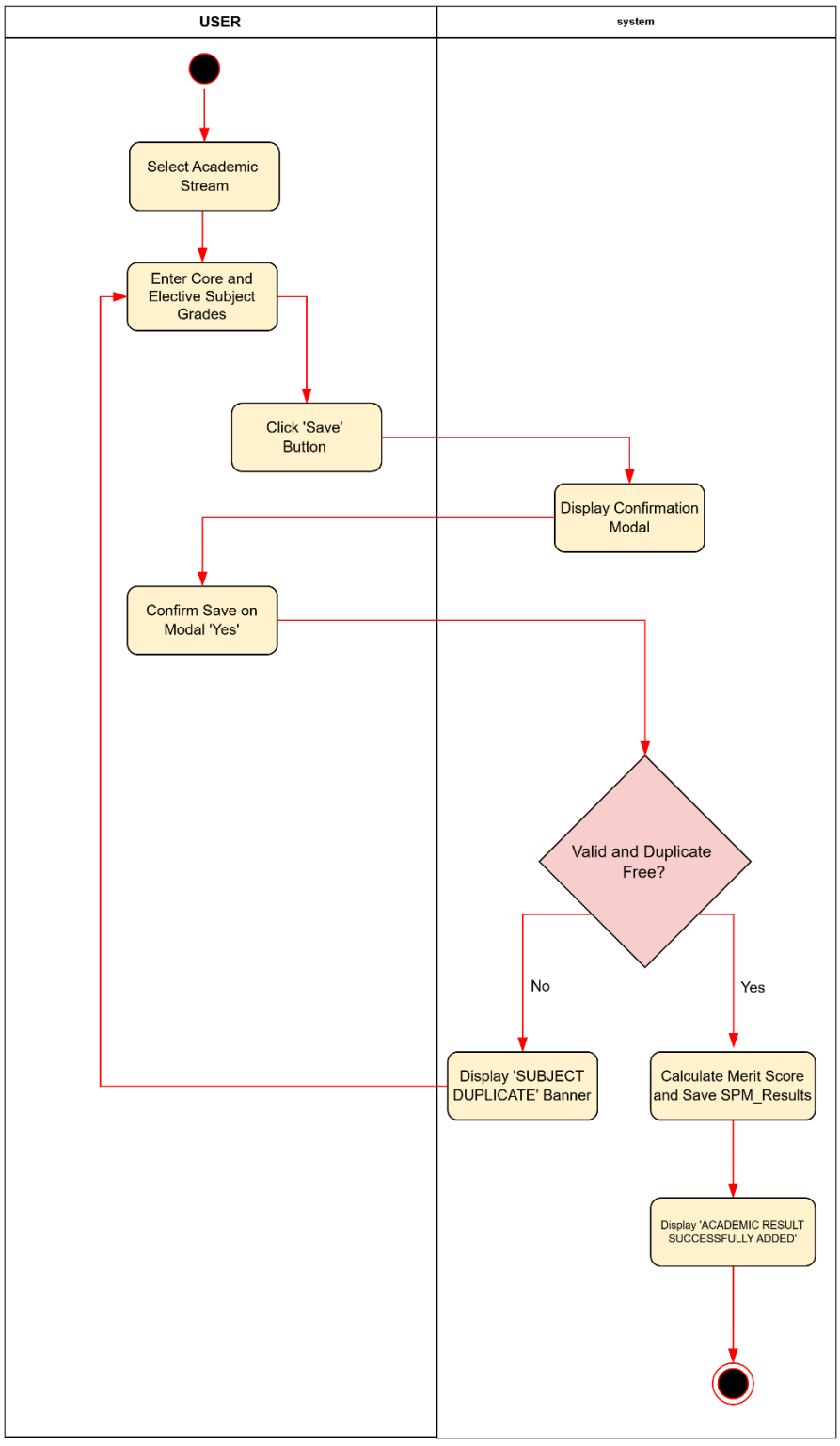
3.5.1.1 Login



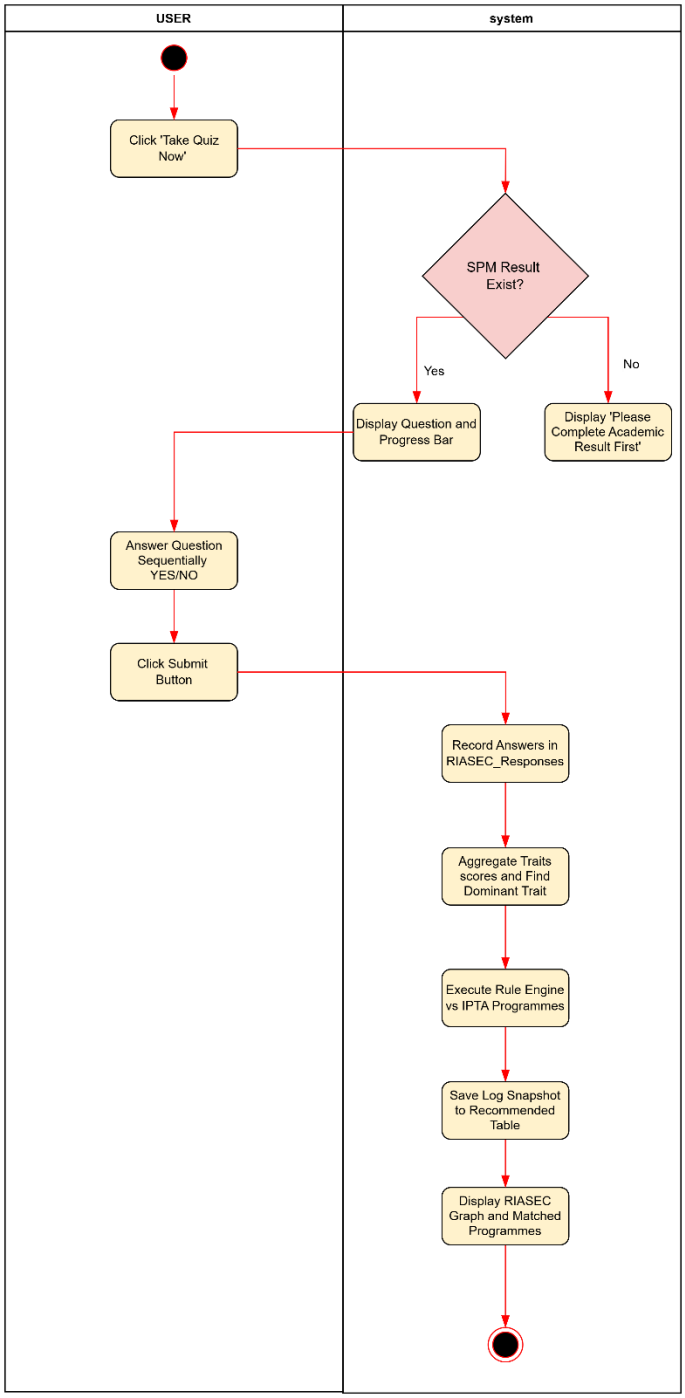
3.5.1.2 Register



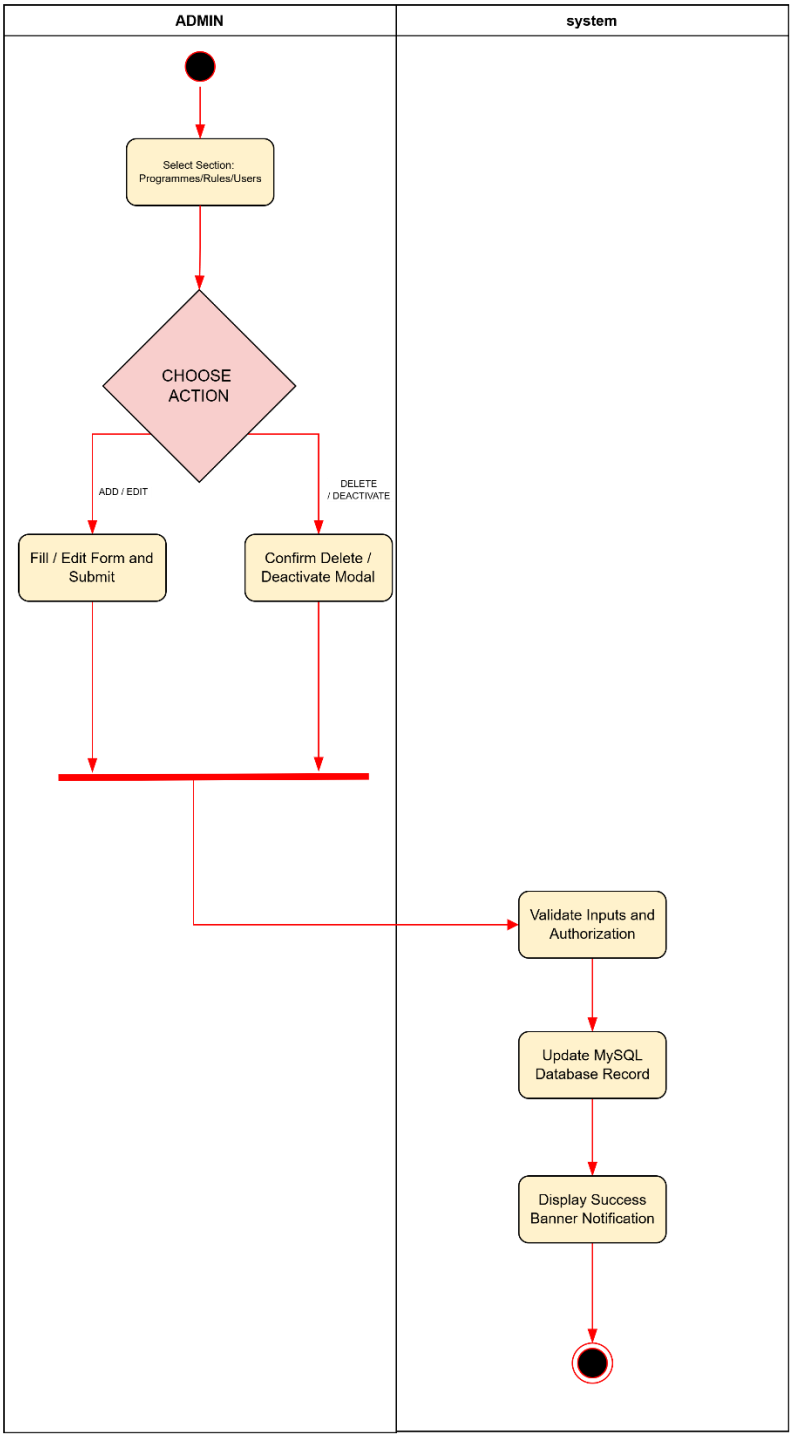
3.5.1.3 Enter and Validate SPM Results



3.5.1.4 Take RIASEC Assessment & Generate Recommendation



3.5.1.5 Admin Management Activity



3.5.2 Class Diagram

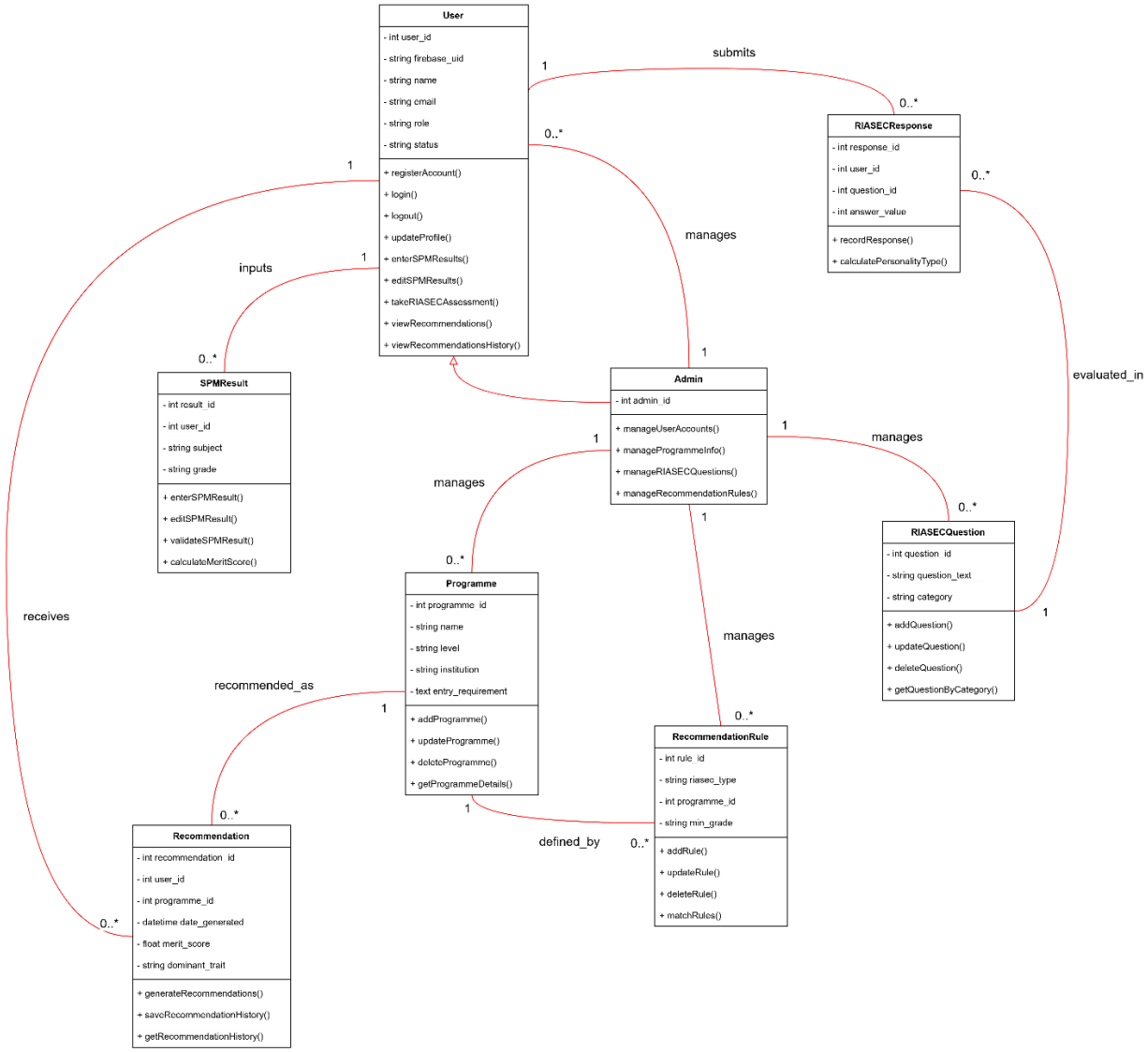


Figure 3.5.2: UniWise Class Diagram

The User class represents the primary user entity within the UniWise system, holding core profile attributes such as user_id, name, email, role, and status, along with authentication operations like registration, login, logout, and profile updates. The Admin class specializes from User via class inheritance, inheriting shared credentials while extending system capabilities to include management functions for user accounts, programme information, RIASEC assessment questions, and recommendation rules. Authentication credentials are managed externally via Firebase Authentication; the User class stores a firebase_uid field as a unique reference linking each MySQL user record to its corresponding Firebase Authentication account.

A User can submit zero or more SPMResult records and zero or more RIASECResponse entries. Each RIASECResponse references a specific RIASECQuestion stored in the database. The rule-based recommendation engine evaluates these inputs against RecommendationRule criteria to generate Recommendation records, which link a User to eligible Programme options offered by public IPTA institutions. The Admin class possesses administrative association links to manage and update records within the Programme, RIASECQuestion, RecommendationRule, and User classes.

3.5.3 Logical Database

The logical database design below summarises the main tables and key fields used to support the system's functions.

Table Name	Key Fields	Description
Users	user_id (PK), firebase_uid (unique), name, email, role	Stores profile information for users and administrators. Authentication is handled externally via Firebase Authentication.
SPM_Results	result_id (PK), user_id (FK), subject, grade	Stores each subject and grade entered by a user.
RIASEC_Questions	question_id (PK), question_text, category	Stores the RIASEC assessment questions maintained by administrators.
RIASEC_Responses	response_id (PK), user_id (FK), question_id (FK), answer_value	Stores a user's answers to the RIASEC assessment.
Programmes	programme_id (PK), name, level, institution, entry_requirement	Stores Diploma, Foundation, and Certificate programme details offered by IPTA.
Recommendation_Rules	rule_id (PK), riasec_type, programme_id (FK), min_grade	Stores the rules that map RIASEC types and academic eligibility to programmes.
Recommendations	recommendation_id (PK), user_id (FK), programme_id (FK), date_generated	Stores each recommendation result generated for a user, forming the recommendation history.

Table.3.5.3 UniWise Logical Database

3.6 Design Constraint Requirements

The following design constraints define the technical and architectural boundaries within which the UniWise system must be developed.

DC ID	Design Constraint
DC001	The system's frontend shall be developed using React.js.
DC002	The system's backend shall be developed using the Laravel framework for managing academic, RIASEC, programme, and recommendation data.
DC003	The system shall use MySQL as its relational database management system for all data except user authentication credentials.
DC004	Communication between the frontend and backend shall follow RESTful API design principles using JSON.
DC005	The backend shall follow the Model-View-Controller (MVC) architectural pattern.
DC006	User authentication (registration, login, and password management) shall be handled via Firebase Authentication, which manages credential storage and hashing independently of the MySQL database.
DC007	The system shall be deployed on a web server that supports PHP and MySQL.
DC008	The user interface shall be built using a responsive design approach to support desktop, tablet, and mobile screens.
DC009	The system shall handle and store personal and academic data in accordance with applicable data protection practices.
DC010	The system shall verify Firebase ID Tokens on the Laravel backend for every authenticated API request to confirm user identity before granting access to protected resources.

Table 3.6 Design Constraint Requirements

3.7 Quality Requirements

This section defines the non-functional quality requirements that the UniWise system must satisfy.

QR ID	Category	Requirement
QR001	Performance	The system shall generate a programme recommendation within 5 seconds of a request under normal load.
QR002	Performance	The system shall support at least 100 concurrent users without significant performance degradation.
QR003	Usability	The system interface shall be simple and intuitive, allowing a first-time user to complete the recommendation process without additional training.
QR004	Reliability	The system shall be available at least 99% of the time, excluding scheduled maintenance.
QR005	Security	The system shall encrypt sensitive data, including passwords and personal information, both in storage and in transit.
QR006	Security	Administrative functions shall be restricted to authenticated administrator accounts only.
QR007	Maintainability	The system shall be built with a modular codebase to allow recommendation rules and programme data to be updated without major code changes.
QR008	Portability	The system shall be accessible through modern web browsers on desktop and mobile devices via responsive design.
QR009	Scalability	The system architecture shall support future growth in the number of users, programmes, and recommendation rules.

Table 3.7 Quality Requirements